COMP 3613 — ASSIGNMENT 2

MyAdvisor

Implementation, Test Planning and Verified Test Reporting

Submission detail	Verified value
Student name	Cameron Ramesar
University ID	816044668
Submission document date	27 October 2026
Repository	https://github.com/Camcoon/MyAdvisorWithTests

Readiness: **NOT SUBMISSION-READY — material audit issues remain open.**

This is the requested single-PDF compilation of the implemented project and retained evidence. It does not certify acceptance. UAT: 1 Pass, 0 Fail, 19 Not Run, including 18 partially verified cases. No application code or test results were regenerated.

Identity source: explicit student statement in the retained Assignment 1 report-export transcript, corroborated by the original PDF cover. The eight-digit ID displayed in demo screens is seeded data. University/campus and full course title are omitted because the reviewed assignment materials do not verify their wording.

2. Table of contents

The numbered page footer is the PDF page number, including the cover. Section 1 is the cover on page 1. Landscape evidence pages continue the same numbering.

1. Cover page 1

2. Table of contents 2

3. Project description 4

3.1 Evidence and audit review 4

4. Use case diagram 5

5. Entity relationship diagram 6

6. Original wireframe and implemented interface 8

7. Test planning 13

7.1 User Acceptance Test Plan 13

7.2. Unit Test Plan 27

7.3. Integration Test Plan 37

8. Test implementation 48

8.1 Environment and CLI 48

8.2 Unit implementation and independent oracles 48

8.3 Integration implementation and database isolation 48

8.4 Authentication, authorization and boundaries 49

8.5 Postman Collection Runner strategy 49

8.6 Test-script example from the actual collection 49

9. Test reporting 50

9.1 Unit Testing Results 51

9.2 Integration Testing Results 58

9.3 Complete Automated Test Suite 66

9.4 Postman API Results 67

9.5 Manual UAT Results 80

10. Requirements traceability 119

10.1 Requirement-level matrix R01–R14 121

10.2 R15 public JSON contracts 126

11. Defects, limitations and audit disposition 127

11.1 Submission identity and PDF assembly update 132

11.2 Environment problems, corrections and retests 132

11.3 Audit reproduction records (not new UAT failures) 133

11.4 Official rubric status at assembly 134

12. Conclusion 138

13. References and GitHub repository 139

13.1 Evidence and submission availability 139

3. Project description

MyAdvisor supports academic advising through three connected workflows: WF1 — Track Degree Progress (Student); WF2 — Plan Semester Course Selections (Student); and WF3 — Review and Approve Course Plans (Advisor). Students inspect recorded academic progress, select eligible courses and submit semester plans. Advisors review a shared Pending queue and return a decision and optional feedback.

Assignment 1 established the Student and Advisor dashboards, degree/program and course models, three-panel wireframe, prerequisite checks, plan submission, decisions, cancellation and resubmission. Assignment 2 adds explicit JSON API contracts, authenticated role and ownership boundaries, isolated unit tests, real-authentication integration tests, a 90-request Postman collection, manual acceptance records and cross-method traceability. The original artifacts remain intact.

Actor	Implemented functionality
Student (regular_user)	Login; Home, Degree Progress, Plan Courses, Messages and Profile; course search/filter and prerequisite eligibility; review/submit selections; revise returned plans; remove Pending courses; cancel permitted plans and create replacements.
Advisor (admin)	Login; Advisor workspace and shared Pending queue; inspect Student/program/course totals; Approve, Reject or Request Changes with optional feedback.

Plans use nine fixed Year 1–3 / semester 1–3 options; semester 3 is summer. Pending and Approved block duplicate active submissions; Changes Requested and Rejected are resubmitted under the same ID. Cancellation permits Pending, Approved and Rejected, retains history and allows a fresh plan ID. The current authorization defect affecting HTML mutations is detailed in Section 11.

Layer	Verified technology / purpose
Backend	Python; FastAPI; Pydantic request/response validation; SQLModel/SQLAlchemy persistence; SQLite in retained local tests.
Frontend	Jinja2 HTML templates; Bootstrap/AdminLTE navigation; CSS and JavaScript selection/review controls.
Authentication	Password hashing with pwdlib/Argon2; signed JWT login cookie; current database role and ownership checks.
Verification	pytest 9.1.1; FastAPI/Starlette TestClient; mocks/autospec; isolated SQLite; Postman Collection Runner and v2.1 JSON export.

Source: docs/report.md; app/models; app/routers; tests; docs/assignment2/api.md.

3.1 Evidence and audit review

The official Assignment 2 wording is the student-supplied brief retained in official-assignment-brief.md, with four equally weighted components: Implementation, Test Plans, Test Implementation and Test Reporting (25% each). The assessment is weighted 15%. Both linked Google templates remain inaccessible on this review; no unverified template requirement is invented. The Assignment 1 rubric and historical competency scores are excluded.

No Critical finding is verified. Open High findings F01–F05 concern authorization, published collection/evidence availability, incomplete UAT and degree requirements. The PDF assembly portion of F11 is addressed here; verified identity was recovered from original materials. This does not close the five High findings.

4. Use case diagram

The original UML diagram below is reproduced from docs/diagrams/use-case.png. Student use cases cover progress, selection, submission and cancellation; Advisor use cases cover review and decisions. The diagram is an Assignment 1 design artifact, not an execution result.

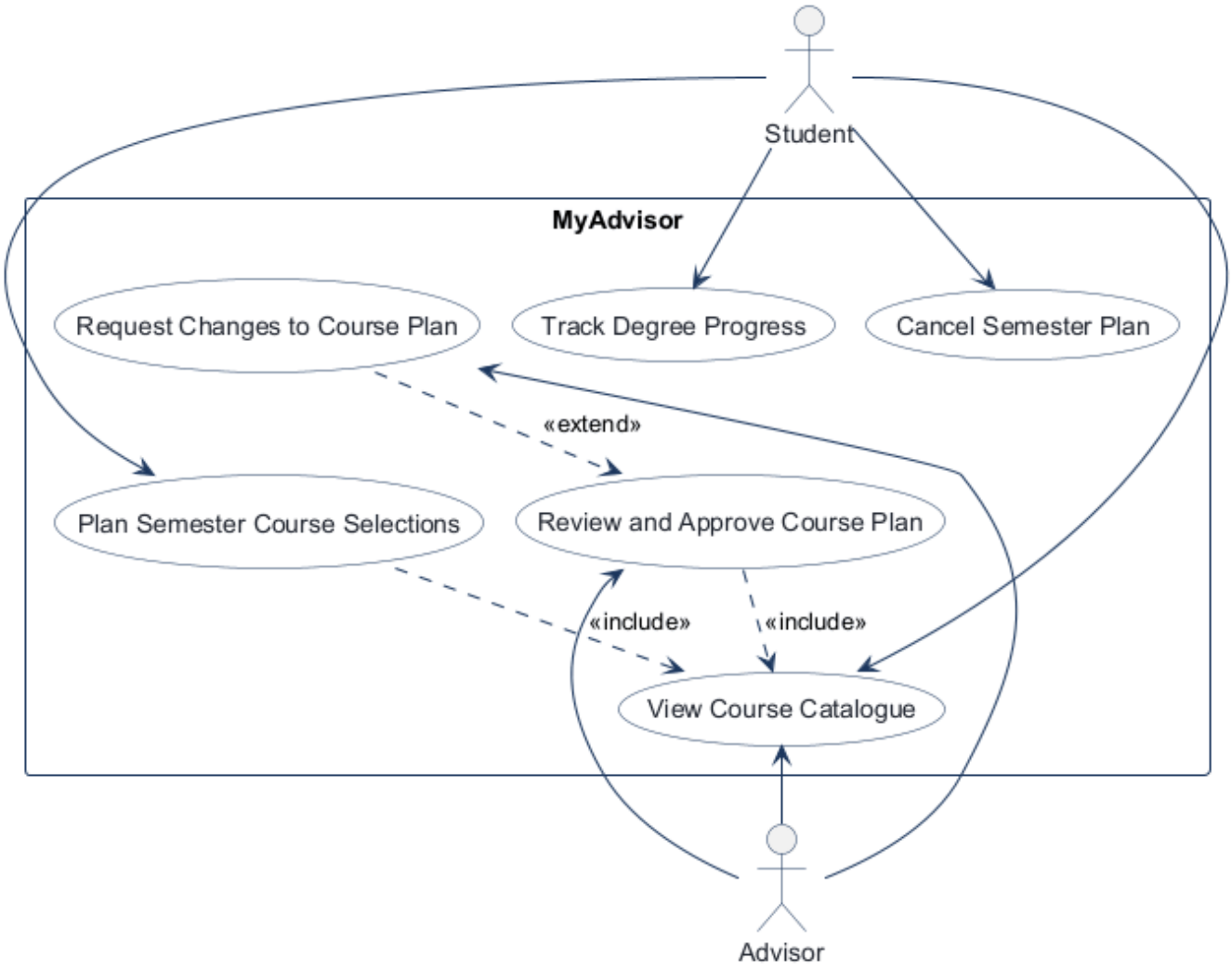


Figure 4.1. Genuine Assignment 1 Use Case Diagram.

Course selection and submission depend on eligibility checks; the Advisor reviews the selected courses before approval/rejection/change requests. Resubmission returns a revised plan to the same review workflow. Login supports both actors; hiding navigation is not sufficient authorization, as the audit demonstrates.

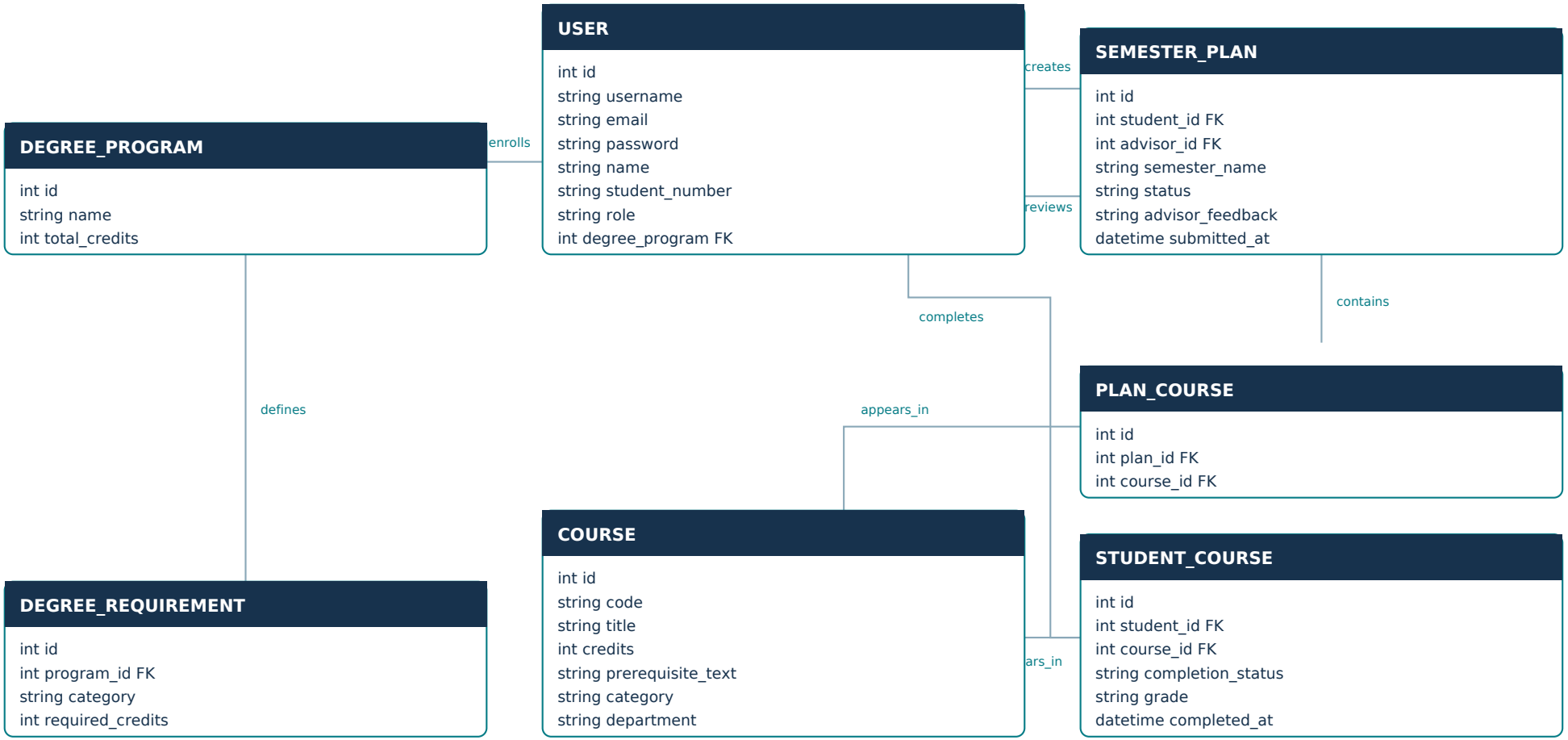
5. Entity relationship diagram

The seven-entity ERD in the original report matches the current SQLModel entities and foreign keys. The next landscape page renders that genuine Mermaid source as vector boxes; the relationship table retains its exact cardinalities and labels. No invented entities or changed degree requirements are introduced.

Relationship from original ERD	Meaning
DEGREE_PROGRAM o|--o{ USER : enrolls	A User may have zero or one program; a program can have many Users.
DEGREE_PROGRAM ||--o{ DEGREE_REQUIREMENT : defines	Each requirement belongs to one program.
USER ||--o{ SEMESTER_PLAN : creates	Each plan has one Student creator.
USER o|--o{ SEMESTER_PLAN : reviews	A plan may have zero or one assigned Advisor; a User may review many plans.
USER ||--o{ STUDENT_COURSE : completes	Academic record joins a Student and a course, with status/grade/date.
COURSE ||--o{ STUDENT_COURSE : appears_in	Each academic record refers to one course.
COURSE ||--o{ PLAN_COURSE : appears_in	Each selected-plan row refers to one course.
SEMESTER_PLAN ||--o{ PLAN_COURSE : contains	PlanCourse implements the plan/course association.

StudentCourse has a unique (student_id, course_id) pair; PlanCourse has a unique (plan_id, course_id) pair; DegreeRequirement is unique by (program_id, category). Plan status is Pending, Approved, Rejected, Changes Requested or Cancelled. The runtime SQLite foreign-key enforcement gap remains F07; declared relationships do not prove runtime enforcement.

Figure 5.1. Original seven-entity ERD



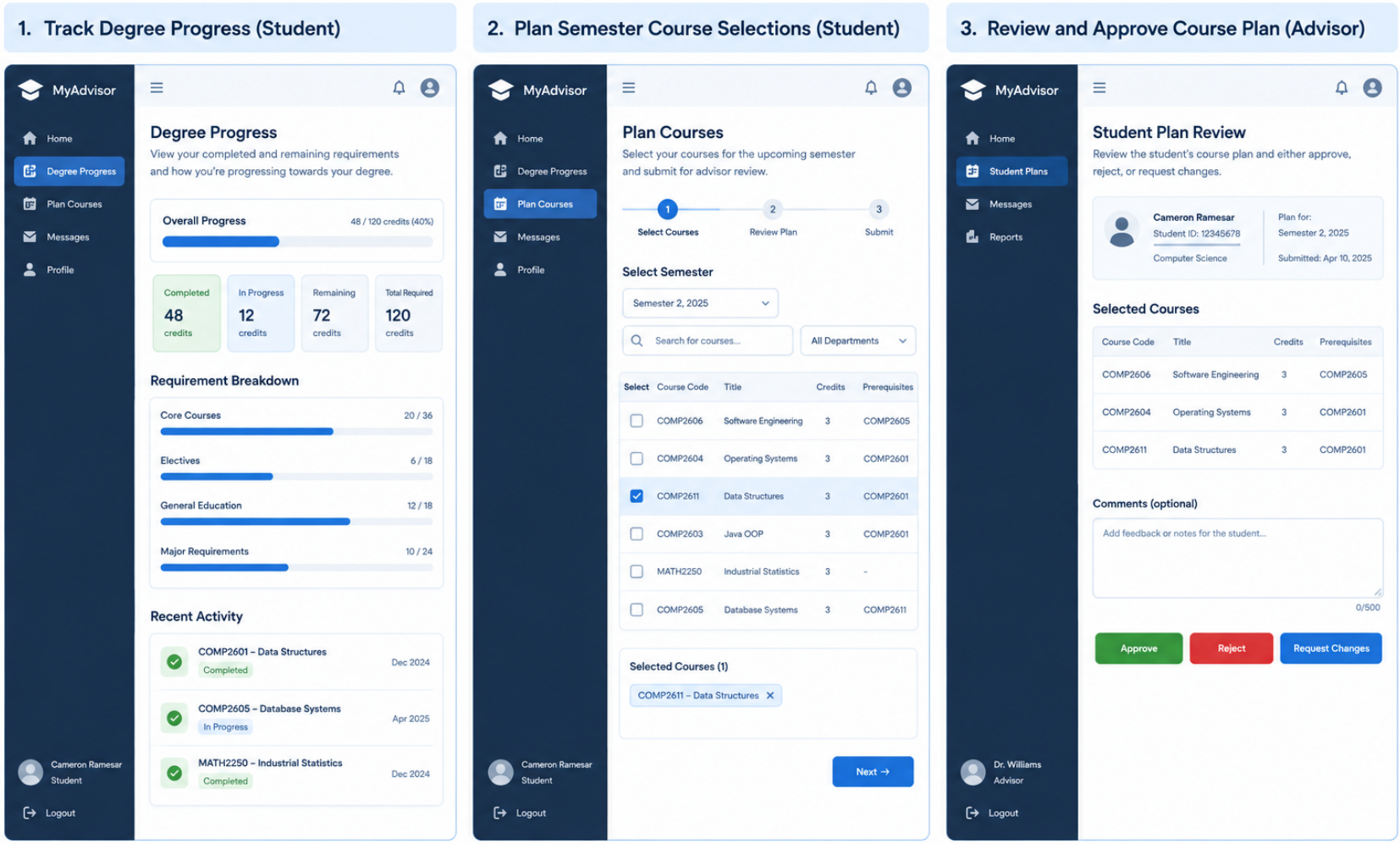
Vector rendering of docs/report.md Mermaid source; all fields retained. Connecting lines indicate related boxes; exact cardinalities and relationship names are in the preceding table. Quoted status/semester constraints are explained there.

6. Original wireframe and implemented interface

The original three-panel Assignment 1 wireframe is preserved below. Panel 1 becomes Degree Progress; panel 2 becomes Plan Courses, selection count and Review Selected Courses; panel 3 becomes Advisor Review with Approve Plan, Reject Plan and Request Changes. Home and Messages expose the latest plan and decision feedback. Cancellation and revision extend the original workflow and are tested explicitly.

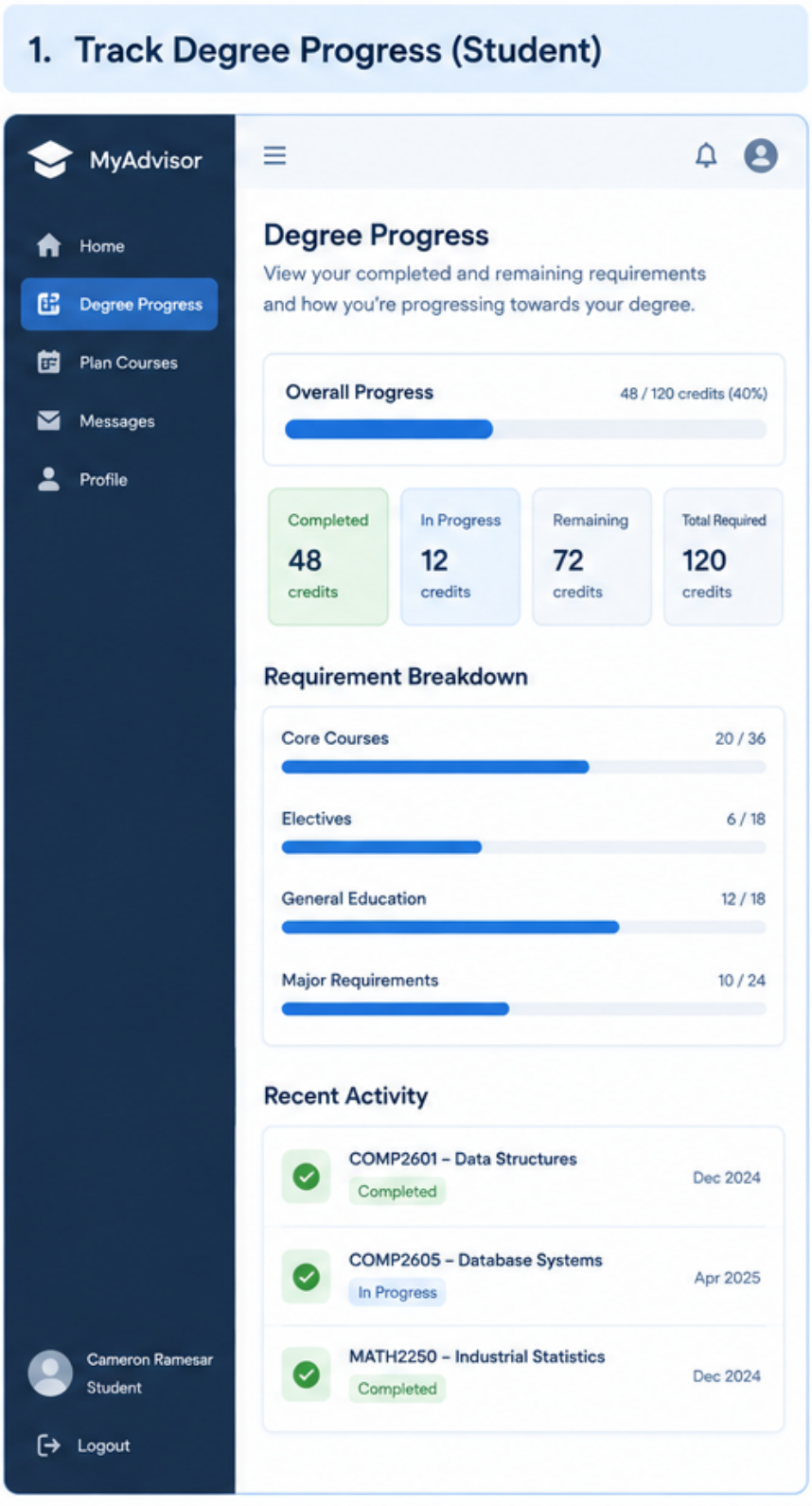
The wireframe’s example 120-credit totals and calendar-year labels are illustrative design data. Executed screens use the current 93-credit degree and nine fixed semester labels. The discrepancy between 93 degree credits and 120 category credits remains unresolved; it is not justified by the mockup.

Figure 6.1. Original Assignment 1 wireframe



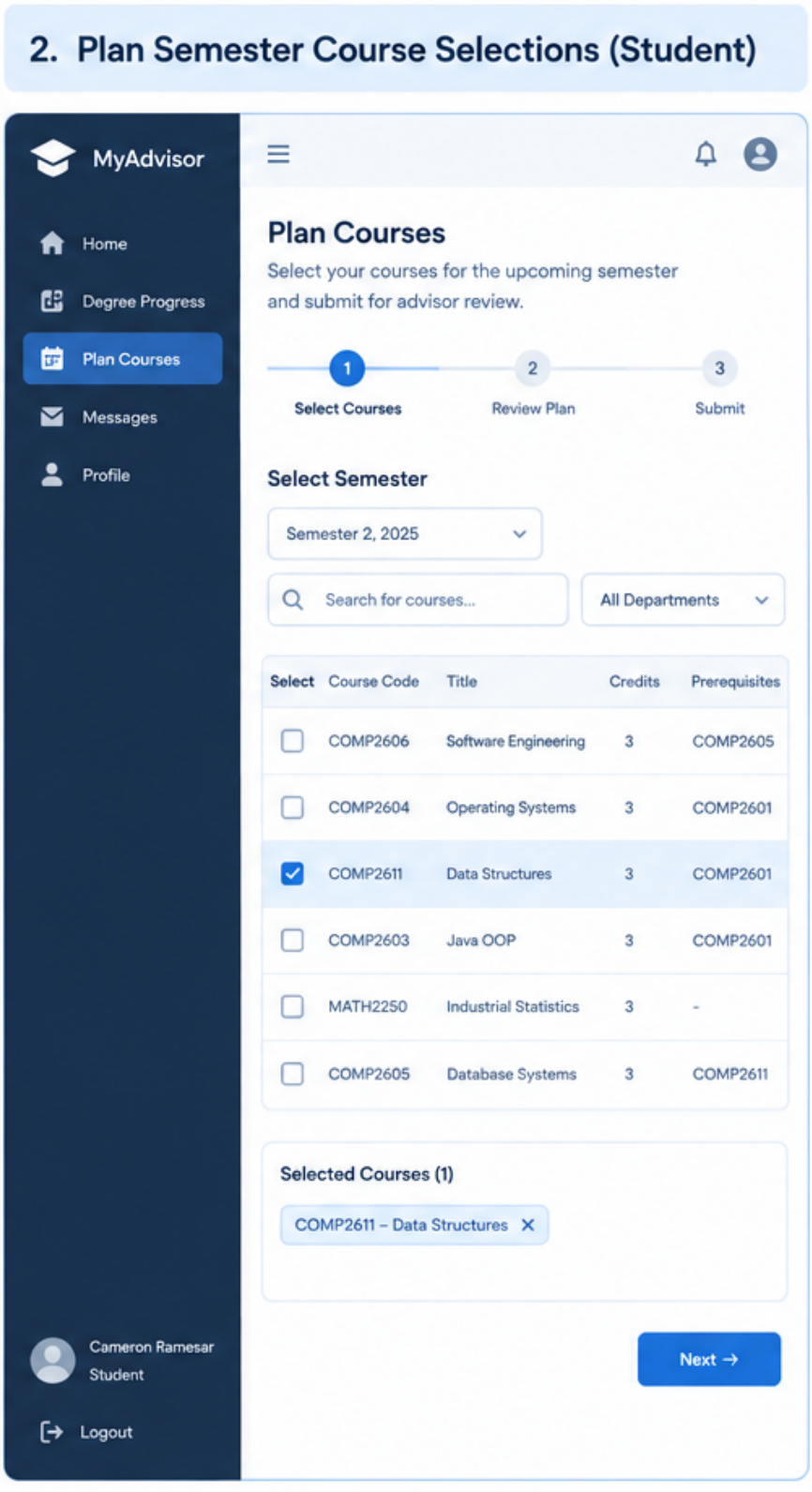
Original complete three-panel image. Source: docs/wireframes/Three-Panel Student Advisor Dashboard Mockup.png.

Figure 6.2. Wireframe panel 1 — enlarged original detail



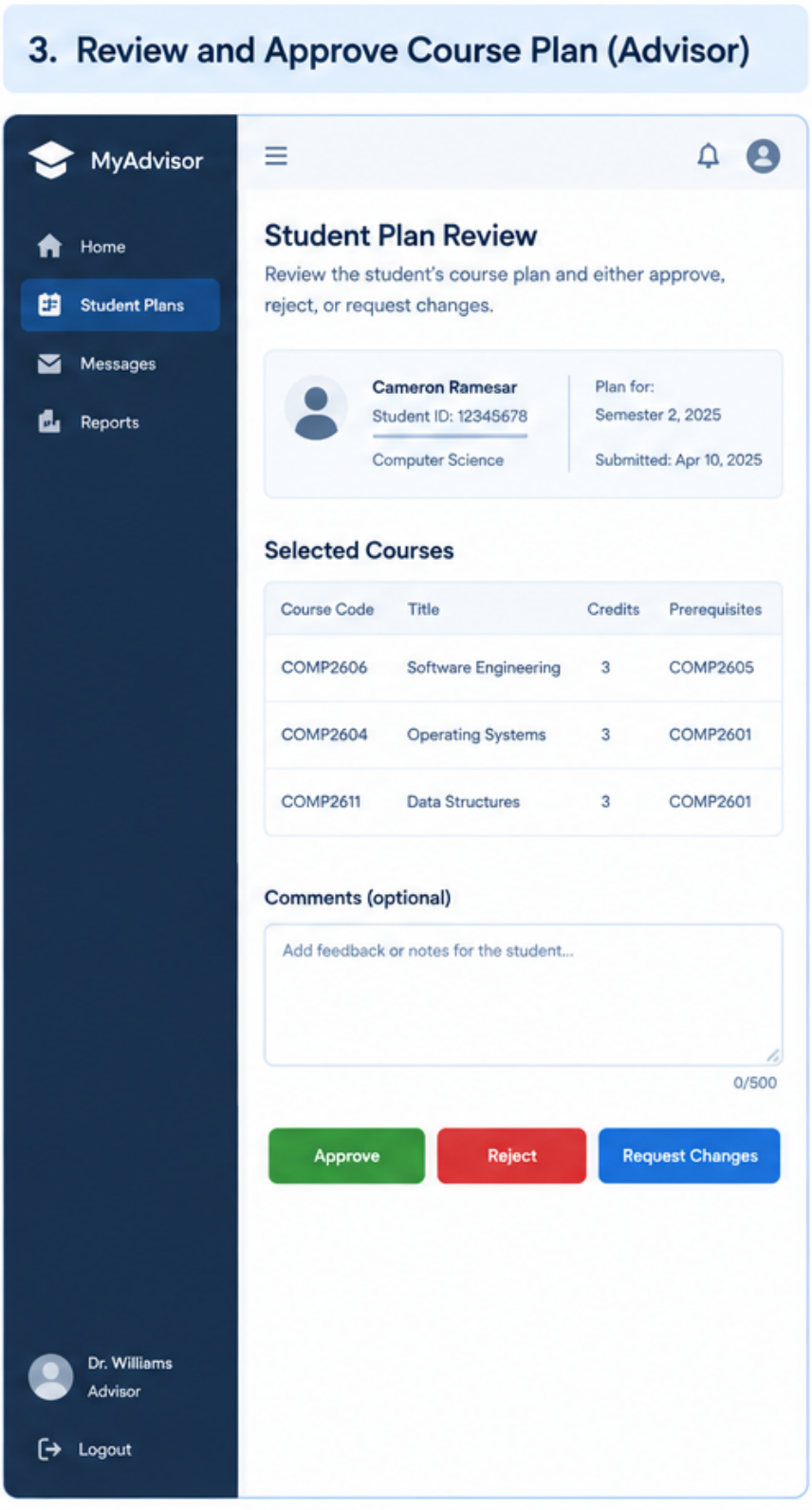
Unmodified pixel excerpt of the original wireframe, enlarged for readability; illustrative values are not observed test results.

Figure 6.3. Wireframe panel 2 — enlarged original detail



Unmodified pixel excerpt of the original wireframe, enlarged for readability; illustrative values are not observed test results.

Figure 6.4. Wireframe panel 3 — enlarged original detail



Unmodified pixel excerpt of the original wireframe, enlarged for readability; illustrative values are not observed test results.

7. Test planning

Detailed specifications below preserve the genuine case IDs, inputs, steps and expected outcomes. Plan preparation records and current execution reports have different dates. Historical Not Run fields are not used as the latest result. Original case inputs remain distinct from the actual three-course UAT substitution. Local R/UT/IT identifiers are project traceability labels, not official rubric IDs.

7.1 User Acceptance Test Plan

Source: docs/assignment2/test-plans/uat-test-plan.md; reports/uat-execution.md.

Verify that a Student can understand degree progress, choose eligible courses, submit and revise semester plans, see advisor decisions, and cancel plans where permitted; and that an Advisor can review submissions and request changes, approve, or reject them. All execution is manual through the browser. Expected results remain the original planned oracles. Actual-result rows below distinguish observed screenshot states from unverified checks.

Original Assignment 1 reference: Three-Panel Student Advisor Dashboard Mockup.

Reference	Original wireframe component	Current browser equivalent
W1	Panel 1: Track Degree Progress (Student); Overall Progress, four credit cards, Requirement Breakdown, Recent Activity	**Degree Progress**; Overall progress, Completed, In progress, Remaining, Total required, Requirement breakdown, Recent course activity
W2	Panel 2: Plan Semester Course Selections (Student); semester dropdown, search, department filter, course checkboxes, Selected Courses, Next, Review Plan, Submit	**Plan Courses**; Semester, Search by code or course name, All departments, course checkboxes, selection count, Review Selected Courses, Back to course selection, Submit Plan for Advisor Review
W3	Panel 3: Review and Approve Course Plan (Advisor); Student Plans, student summary, selected-course table, Comments (optional), Approve, Reject, Request Changes	**Advisor Review**; Student plan review, student summary, Selected courses, Comments (optional), Approve Plan, Reject Plan, Request Changes
W4	Student Messages navigation in panels 1–2; feedback implied by panel 3 decisions	**Messages** and Plan Courses status/feedback; Home latest-plan summary
W5	Extension of panel 2 planning and panel 3 decision workflow; cancellation/resubmission controls are not drawn explicitly	**Cancel Plan**, browser confirmation, editable returned plan, Resubmit Plan for Advisor Review

Implementation references: student template, advisor template, navigation, workflow rules and demo data, student routes, and advisor routes. Labels and sample values in the original mockup are illustrative: use the actual seeded course codes, semesters, and degree totals below.

Reviewed the original wireframe visually, current Student/Advisor templates and sidebar/login, HTML workflow routes, auth/error handling, service/repository transitions and seed data, Assignment 1 workflow definitions, project setup, API documentation, and Postman isolation instructions.

• The three original workflows are WF1 Track Degree Progress (Student), WF2 Plan Semester Course Selections (Student), WF3 Review and Approve Course Plan (Advisor).

• The wireframe's **Next**, **Student Plans**, **Approve** and **Reject** correspond to **Review Selected Courses**, **Advisor Review**, **Approve Plan** and **Reject Plan**. Use current labels. Login is supporting authentication, not a drawn wireframe panel. Cancellation and resubmission extend the original workflows; their controls are not drawn explicitly.

• No Reports screen, bell notifications, inline selected-course chips, academic-record editor, separate Advisor login page, or semester-year text entry is required by these tests. Use the implemented sidebar, checkboxes, selection count and review step.

• Semesters are the nine fixed **Year 1–3 semester 1–3** choices, not the mockup's calendar year. Semester 3 is summer. Seeded academic values replace the mockup's illustrative values.

• Only Pending plans appear in Advisor Review. Pending and Approved plans lock standard selection. Rejected and Changes Requested plans are editable. Cancellation is allowed for Pending, Rejected and Approved, and is absent for Changes Requested and Cancelled.

• Resubmitting a returned plan clears its previous feedback and Advisor assignment. Cancelling retains history but clears the active selection; the next submission creates a new plan even though the button says **Resubmit Plan for Advisor Review**.

• Source review identifies category requirements totalling 120 versus degree total 93. This is an unresolved acceptance concern, not an observed UAT failure. Do not silently change the expected total or mark acceptance complete merely because the seed matches.

• Browser role coverage is limited to real navigation and direct GET access. Hidden links alone do not prove backend security; no browser-only claim is made for cross-student ownership or all write endpoints. Automated/API checks are separate.

Test data baseline

These values come from the existing demo-data implementation and must be confirmed for the chosen test environment before execution. A mismatch is a setup discrepancy, not evidence of a passed test.

Data	Expected baseline
Student program	Computer Science, 93 total credits
Recorded courses	COMP 1101 Introduction to Programming: Completed, 3 credits; MATH 1200 Discrete Mathematics: Completed, 3 credits; COMP 1600 Data Structures: In Progress, 3 credits
Progress	Completed 6; In progress 3; Remaining 84; Total required 93; Overall progress 6 / 93 (6%)
Category breakdown	Core Courses 3/36, 33 remaining; General Education 3/18, 15 remaining; Electives 0/18, 18 remaining; Major Requirements 0/24, 24 remaining; Additional Requirements 0/24, 24 remaining
E1	COMP 1201 Object-Oriented Programming, 3 credits, Computer Science, prerequisite COMP 1101; eligible
E2	MATH 2250 Industrial Statistics, 3 credits, Mathematics, no prerequisites; eligible
E3	ENGL 1000 Academic Writing, 3 credits, English, no prerequisites; eligible
Ineligible examples	COMP 1101: Already completed; COMP 1600: Already in progress; COMP 2100 Database Design: Missing prerequisite(s): COMP 1600

The seeded category requirements total 120 while the degree total is 93. This is a source/data inconsistency identified during planning, not an executed failure. Record whether the resulting breakdown is understandable and raise it for review; do not replace the degree total with the mockup's 120 or invent corrected data. If acceptance requires reconciled category totals, this remains an acceptance issue even if the screen matches the current seed.

Planned setup: disposable SQLite, dedicated local port 5002, separate Student/Advisor browser profiles and JavaScript enabled. Existing databases must not be reset. Exact actual browser, database path, build and base URL were not supplied. Test account secrets are deliberately omitted. A/B/C mean Year 1 semesters 1/2/3. Execution order: UAT-18, UAT-01–05, UAT-19, UAT-06–17, UAT-20.

UAT-18 — Student login and clean-run baseline (execute first)

Field	Detail
UAT ID	UAT-18
Objective	Verify Student login, identity, empty baseline and persistent session.
Original workflow covered	Supporting Student authentication for WF1 Track Degree Progress and WF2 Plan Semester Course Selections
Actor	Student
Wireframe component	Student identity/sidebar in panels 1–2; login form is outside the original wireframe
Preconditions	Safe local setup complete; fresh database; dedicated Student browser profile signed out; server running
Test data	[Student test account] / [Student test password]; invalid password [invalid Student test password]; semesters A/B/C

Exact browser steps

• In the Student window open http://127.0.0.1:5002/login.

• Enter [Student test account] in **Username**, [invalid Student test password] in **Password**, then click **Sign in**. Inspect the response.

• Enter [Student test account] and [Student test password] and click **Sign in**.

• Confirm **Student workspace**, Home, Cameron Ramesar and Student ID 12345678/Computer Science. Click **Profile** if needed to read **Username**, **Student ID** and degree program.

• Click **Home** and confirm the no-submitted-plan message in **Latest semester plan**. Click **Messages** and confirm **No advisor updates yet**.

• Click **Plan Courses**. Choose **Year 1 semester 1**, then **Year 1 semester 2**, then **Year 1 semester 3**, waiting for each reload. Confirm each has zero selected courses and no existing plan status banner or selected-course list. Return to **Year 1 semester 1**.

• Click **Home** and refresh. Confirm the Student remains signed in. Continue with UAT-01.

Expected result: Invalid credentials show **Incorrect username or password** on the login page without opening the workspace. Valid credentials open /app as the Student, with the seeded identity, Student navigation and no Advisor Review link. Home, Messages and A/B/C have the fresh empty baseline. Refresh preserves the authenticated Student session. If the baseline is occupied, record a setup blocker and do not begin dependent cases.

UAT-01 — View degree progress

Field	Detail
UAT ID	UAT-01
Actor	Student
Original workflow covered	WF1 — Track Degree Progress (Student)
Objective	Understand completed, current, and remaining degree requirements.
Relevant wireframe component	W1: Overall Progress, credit cards, Requirement Breakdown, Recent Activity.
Preconditions	Local setup and UAT-18 completed; Student signed in; compare displayed data with the documented seed baseline.
Test data	Student demo account; baseline progress and category values above.

Exact browser steps

• In the signed-in Student window click **Degree Progress** in the sidebar.

• Read Overall progress, all four credit cards, every Requirement breakdown row, and Recent course activity. Capture the displayed values.

• Refresh the page and compare the values again.

Expected result: Student workspace opens. Credits are 6 completed, 3 in progress, 84 remaining, and 93 required; overall progress is 6 / 93 (6%). Breakdown matches the baseline and distinguishes completed from in-progress courses. Recent activity shows the three recorded courses with their correct statuses. Refresh preserves the values. Record the category-total inconsistency noted above for acceptance review.

UAT-02 — Browse, search, and filter available courses

Field	Detail
UAT ID	UAT-02
Actor	Student
Original workflow covered	WF2 — Plan Semester Course Selections (Student), including cancellation/resubmission; queue checks support WF3
Objective	Find courses and understand their credits, prerequisites, and eligibility.
Relevant wireframe component	W2: semester dropdown, search, department filter, course list.
Preconditions	UAT-01 completed; Student signed in; A has no plan.
Test data	A; E1/E2; search COMP 1201, Industrial Statistics, and ZZZ-UAT-NO-COURSE.

Exact browser steps

• Click **Plan Courses**. Open **Semester** and inspect the nine Year 1–3/semester 1–3 options; choose **Year 1 semester 1**.

• Scroll the course list and inspect course codes, titles, credits, departments, prerequisites, and eligibility messages.

• Enter COMP 1201 in **Search by code or course name**; inspect the matching row.

• Replace the search with Industrial Statistics; inspect MATH 2250.

• Clear the search and choose **Mathematics** in the department dropdown; inspect the remaining rows.

• With Mathematics still selected, search COMP 1201; inspect the no-match message, then click **Clear Filters**.

• Search ZZZ-UAT-NO-COURSE; inspect the no-match message and click **Clear Filters** again.

Expected result: The chosen semester loads, and semester 3 is identified as summer. E1 and E2 show their documented details. Search matches code or title, and the department filter restricts results. Combined incompatible filters and the nonexistent search show “No courses match your search and department filters.” Clear Filters restores the catalog and All departments. No plan is submitted.

UAT-03 — Select only eligible courses and review the selection

Field	Detail
UAT ID	UAT-03
Actor	Student
Original workflow covered	WF2 — Plan Semester Course Selections (Student), including cancellation/resubmission; queue checks support WF3
Objective	Prevent an empty or ineligible selection and check the proposed credit load.
Relevant wireframe component	W2: checkboxes, Selected Courses, Next/Review Plan.
Preconditions	UAT-02 completed; A selected; filters clear; no boxes selected.
Test data	E1 and E2 (6 credits); ineligible baseline examples.

Exact browser steps

• Observe **0 courses selected** and attempt to click **Review Selected Courses**.

• Locate COMP 1101, COMP 1600, and COMP 2100, scrolling as necessary. Read their eligibility messages and attempt to select each checkbox.

• Select COMP 1201 and MATH 2250. Check the count, then click **Review Selected Courses**.

• Read the two reviewed courses, their prerequisites, and **Total selected credits**.

• Click **Back to course selection**, uncheck MATH 2250, and check the count. Recheck it and click **Review Selected Courses** again.

Expected result: Empty selection cannot advance. The three ineligible checkboxes are disabled with the documented reasons; an in-progress prerequisite does not count as completed. Only E1/E2 are selected. The count moves 0 → 2 → 1 → 2; review shows exactly E1/E2 and 6 credits. Returning to selection retains choices. Nothing is submitted yet; leave the browser on review for UAT-04.

UAT-04 — Submit semester plan A

Field	Detail
UAT ID	UAT-04
Actor	Student
Original workflow covered	WF2 — Plan Semester Course Selections (Student), including cancellation/resubmission; queue checks support WF3
Objective	Send the reviewed selection to an Advisor.
Relevant wireframe component	W2: Review Plan and Submit.
Preconditions	UAT-03 left E1/E2 on the review step for A; no existing A plan.
Test data	A; E1/E2, 6 credits.

Exact browser steps

• Confirm E1/E2 and 6 credits on **Review your selection**.

• Click **Submit Plan for Advisor Review** once.

• Read the confirmation, semester/status banner, and **Selected courses for Year 1 semester 1** list.

• Refresh the page. Attempt to alter a course checkbox and click **Review Selected Courses**.

Expected result: “Semester plan submitted for advisor review.” appears. A is Pending with exactly E1/E2 and is awaiting advisor review. Refresh preserves the submission. Standard selection controls and the review button are disabled for the submitted semester, preventing another submission through this form. Pending-plan Remove/Cancel controls remain available as separate permitted actions; leave them untouched here.

UAT-05 — View submission status

Field	Detail
UAT ID	UAT-05
Actor	Student
Original workflow covered	WF2 — Plan Semester Course Selections (Student), including cancellation/resubmission; queue checks support WF3
Objective	Find the current submission state after navigating away and signing in again.
Relevant wireframe component	W2/W4: submitted plan, Home, Messages navigation.
Preconditions	UAT-04 completed; only A has been submitted.
Test data	A Pending; E1/E2, 6 credits.

Exact browser steps

• Click **Home** and read **Latest semester plan**.

• Click **Messages** and find the **Year 1 semester 1 plan** entry.

• Click **Sign out**, open /login, enter the Student credentials, and click **Sign in**.

• Click **Plan Courses**, choose **Year 1 semester 1**, and inspect its status and selected courses.

Expected result: Home shows A Pending, 2 courses, 6 credits, and a waiting-for-review message. Messages identifies the same semester as Pending. Signing in again preserves A and its selection; no decision is implied before the Advisor acts.

UAT-19 — Advisor login (execute after UAT-05, before UAT-06)

Field	Detail
UAT ID	UAT-19
Objective	Verify Advisor login, identity and isolated persistent session.
Original workflow covered	Supporting Advisor authentication for WF3 Review and Approve Course Plan
Actor	Advisor
Wireframe component	Advisor identity/navigation in panel 3; shared login form is outside the original wireframe
Preconditions	UAT-05 complete; A Pending; separate Advisor profile/private session signed out; Student window stays open
Test data	[Advisor test account] / [Advisor test password]; invalid password [invalid Advisor test password]

Exact browser steps

• In the isolated Advisor window open http://127.0.0.1:5002/login.

• Enter [Advisor test account] in **Username**, [invalid Advisor test password] in **Password**, and click **Sign in**.

• Check the login error, then enter [Advisor test account] / [Advisor test password] and click **Sign in**.

• Inspect **Advisor workspace**, Home, Dr. Williams and the **Advisor Review** sidebar link. Click **Profile** and check Username supplied for the Advisor test account and Role **Advisor administrator**.

• Click **Home** and refresh. Confirm the Advisor session remains active. Refresh Student **Home** in its separate window and confirm it still shows the Student identity. Continue with UAT-06 in the Advisor window.

Expected result: Invalid credentials show **Incorrect username or password**. Valid login opens Advisor Home at /app, with Advisor navigation and no Degree Progress or Plan Courses sidebar links. It does not automatically open the review queue. Refresh retains the Advisor session, and the separate Student session is unaffected. If Student identity changes, stop and correct profile isolation before making decisions.

UAT-06 — Advisor reviews the submitted plan

Field	Detail
UAT ID	UAT-06
Actor	Advisor
Original workflow covered	WF3 — Review and Approve Course Plan (Advisor); Student status/feedback supports WF2
Objective	Inspect the correct Student's submission and credit load before deciding.
Relevant wireframe component	W3: Student Plans, student summary, Selected Courses table.
Preconditions	UAT-05 and UAT-19 completed; A Pending; Advisor signed in in its separate browser profile.
Test data	Advisor demo account; Student identity recorded during setup; A; E1/E2.

Exact browser steps

• Confirm the separate Advisor window shows Dr. Williams and Advisor workspace after UAT-19.

• Click **Advisor Review** in the sidebar.

• Find the card matching the recorded Student identity and **Year 1 semester 1**.

• Inspect student ID, program, submitted date, Pending status, course count, each course row, prerequisites, and **Total requested credits**.

• Locate **Comments (optional)**, its 0/500 counter, **Approve Plan**, **Reject Plan**, and **Request Changes**. Do not click a decision yet.

Expected result: The Advisor sees A in the queue with the correct Student identity, Computer Science program, submission date matching execution, E1/E2, 2 courses, and 6 total credits. All three decision controls are present. Reading the card leaves the plan Pending. Match by Student plus semester because the queue may contain other students.

UAT-07 — Request changes and deliver feedback

Field	Detail
UAT ID	UAT-07
Actor	Advisor (Student verifies delivery in the named Student steps)
Original workflow covered	WF3 — Review and Approve Course Plan (Advisor); Student status/feedback supports WF2
Objective	Return a plan with clear revision instructions that the Student can read.
Relevant wireframe component	W3: Comments and Request Changes; W4/W5: feedback and returned plan.
Preconditions	UAT-06 completed; A still Pending.
Test data	Feedback: Replace MATH 2250 with ENGL 1000, then resubmit.

Exact browser steps

• In the Advisor's A card, enter the exact feedback above into **Comments (optional)**. Observe the character counter.

• Click **Request Changes** and read the confirmation; refresh **Advisor Review**.

• Click Advisor **Messages** and locate the Student/A decision.

• In the Student window click **Messages**, refreshing if already there, and locate A.

• Read the feedback and click **Review and update plan**. Inspect the semester, status, selected courses, and available actions.

Expected result: Advisor confirmation says changes were requested. A disappears from the Pending queue and appears in Advisor Messages as Changes Requested with the entered feedback. The Student sees the same status/feedback and opens editable A with E1/E2 preselected. **Cancel Plan** is absent for Changes Requested; the instruction to update and resubmit is visible. No approval or rejection has occurred.

UAT-08 — Modify and resubmit the returned plan

Field	Detail
UAT ID	UAT-08
Actor	Student (Advisor verifies the queue in the named Advisor steps)
Original workflow covered	WF2 — Plan Semester Course Selections (Student), including cancellation/resubmission; queue checks support WF3
Objective	Apply the requested change and return the revised selection for review.
Relevant wireframe component	W2/W5: editable selection, Review Plan, resubmission; W3: revised queue card.
Preconditions	UAT-07 completed; Student viewing editable A with E1/E2 selected.
Test data	Keep E1; replace E2 with E3; revised total 6 credits.

Exact browser steps

• In Student **Plan Courses** for A, uncheck MATH 2250 and check ENGL 1000, scrolling the list as necessary.

• Click **Review Selected Courses** and confirm COMP 1201 plus ENGL 1000 and 6 credits.

• Click **Resubmit Plan for Advisor Review** and inspect the resulting status and selected-course list.

• Click Student **Messages** and inspect A's current update.

• In the Advisor window click **Advisor Review**, refresh, and inspect the Student/A card.

Expected result: A returns to Pending with exactly E1/E3. The old request-changes feedback is cleared on resubmission; Messages reflects the current Pending state rather than promising immutable feedback history. Advisor queue contains one A card for this Student with the revised courses and 6 credits, ready for another decision. Capture the previous feedback in UAT-07 before this step replaces it.

UAT-09 — Approve the revised plan

Field	Detail
UAT ID	UAT-09
Actor	Advisor (Student verifies delivery in the named Student steps)
Original workflow covered	WF3 — Review and Approve Course Plan (Advisor); Student status/feedback supports WF2
Objective	Approve the revised plan and let the Student see the decision.
Relevant wireframe component	W3: Comments and Approve; W2/W4: plan status and Messages.
Preconditions	UAT-08 completed; revised A Pending.
Test data	A; E1/E3; feedback Revised selection approved.

Exact browser steps

• In Advisor **Advisor Review**, find the Student/A card and confirm E1/E3.

• Enter Revised selection approved. into its **Comments (optional)** and click **Approve Plan**.

• Read the confirmation and refresh the review queue. Click Advisor **Messages** and inspect A.

• In the Student window click **Messages** and read A's status/feedback.

• Click **Plan Courses**, choose A, and inspect the selected courses and editing controls. Refresh once.

Expected result: “Plan approved.” appears and A leaves the Pending queue. Both actors see Approved and the exact feedback. Student A retains E1/E3 and remains Approved after refresh. Course checkboxes/review button are locked; Pending-only Remove buttons are absent. **Cancel Plan** remains available because the current implementation permits cancellation of Approved plans; do not cancel until UAT-14.

UAT-10 — Submit a separate plan for rejection

Field	Detail
UAT ID	UAT-10
Actor	Student
Original workflow covered	WF2 — Plan Semester Course Selections (Student), including cancellation/resubmission; queue checks support WF3
Objective	Submit a different semester without altering the approved plan.
Relevant wireframe component	W2: semester, selection, review, submit.
Preconditions	UAT-09 completed; A Approved; B has no plan.
Test data	B = Year 1 semester 2; E2 only, 3 credits.

Exact browser steps

• Click Student **Plan Courses** and choose **Year 1 semester 2**.

• Select only MATH 2250; click **Review Selected Courses** and confirm one course, 3 credits.

• Click **Submit Plan for Advisor Review** and inspect B's status/list.

• Choose **Year 1 semester 1** to inspect A, then switch back to **Year 1 semester 2**.

Expected result: B is Pending with only E2; A remains Approved with E1/E3. Switching semesters displays the selected semester's own plan and status without mixing the two selections.

UAT-11 — Reject the separate plan

Field	Detail
UAT ID	UAT-11
Actor	Advisor (Student verifies delivery in the named Student steps)
Original workflow covered	WF3 — Review and Approve Course Plan (Advisor); Student status/feedback supports WF2
Objective	Reject B and communicate the reason to the Student.
Relevant wireframe component	W3: Comments and Reject; W4/W5: rejected status and revision option.
Preconditions	UAT-10 completed; B Pending.
Test data	B; feedback Plan rejected for this UAT scenario; choose a different course mix.

Exact browser steps

• In the Advisor window click **Advisor Review**, refresh, and find the Student's **Year 1 semester 2** card containing only MATH 2250.

• Enter the exact feedback above into that card's **Comments (optional)** and click **Reject Plan**.

• Read the confirmation, refresh the queue, then inspect B in Advisor **Messages**.

• In the Student window click **Messages** and inspect B's status and feedback.

• Click B's **Review and update plan** link; inspect the editable course selection and **Cancel Plan** control without using either yet.

Expected result: “Plan rejected.” appears; B leaves the Pending queue. Both actors see Rejected with the exact reason. Student B is editable for resubmission and cancellation is available. A's approval remains unchanged. Leave B Rejected for UAT-15.

UAT-12 — Submit a pending plan for cancellation checks

Field	Detail
UAT ID	UAT-12
Actor	Student (Advisor verifies the queue in the named Advisor steps)
Original workflow covered	WF2 — Plan Semester Course Selections (Student), including cancellation/resubmission; queue checks support WF3
Objective	Prepare a Pending plan through normal browser submission.
Relevant wireframe component	W2: semester, selection, review, submit; W3: queue.
Preconditions	UAT-11 completed; C has no plan.
Test data	C = Year 1 semester 3; E1/E2, 6 credits.

Exact browser steps

• Click Student **Plan Courses**, choose **Year 1 semester 3**, and select COMP 1201 and MATH 2250.

• Click **Review Selected Courses**, verify 6 credits, and click **Submit Plan for Advisor Review**.

• Inspect C's Pending status and its selected-course list.

• In the Advisor window click **Advisor Review**, refresh, and find the Student/C card; make no decision.

Expected result: C is Pending with E1/E2 and is visible to the Advisor with 6 credits. A remains Approved and B remains Rejected. C stays Pending for UAT-13 and UAT-16.

UAT-13 — Remove a course from a Pending plan and protect the last course

Field	Detail
UAT ID	UAT-13
Actor	Student (Advisor verifies the queue in the named Advisor steps)
Original workflow covered	WF2 — Plan Semester Course Selections (Student), including cancellation/resubmission; queue checks support WF3
Objective	Adjust a submitted Pending plan without leaving an empty plan.
Relevant wireframe component	W2/W5: selected-course removal; submitted-plan Remove is an implementation extension.
Preconditions	UAT-12 completed; C Pending with E1/E2.
Test data	Remove E2; retain E1, 3 credits.

Exact browser steps

• In Student **Plan Courses**, choose C and click **Remove** beside MATH 2250 in **Selected courses for Year 1 semester 3**.

• Dismiss the browser confirmation with **Cancel** and verify both courses remain.

• Click the same **Remove** again and accept the confirmation with **OK**.

• Read the confirmation and inspect the remaining course and available removal action. Refresh.

• Refresh Advisor **Advisor Review** and inspect C's course table and total.

Expected result: Dismissing the dialog changes nothing. Confirming removes only E2 and shows “Course removed from the submitted plan.” C stays Pending with E1 and 3 credits. The remaining course has no Remove button and displays “Cancel the plan to remove its last course.” Advisor queue reflects E1/3 credits after refresh.

UAT-14 — Cancel an Approved plan

Field	Detail
UAT ID	UAT-14
Actor	Student
Original workflow covered	WF2 — Plan Semester Course Selections (Student), including cancellation/resubmission; queue checks support WF3
Objective	Cancel A using the implemented permission for Approved plans.
Relevant wireframe component	W5: cancellation extension to panel 2.
Preconditions	UAT-09 approval retained; A Approved; UAT-12/13 completed.
Test data	A; Approved → Cancelled.

Exact browser steps

• Click Student **Plan Courses**, choose A, and confirm its current status is Approved.

• Click **Cancel Plan** and dismiss the confirmation with **Cancel**; verify A remains Approved with E1/E3.

• Click **Cancel Plan** again and accept the confirmation with **OK**.

• Read the success message; refresh and inspect status, selected courses, and available actions.

• Click **Messages** and find A's status.

Expected result: Dismissing cancellation leaves approval unchanged. Confirming shows “Semester plan cancelled. You can edit it and resubmit.” A becomes Cancelled and remains so after refresh; Messages also shows Cancelled. The planning screen clears the active selection, permits a new choice, and hides Cancel Plan. Previous feedback may remain attached to the cancelled record; it must not make the current status appear Approved.

UAT-15 — Cancel a Rejected plan

Field	Detail
UAT ID	UAT-15
Actor	Student
Original workflow covered	WF2 — Plan Semester Course Selections (Student), including cancellation/resubmission; queue checks support WF3
Objective	Cancel B instead of resubmitting the rejected selection.
Relevant wireframe component	W5: cancellation extension to returned plan.
Preconditions	UAT-11 rejection retained; B Rejected.
Test data	B; Rejected → Cancelled.

Exact browser steps

• Click Student **Plan Courses** and choose B; confirm Rejected and its rejection feedback.

• Click **Cancel Plan** and accept the confirmation with **OK**.

• Read the success message, refresh, and inspect the status, selection count, and actions.

• Click **Messages** and find B's entry.

Expected result: B becomes Cancelled and persists after refresh. No active courses are selected; a new selection is possible and Cancel Plan is absent. Messages reports Cancelled. A's cancellation and C's Pending state are unaffected. Retained rejection feedback does not override the current Cancelled status.

UAT-16 — Cancel a Pending plan and remove it from the review queue

Field	Detail
UAT ID	UAT-16
Actor	Student (Advisor verifies the queue in the named Advisor steps)
Original workflow covered	WF2 — Plan Semester Course Selections (Student), including cancellation/resubmission; queue checks support WF3
Objective	Withdraw C before the Advisor makes a decision.
Relevant wireframe component	W5: Cancel Plan; W3: Pending review queue.
Preconditions	UAT-13 completed; C Pending with E1; no Advisor decision on C.
Test data	C; Pending → Cancelled.

Exact browser steps

• Click Student **Plan Courses**, choose C, and confirm Pending with only COMP 1201.

• Click **Cancel Plan** and accept the confirmation with **OK**.

• Read the success message and refresh; inspect the Cancelled banner, selection count, and available actions.

• Click Student **Messages** and inspect C's status.

• In the Advisor window click **Advisor Review**, refresh, and look for the Student/C card.

Expected result: C becomes Cancelled, the active selection clears, and no repeat cancellation control appears. Student Messages shows Cancelled. The Student/C card is absent from the refreshed Pending queue, so it cannot be reviewed there. Other students' Pending submissions, if any, remain available.

UAT-17 — Start a new plan after cancellation

Field	Detail
UAT ID	UAT-17
Actor	Student (Advisor verifies the queue in the named Advisor steps)
Original workflow covered	WF2 — Plan Semester Course Selections (Student), including cancellation/resubmission; queue checks support WF3
Objective	Submit a fresh selection for the same semester after withdrawing its previous plan.
Relevant wireframe component	W2/W5: new selection after cancellation, review and submit; W3: queue.
Preconditions	UAT-16 completed; C Cancelled with no active selection.
Test data	C; E3 only, 3 credits.

Exact browser steps

• Click Student **Plan Courses**, choose C, and confirm no course is selected.

• Select only ENGL 1000 and click **Review Selected Courses**.

• Confirm ENGL 1000 and 3 credits, then click **Resubmit Plan for Advisor Review** (the implemented label on the cancelled-plan screen).

• Inspect the resulting Pending status and selected-course list; refresh once.

• In the Advisor window click **Advisor Review**, refresh, and inspect the Student/C card.

Expected result: C's current plan is Pending with only E3 and 3 credits; the previously cancelled E1 selection does not reappear. Advisor queue contains one active Pending C card for this Student with E3. The screen describes this as a brand new plan even though the action label says Resubmit. Historical Cancelled entries may remain in Messages. Do not infer database identifiers from the browser or require deletion of history.

UAT-20 — Browser-visible role and signed-out restrictions (execute last)

Field	Detail
UAT ID	UAT-20
Objective	Verify browser role restrictions and signed-out access denial.
Original workflow covered	Role separation supporting WF1/WF2 Student and WF3 Advisor
Actor	Student (Advisor checks use the isolated Advisor window in steps 4–6)
Wireframe component	Different Student navigation in panels 1–2 and Advisor navigation in panel 3; access-denied pages are supporting implementation
Preconditions	UAT-17 complete; both role sessions authenticated and isolated; A/B Cancelled, new C Pending with E3
Test data	Direct URLs /admin, /app?tab=plan-courses and /app?tab=degree-progress; no plan IDs or request tampering

Exact browser steps

• In Student **Home**, confirm **Degree Progress** and **Plan Courses** links exist and **Advisor Review** is absent.

• In that Student window type http://127.0.0.1:5002/admin directly into the address bar and load it.

• Record the response. Return by typing http://127.0.0.1:5002/app?tab=home; confirm Student Home still opens.

• In the Advisor window confirm **Advisor Review** exists while **Degree Progress** and **Plan Courses** sidebar links are absent.

• In the Advisor address bar load http://127.0.0.1:5002/app?tab=plan-courses, then http://127.0.0.1:5002/app?tab=degree-progress. Inspect the content after each load.

• Click **Advisor Review** and verify the Student/C Pending card still contains only ENGL 1000, 3 credits. Make no decision.

• In the Student window click **Sign out**. Type http://127.0.0.1:5002/app in the address bar and load it afresh (do not rely on a cached Back page). Inspect the response, then click **Return to Login**.

• In the Advisor window click **Sign out**, then directly load http://127.0.0.1:5002/admin. Inspect the response and click **Return to Login**.

Expected result: Student direct access to /admin is denied with **You are not authorized to access this resource** (HTTP 403; the browser can display JSON, not a custom denial dashboard). No queue or review controls are exposed. Advisor attempts to use Student-only tab query values render **Advisor Home** at /app, because unsupported tabs fall back to Home; they do not display a Student planning/progress screen and do not promise HTTP 403. The valid Advisor queue remains accessible with unchanged C. After sign-out, direct /app and /admin loads show the implemented **401 Unauthorized / Session Expired** page with **Return to Login**. No plan state changes occur.

This case demonstrates these specific browser restrictions only. It does not test forged POSTs, all API permissions, or another Student's ownership. Do not mark those as manually passed by inference.

7.2. Unit Test Plan

Source: docs/assignment2/test-plans/unit-test-plan.md.

The following are the original specifications. Historical planning statements about absent API routes or HTML/200 authorization are superseded by the current API and report evidence in Sections 8–11. Planned inputs and expected-result text remain verbatim; the original plan's out-of-date execution fields and proposed evidence filenames are excluded from these specification tables. Current execution mappings follow in Section 9.

Shared rule interpretation

• Remaining credits subtract completed **and in-progress** credits; overall progress percentage uses **completed only**. Category credits also use completed only.

• Semester names are exactly Year 1 semester 1 through Year 3 semester 3; submission strips outer whitespace and then validates case-sensitively. Invalid dashboard semester queries fall back to a valid latest plan or the first option.

• Pending/Approved block same-semester submission. Rejected/Changes Requested resubmit the same plan. Cancelled permits a new plan. Cancellation itself permits only Pending/Approved/Rejected.

• All three advisor decisions allow empty feedback. The 500-character limit checks raw string length **before** stripping whitespace.

• Current advisor review is a global Pending queue; no advisor-student assignment restriction is defined. Do not invent an assignment-ownership rule.

• Student mutation routes currently use AuthDep rather than a student-role guard. The specified ownership tests check authenticated identity and foreign-plan denial; they do not claim the application prevents every advisor from creating their own plan.

• Business ValueError failures become warning flashes and 303 redirects, not generic 400/422 API responses. The review route accepts JSON or form input; student plan submission accepts form fields. No separate /api/plans endpoint is assumed.

• **Historical H3 (resolved before this audit):** the earlier HTML 401 handler defaulted to HTTP 200. Current app/main.py explicitly returns HTTP 401; authenticated wrong-role guards return HTTP 403. Existing integration tests verify status, denial content, no private data and no writes. This historical issue is distinct from the open HTML planning role gap in the final quality audit.

Objective and isolation strategy

Verify business decisions independently of HTTP, template rendering and SQLModel persistence. Instantiate real AdvisingService with repository constructors replaced in app.services.advising_service, supplying autospecced repository doubles; use a nonfunctional session sentinel so accidental database access fails. Alternatively replace both service repository attributes before calling business methods. Return in-memory SQLModel objects with explicit IDs; mutation methods are spies/sentinels. Assert results, domain error messages and whether create/resubmit/update/delete was invoked. Do not mock the method being tested or compute expected answers using that method.

AuthService receives a mocked UserRepository; patch hashing/token collaborators at their import location in app.services.auth_service for orchestration tests. Dependency tests use synthetic Requests and mocked decoder/repository lookups. UT-23 deliberately uses real local cryptography with test-only settings and controlled time; it needs no database/network. SQL constraints, commit/rollback and actual cookie delivery belong to the integration plan.

F1 — in-memory fixture vocabulary

S1 is a regular_user student, S2 another student, and A1 an admin advisor; use explicit distinct integer IDs. T1=Year 1 semester 1, T2=Year 1 semester 2. Catalogue: C1 (ID 101, COMP 1100, 3 credits, Core Courses), C2 (102, MATH 1000, 3 credits, General Education), C3 (103, COMP 2000, 3 credits, prerequisite COMP 1100, Core Courses), C4 (104, MATH 2000, 3 credits, no prerequisites, Electives), C5 (105, COMP 3000, 3 credits, prerequisites COMP 1100 and MATH 9999). Default S1 records: C1 Completed/A, C2 In Progress; C3/C4 untracked. Program total 93; Core required 30. Cases explicitly override these defaults. Empty plan history is the default. Mock every repository method reached by dashboard construction, including message/history lookups; do not leave implicit MagicMock return values.

Existing baseline tests E01–E12 used by both plans

Existing code is coverage evidence, not execution evidence. References below identify tests already present; none is marked passed without a retained run record. All E references except E12 are in test_advising_service.py; E12 is in test_login_routing.py.

Ref	Existing test function	Coverage and reuse decision
E01	test_degree_progress_uses_completed_and_in_progress_course_records	Service + real SQLite; baseline credits/category totals.
E02	test_plan_submission_review_request_and_student_resubmission	Service + SQLite; change request and same-ID resubmission.
E03	test_json_request_changes_includes_decision_and_feedback	Template assertions + direct route call; bypasses HTTP dependency resolution.
E04	test_request_changes_json_http_route	HTTP + SQLite; overrides is_admin_dep, so does not verify authentication.
E05	test_plan_validation_blocks_ineligible_courses_and_duplicate_plans	Service + SQLite; in-progress course and existing Pending plan.
E06	test_semester_options_cover_three_years_and_summer_terms	Service + SQLite; nine semester choices.
E07	test_plan_submission_rejects_legacy_semester_names	Service + SQLite; Fall 2026 rejected.
E08	test_plan_submission_names_the_missing_prerequisite	Service + SQLite; missing prerequisite named.
E09	test_student_can_remove_submitted_course_cancel_and_resubmit_plan	Service + SQLite; removal/last-course guard, ownership, Pending/Approved cancellation, replacement.
E10	test_rejected_plan_can_be_cancelled_and_replaced	Service + SQLite; Rejected cancellation and new plan ID.
E11	test_student_registration_stores_profile_and_default_degree	AuthService + SQLite; profile and default program.
E12	test_admin_login_redirects_to_home	Mocked direct login handler; redirect only, no real credential/token validation.

The advising file also has template/status/navigation rendering, demo-seed idempotence, registration, favicon and schema-upgrade tests. Retain those; do not duplicate their presentation or migration assertions here. The nested comp3613a1/ checkout contains another tests tree; these plans target the workspace-root app/ and tests/, not a second product or a second test run. Existing database-backed service tests are integration-level coverage even though their filename says service. When implementing later, extend/parameterize a

matching test or reuse its fixture before creating a duplicate. Planned unit variants add repository isolation; planned HTTP variants add real authentication, request parsing and fresh-session persistence.

UT-01 — Degree totals and category calculation

Field	Specification
Test ID	UT-01
Objective	Degree totals and category calculation.
Source Function or Endpoint	AdvisingService.get_student_dashboard
Preconditions	Mock repositories return fixture S1 and no plans.
Test Data	F1 baseline: completed C1=3; in-progress C2=3; program=93; Core requirement=30.
Test Steps	Call dashboard(S1,T1); inspect totals, percentage and category output.
Expected Result	Completed=3; in-progress=3; remaining=87; total=93; progress=3%; Core earned=3, remaining=27, percent=10. E01 already checks baseline persistence: add only isolated calculation assertions.

UT-02 — Empty academic record and missing program fallback

Field	Specification
Test ID	UT-02
Objective	Empty academic record and missing program fallback.
Source Function or Endpoint	AdvisingService.get_student_dashboard
Preconditions	Mock user/program/record lookups; no plans.
Test Data	Variants: existing 93-credit program with no records; missing degree program.
Test Steps	Call each variant with empty completed/in-progress records.
Expected Result	Credits=0/0, remaining=93, progress=0; category earned=0 for configured requirements; absent program uses 93 and category_progress=[].

UT-03 — Progress limits, zero denominators and duplicate records

Field	Specification
Test ID	UT-03
Objective	Progress limits, zero denominators and duplicate records.
Source Function or Endpoint	AdvisingService.get_student_dashboard
Preconditions	Repository doubles only; deliberately synthetic boundary data.
Test Data	Variants: total=6, completed=9, in-progress=3; total=0; category required=0; repeated completed record for C1.
Test Steps	Call each variant; compare arithmetic independently.
Expected Result	Remaining never negative; progress and category percent capped at 100; zero denominator yields 0; duplicate same-status course records count its credits once. Do not claim service sanitizes all inconsistent records.

UT-04 — Prerequisites satisfied or absent

Field	Specification
Test ID	UT-04
Objective	Prerequisites satisfied or absent.
Source Function or Endpoint	AdvisingService.prerequisite_ok; get_course_availability
Preconditions	Mock catalogue, completed IDs and course records.
Test Data	C3 requires C1; C4 requires none; advanced course requires ' COMP 1100, MATH 1000 '; both completed in last variant.
Test Steps	Call both functions for each target; compare bool and availability message.
Expected Result	Untracked targets eligible if all named prerequisites completed; whitespace ignored; no prerequisites eligible; availability=(True, 'Eligible').

UT-05 — Missing and in-progress prerequisites

Field	Specification
Test ID	UT-05
Objective	Missing and in-progress prerequisites.
Source Function or Endpoint	AdvisingService.prerequisite_ok; get_course_availability
Preconditions	Mock completed IDs exclude C2; C2 is In Progress.
Test Data	C3 requiring C2; advanced course requiring C1 and unknown MATH 9999; blank comma-separated tokens.
Test Steps	Evaluate both eligibility functions per variant.
Expected Result	In-progress prerequisite does not satisfy completion; unknown code fails; availability names only unmet codes; empty comma tokens ignored. E08 supplies DB-backed missing-code baseline.

UT-06 — Tracked or unknown target courses

Field	Specification
Test ID	UT-06
Objective	Tracked or unknown target courses.
Source Function or Endpoint	AdvisingService.prerequisite_ok; get_course_availability
Preconditions	Mock repository returns completed C1, in-progress C2 and absent ID 999999.
Test Data	Targets C1, C2, 999999.
Test Steps	Evaluate prerequisite_ok for all; inspect catalogue availability for C1/C2.
Expected Result	False for all three; C1 says Already completed; C2 says Already in progress; nonexistent course absent from availability map.

UT-07 — Valid semester names and trimming

Field	Specification
Test ID	UT-07
Objective	Valid semester names and trimming.
Source Function or Endpoint	AdvisingService.submit_plan; SEMESTER_OPTIONS
Preconditions	Mock eligible C3, no existing plan, create_plan sentinel.
Test Data	All nine Year {1..3} semester {1..3}; padded ' Year 1 semester 1 '.
Test Steps	Parameterize valid values; submit [C3]; inspect create_plan arguments.
Expected Result	Each accepted; exactly nine options including semester 3; padded input stored trimmed. E06 baseline reused rather than copied.

UT-08 — Invalid semester names

Field	Specification
Test ID	UT-08
Objective	Invalid semester names.
Source Function or Endpoint	AdvisingService.submit_plan
Preconditions	Mock repositories; mutation methods are spies.
Test Data	'', whitespace, Fall 2026, Year 0 semester 1, Year 4 semester 1, Year 1 semester 4, Year 1 Semester 1.
Test Steps	Submit [C3] for each input; capture exception and calls.
Expected Result	ValueError 'Choose a valid year and semester before submitting a plan.'; no create/resubmit call. E07 reused for its legacy-name variant.

UT-09 — Empty selection

Field	Specification
Test ID	UT-09
Objective	Empty selection.
Source Function or Endpoint	AdvisingService.submit_plan
Preconditions	No database; repository spies.
Test Data	[] with valid T1.
Test Steps	Call submit_plan(S1,[],T1).
Expected Result	ValueError 'Select at least one course for the semester plan.'; no persistence or eligibility lookup.

UT-10 — Duplicate selected course IDs

Field	Specification
Test ID	UT-10
Objective	Duplicate selected course IDs.
Source Function or Endpoint	AdvisingService.submit_plan
Preconditions	No database; repository spies.
Test Data	[C3,C3] and [C3,C4,C3] with T1.
Test Steps	Submit each duplicate list.
Expected Result	ValueError 'A course may only appear once in a semester plan.'; no create/resubmit call. Distinct from duplicate semester plan in E05.

UT-11 — Missing or ineligible selection aborts submission

Field	Specification
Test ID	UT-11
Objective	Missing or ineligible selection aborts submission.
Source Function or Endpoint	AdvisingService.submit_plan
Preconditions	Mock availability and course lookup, no real repository operations.
Test Data	[C3,999999]; [C1]; [C2]; [advanced course missing MATH 9999].
Test Steps	Submit each selection; inspect all mutation spies.
Expected Result	Missing ID reports no longer available; ineligible selection includes course code and reason; no create/resubmit call even when earlier list item was valid.

UT-12 — Submission status decision table

Field	Specification
Test ID	UT-12
Objective	Submission status decision table.
Source Function or Endpoint	AdvisingService.submit_plan
Preconditions	Mock existing-plan lookup and mutation return sentinels.
Test Data	Existing status: none, Pending, Approved, Rejected, Changes Requested, Cancelled; valid [C3],T1.
Test Steps	Submit each variant separately and assert exactly which repository method is called.
Expected Result	None/Cancelled -> create_plan; Rejected/Changes Requested -> resubmit_plan(existing,[C3]); Pending/Approved -> ValueError already under review or approved and no mutation. Database replacement/reset details belong to integration tests.

UT-13 — Dashboard edit and cancelled-selection rules

Field	Specification
Test ID	UT-13
Objective	Dashboard edit and cancelled-selection rules.
Source Function or Endpoint	AdvisingService.get_student_dashboard
Preconditions	Mock current plan and its C3 link, no DB.
Test Data	Each status Pending, Approved, Rejected, Changes Requested, Cancelled; also no plan.
Test Steps	Build dashboard per variant; inspect selected IDs, planned courses, can_edit_plan.
Expected Result	Edit true only Rejected/Changes Requested/Cancelled; false Pending/Approved/no plan. Cancelled hides saved course links; other existing statuses retain selected C3.

UT-14 — Advisor decisions and feedback normalization

Field	Specification
Test ID	UT-14
Objective	Advisor decisions and feedback normalization.
Source Function or Endpoint	AdvisingService.review_plan
Preconditions	Mock Pending plan, update_plan sentinel.
Test Data	approve, reject, request_changes; feedback=' Review note '; advisor=A1.
Test Steps	Call each decision separately; inspect update_plan keywords.
Expected Result	Mapped statuses Approved, Rejected, Changes Requested; feedback='Review note'; advisor_id=A1; exactly one update call. Empty feedback is allowed for every decision.

UT-15 — Invalid advisor decision

Field	Specification
Test ID	UT-15
Objective	Invalid advisor decision.
Source Function or Endpoint	AdvisingService.review_plan
Preconditions	Repository lookup/update spies.
Test Data	'', APPROVE, cancel, unknown; valid short feedback.
Test Steps	Call review_plan per decision.
Expected Result	ValueError 'Choose approve, reject, or request changes.' before persistence; no update call.

UT-16 — Feedback length boundary before trimming

Field	Specification
Test ID	UT-16
Objective	Feedback length boundary before trimming.
Source Function or Endpoint	AdvisingService.review_plan
Preconditions	Mock Pending plan and update spy.
Test Data	For each legal decision: lengths 0,499,500,501; 501 spaces; ' '+499 characters+' ' (501 total).
Test Steps	Call review_plan per variant; compare raw and stripped lengths.
Expected Result	0/499/500 accepted, stored stripped; 501 rejected with 'Advisor feedback must be 500 characters or fewer.' even if trimming would shorten it; no update on rejection.

UT-17 — Review only existing Pending plans

Field	Specification
Test ID	UT-17
Objective	Review only existing Pending plans.
Source Function or Endpoint	AdvisingService.review_plan
Preconditions	Mock lookup absent or returning status variant.
Test Data	Missing plan; Approved, Rejected, Changes Requested, Cancelled; valid approve/feedback.
Test Steps	Call review_plan for each variant.
Expected Result	Missing -> ValueError 'Plan not found'; each non-Pending -> 'Only pending plans can be reviewed.'; no update.

UT-18 — Cancellation status and ownership matrix

Field	Specification
Test ID	UT-18
Objective	Cancellation status and ownership matrix.
Source Function or Endpoint	AdvisingService.cancel_plan
Preconditions	Mock owned/foreign/missing plan and cancel spy.
Test Data	Owned Pending, Approved, Rejected, Changes Requested, Cancelled; foreign S2 plan; missing plan.
Test Steps	Cancel as S1 per independent variant.
Expected Result	Only owned Pending/Approved/Rejected delegated to repo.cancel_plan; other owned statuses reject; foreign/missing -> 'Plan not found.'; no mutation on denial.

UT-19 — Remove course guard matrix

Field	Specification
Test ID	UT-19
Objective	Remove course guard matrix.
Source Function or Endpoint	AdvisingService.remove_submitted_course
Preconditions	Mock owned Pending plan with C3/C4 links and remove spy.
Test Data	Remove C4 from two links; last C3; absent course; non-Pending statuses; foreign or absent plan.
Test Steps	Call each variant independently; inspect remove_plan_course call.
Expected Result	Only an owned Pending plan with matching course and more than one link permits deletion; others reject with existing guard messages; no deletion on failure.

UT-20 — Authentication success

Field	Specification
Test ID	UT-20
Objective	Authentication success.
Source Function or Endpoint	AuthService.authenticate_user
Preconditions	Mock UserRepository; patch verify_password and create_access_token in app.services.auth_service.
Test Data	Stored user id=S1, role=regular_user, password='stored-hash'; verifier=True.
Test Steps	Authenticate with username and plaintext; inspect calls.
Expected Result	Returns token sentinel; verifier receives plaintext and stored hash; token data={'sub': str(S1), 'role':'regular_user'}; no real DB/hash/JWT needed.

UT-21 — Authentication failures and short circuit

Field	Specification
Test ID	UT-21
Objective	Authentication failures and short circuit.
Source Function or Endpoint	AuthService.authenticate_user
Preconditions	Mock repository and security collaborators.
Test Data	Unknown username; known user with verifier=False.
Test Steps	Authenticate each case; inspect verifier/token calls.
Expected Result	Returns None; missing user never invokes verifier; neither case issues token. No token or password material recorded as evidence.

UT-22 — Cookie token resolution and admin role

Field	Specification
Test ID	UT-22
Objective	Cookie token resolution and admin role.
Source Function or Endpoint	get_current_user; is_admin_dep in app.dependencies.auth
Preconditions	Synthetic Request, mocked jwt.decode/settings/UserRepository; async invocation.
Test Data	Missing cookie; decoder InvalidTokenError; nonexistent subject user; valid S1; valid A1.
Test Steps	Call current-user resolver per variant; call admin guard with resolved S1 and A1.
Expected Result	Missing/invalid/unresolved -> HTTPException 401; valid user returned from repository; guard denies regular_user with 403 and returns admin. The original 401 expectation is superseded by the current require_roles contract (see execution addendum). HTTP handler behavior tested separately in IT-12/13.

UT-23 — Password hash and JWT utility contracts

Field	Specification
Test ID	UT-23
Objective	Password hash and JWT utility contracts.
Source Function or Endpoint	encrypt_password; verify_password; create_access_token
Preconditions	No DB; actual local hash/JWT utilities, fixed test settings and controlled clock.
Test Data	Test-only password; correct/wrong password; claims sub=S1,role=regular_user; explicit 5-minute lifetime.
Test Steps	Hash/verify locally; create token at fixed UTC time and decode with test key; compare input claims after call.
Expected Result	Hash differs from plaintext; correct verifies true, wrong false; signed token carries sub/role and exp at fixed time +5 minutes; input dict not mutated. Patch time/settings only, not functions under test.

7.3. Integration Test Plan

Source: docs/assignment2/test-plans/integration-test-plan.md.

The following are the original specifications. Historical planning statements about absent API routes or HTML/200 authorization are superseded by the current API and report evidence in Sections 8–11. Planned inputs and expected-result text remain verbatim; the original plan's out-of-date execution fields and proposed evidence filenames are excluded from these specification tables. Current execution mappings follow in Section 9.

Shared rule interpretation

• Remaining credits subtract completed **and in-progress** credits; overall progress percentage uses **completed only**. Category credits also use completed only.

• Semester names are exactly Year 1 semester 1 through Year 3 semester 3; submission strips outer whitespace and then validates case-sensitively. Invalid dashboard semester queries fall back to a valid latest plan or the first option.

• Pending/Approved block same-semester submission. Rejected/Changes Requested resubmit the same plan. Cancelled permits a new plan. Cancellation itself permits only Pending/Approved/Rejected.

• All three advisor decisions allow empty feedback. The 500-character limit checks raw string length **before** stripping whitespace.

• Current advisor review is a global Pending queue; no advisor-student assignment restriction is defined. Do not invent an assignment-ownership rule.

• Student mutation routes currently use AuthDep rather than a student-role guard. The specified ownership tests check authenticated identity and foreign-plan denial; they do not claim the application prevents every advisor from creating their own plan.

• Business ValueError failures become warning flashes and 303 redirects, not generic 400/422 API responses. The review route accepts JSON or form input; student plan submission accepts form fields. No separate /api/plans endpoint is assumed.

• **Historical H3 (resolved before this audit):** the earlier HTML 401 handler defaulted to HTTP 200. Current app/main.py explicitly returns HTTP 401; authenticated wrong-role guards return HTTP 403. Existing integration tests verify status, denial content, no private data and no writes. This historical issue is distinct from the open HTML planning role gap in the final quality audit.

Objective and integration boundary

Exercise real FastAPI routing, form/JSON parsing, cookie authentication, role dependencies, services, repositories and SQLModel persistence using an isolated database. Use the real application/router/middleware stack and real password hashing/JWT for HTTP cases. Do not override get_current_user, is_admin_dep, AuthService, AdvisingService or repository CRUD: that would conceal the interaction being tested. IT-19/20 are intentionally repository/database integration cases for constraint/rollback paths the service rejects before SQL.

F2 — isolated SQLModel environment and data

Use a fresh sqlite:// engine per independent test variant with StaticPool and check_same_thread=False. Import all model modules, enable SQLite PRAGMA foreign_keys=ON on every connection, and create SQLModel metadata. Use deterministic F1-equivalent data, **not** existing database.db, deployment data or demo accounts:

• S1 and S2: distinct usernames/emails/student numbers, role regular_user; A1: role admin. Store real test-password hashes and login via POST /login with separate cookie jars for each actor.

• Computer Science program: 93 credits; Core Courses requirement: 30 credits.

• C1: COMP 1100, Core, 3 credits, no prerequisite; C2: MATH 1000, General Education, 3 credits, no prerequisite; C3: COMP 2000, Core, 3 credits, prerequisite COMP 1100; C4: MATH 2000, Electives, 3 credits, no prerequisite; C5: COMP 3000, Electives, 3 credits, prerequisites COMP 1100 and MATH 9999.

• S1 has C1 Completed/A and C2 In Progress; C3/C4 eligible and untracked; C5 blocked. Seed distinctive S2-only activity/message data for privacy assertions. Resolve generated IDs from fixture objects; symbolic C1–C5/S1/S2/A1 names below represent those IDs.

• T1=Year 1 semester 1, T2=Year 1 semester 2. Independent variants reset their own plan histories and fixture state. Setup may insert plan statuses directly when the transition is not itself under test.

Override the exact app.database.get_session callable used by SessionDep, yielding a **new Session on the test engine for each request**. Close it after each request. Fresh-session assertions verify commits instead of relying on shared identity-map objects. Configure test-only settings/key, local HTTP cookie behavior and database URI before importing app/database; never print secrets.

Isolation must cover startup and database recovery too: app.main.lifespan calls ensure_db_and_tables/get_cli_session, and recovery middleware may initialize the configured engine. A request dependency override alone does not isolate those paths. Before starting a context-managed TestClient, rebind app.database.engine to the isolated engine and arrange test-only startup without demo seeding (replace only app.main.lifespan's ensure_db_and_tables/get_cli_session initialization collaborators or provide a no-seed test lifespan). Keep actual request handlers, authentication and persistence real. With a test lifespan, recovery must still reference the isolated engine. Never enter the production initializer against the workspace/deployed database. Record the selected fixture approach when later implementing.

Use TestClient with follow_redirects=False, then follow only the intended Location to inspect flash/HTML. Send list-valued form course_ids as repeated form keys. Send malformed JSON as raw content. On teardown close clients/sessions, restore all overrides/settings/engine/lifespan changes in finally, and dispose the test engine. SQLite does not prove PostgreSQL-specific behavior or concurrent duplicate-plan prevention; those are outside these plans.

HTTP response and persistence oracles

• Successful HTML mutations return 303 to the named page; business rejections also return 303 but with warning flashes and unchanged DB. Check message/state, not the redirect status alone.

• Framework type/path validation returns 422; malformed advisor JSON returns 400; invalid review object/field types return 422.

• Authorization follows H3 above: record the dependency's 401 intent and the current HTML/200 response separately. Denial requires no protected content and no writes.

• Snapshot relevant plan rows, links, advisor/feedback/status/timestamps before negative operations and compare in a newly opened Session afterward. A rejected mixed selection must not leave a partially inserted plan.

• Course selection before submission is browser-side behavior. HTTP tests verify catalogue eligibility and posted selections; they do not claim to exercise JavaScript selection controls.

IT-01 — Real login, cookie and logout

Field	Specification
Test ID	IT-01
Objective	Real login, cookie and logout.
Source Function or Endpoint	POST /login; GET /app; GET /logout
Preconditions	F2 database, separate unauthenticated client, real hash/JWT/auth dependencies.
Test Data	S1 correct password; wrong password; unknown username.
Test Steps	POST form credentials for each variant; capture cookie/redirect; with successful client GET /app; GET /logout then protected GET.
Expected Result	Success ->303 /app and access_token cookie, authenticated page 200; wrong/unknown ->303 /login, Incorrect username or password flash, no new access_token on clean client; logout ->303 /login, cookie removed, protected access denied under H3.

IT-02 — Student degree progress through HTTP

Field	Specification
Test ID	IT-02
Objective	Student degree progress through HTTP.
Source Function or Endpoint	GET /app?tab=degree-progress&semester=T1
Preconditions	F2; login S1 using IT-01 setup.
Test Data	C1 Completed=3; C2 In Progress=3; 93-credit program; Core required=30; S2 distinct records.
Test Steps	GET page; inspect rendered totals and category; query records in fresh session.
Expected Result	200; completed=3, in-progress=3, remaining=87, overall=3%; Core earned=3/30 and remaining=27; no S2 activity or profile; no database mutation.

IT-03 — Eligible selection, submission and advisor queue

Field	Specification
Test ID	IT-03
Objective	Eligible selection, submission and advisor queue.
Source Function or Endpoint	GET /app?tab=plan-courses; POST /app/plan; GET /admin
Preconditions	F2; S1 and A1 authenticated in separate clients; no T1 plan.
Test Data	Repeated form course_ids=C3 and C4; semester_name=T1.
Test Steps	GET catalogue; inspect eligible/disabled options; POST form selection; follow redirect manually; GET /admin as A1; query saved plan/links.
Expected Result	Catalogue eligibility reflects completed C1 and in-progress C2; POST ->303 with semester and tab=plan-courses; success flash; exactly one Pending S1/T1 plan, two unique links, submitted_at set, advisor_id=None; appears in real advisor queue.

IT-04 — Persistence across requests and sessions

Field	Specification
Test ID	IT-04
Objective	Persistence across requests and sessions.
Source Function or Endpoint	POST /app/plan; GET /app; AdvisingRepository.get_plan_for_student/get_plan_courses
Preconditions	F2; S1 authenticated; no T1 plan.
Test Data	[C3,C4],T1.
Test Steps	Submit through HTTP; close request-scoped session; open a fresh Session on same isolated engine; read plan/links; issue another GET.
Expected Result	Committed plan ID, S1, T1, Pending and both links survive session boundaries and render as selected; no reliance on Python object identity. E02/E09 do not prove full authenticated request chain.

IT-05 — Advisor approval and rejection

Field	Specification
Test ID	IT-05
Objective	Advisor approval and rejection.
Source Function or Endpoint	POST /admin/review/{plan_id}; GET /admin; GET /app?tab=messages
Preconditions	F2; seed a separate Pending S1 plan for each decision; S1 and A1 logged in.
Test Data	approve with ''; reject with 'Please choose another course.'; JSON and form variants.
Test Steps	A1 POST each review; inspect redirect/flash; read row in fresh session; GET advisor queue and student messages.
Expected Result	303 /admin; matching success flash; Approved or Rejected stored with A1 and stripped feedback; reviewed plan leaves Pending queue; student sees own decision/message; plan course links preserved.

IT-06 — Request changes and resubmit

Field	Specification
Test ID	IT-06
Objective	Request changes and resubmit.
Source Function or Endpoint	POST /admin/review/{id}; POST /app/plan; GET /app and /admin
Preconditions	F2; Pending S1/T1 plan containing C3, A1/S1 authenticated.
Test Data	request_changes, feedback=' Choose another course. '; replacement=[C4].
Test Steps	A1 request changes through HTTP; verify persisted status/student message; S1 submit replacement for same semester; query fresh session and advisor queue.
Expected Result	Changes Requested, A1, trimmed feedback initially; resubmission ->303; same plan ID becomes Pending; links exactly [C4]; old C3 link removed; advisor_id=None, feedback=''; submitted_at refreshed; returns to Pending queue.

IT-07 — Rejected plan resubmission without cancellation

Field	Specification
Test ID	IT-07
Objective	Rejected plan resubmission without cancellation.
Source Function or Endpoint	POST /admin/review/{id}; POST /app/plan
Preconditions	F2; S1/A1 authenticated; S1 Pending T1 plan containing C3.
Test Data	reject with feedback; replacement=[C4].
Test Steps	Reject via HTTP; submit C4 for same semester without cancel; read fresh-session plan and links.
Expected Result	303 for both requests; same ID reset to Pending; C4 replaces C3; feedback and advisor cleared; timestamp refreshed. Distinct from E10 cancellation/new-ID behavior.

IT-08 — Allowed cancellation and replacement

Field	Specification
Test ID	IT-08
Objective	Allowed cancellation and replacement.
Source Function or Endpoint	POST /app/plan/{id}/cancel; POST /app/plan; GET /admin and /app
Preconditions	F2; independently owned Pending, Approved and Rejected plans; S1 logged in.
Test Data	Each allowed status with existing [C3] links; replacement [C4], same semester.
Test Steps	Cancel per variant; inspect row, links, UI and advisor queue; submit replacement; query all matching plan rows.
Expected Result	303; original row Cancelled and retains links/history; hidden from Pending queue; cancelled plan displays no selected/planned courses; replacement has new ID and Pending status, old Cancelled row remains.

IT-09 — Forbidden cancellation statuses

Field	Specification
Test ID	IT-09
Objective	Forbidden cancellation statuses.
Source Function or Endpoint	POST /app/plan/{id}/cancel
Preconditions	F2; S1 authenticated; independently owned Changes Requested and Cancelled plans.
Test Data	Each disallowed status; unchanged links/feedback snapshots.
Test Steps	POST cancel per variant; follow redirect for warning; query fresh state.
Expected Result	303 tab=plan-courses with 'Only pending, rejected, or approved plans can be cancelled.' warning; status, links and feedback unchanged. Changes Requested permits resubmission, not cancellation.

IT-10 — Submitted course removal and last-course rule

Field	Specification
Test ID	IT-10
Objective	Submitted course removal and last-course rule.
Source Function or Endpoint	POST /app/plan/{id}/courses/{course_id}/remove
Preconditions	F2; S1 authenticated, owned Pending plan [C3,C4]; separate non-Pending variants.
Test Data	C4 then C3; absent course ID; each Approved/Rejected/Changes Requested/Cancelled.
Test Steps	Remove C4; try last C3; test absent and non-Pending variants independently; query DB after each.
Expected Result	First request 303/success, one C3 link remains; last-course, absent-course and non-Pending requests 303/warning without state change; no orphaned/deleted plan.

IT-11 — Cross-student ownership and data visibility

Field	Specification
Test ID	IT-11
Objective	Cross-student ownership and data visibility.
Source Function or Endpoint	GET /app?tab=degree-progress; GET /app?tab=messages; POST /app/plan/{id}/cancel; POST /app/plan/{id}/courses/{course_id}/remove; POST /app/plan
Preconditions	F2; S1 and S2 logged in separately; S2 owns Pending [C3,C4] and distinctive records/feedback.
Test Data	S1 uses S2 plan ID for cancel/remove; submits valid own selection with extra student_id=S2 field.
Test Steps	S1 GET dashboard/messages; POST foreign cancel/remove; POST own plan with forged student_id field; compare DB before/after.
Expected Result	Foreign operations ->303 warning 'Plan not found.' with S2 row/links untouched; no S2 private record/message in S1 output; new submission belongs to cookie-authenticated S1, extra form student_id does not override identity.

IT-12 — Unauthenticated and invalid-token access

Field	Specification
Test ID	IT-12
Objective	Unauthenticated and invalid-token access.
Source Function or Endpoint	GET /app; GET /admin; POST /app/plan; POST /admin/review/{id}; POST /app/plan/{id}/cancel; POST /app/plan/{id}/courses/{course_id}/remove
Preconditions	F2; valid target plan; clean client per token variant; no auth override.
Test Data	No cookie; malformed token; expired correctly signed token; wrong-signature token; valid token for nonexistent user.
Test Steps	Send each protected request with otherwise valid payload under each variant; compare entire relevant DB state.
Expected Result	Access denied, no private student/advisor data and no writes. Dependencies raise 401; current application handler returns denial HTML with HTTP 200 (H3). Record HTTP status mismatch as a known contract issue, never count 200 alone as authorization success.

Current contract clarification: historical H3/HTTP-200 text in the preserved expectation above is superseded. Current unauthenticated HTML/JSON requests return 401; logged-in wrong-role requests guarded by the role dependency return 403. Current execution evidence is in Section 9.2; the separate HTML Student-mutation role gap remains F01.

IT-13 — Student cannot use advisor operations

Field	Specification
Test ID	IT-13
Objective	Student cannot use advisor operations.
Source Function or Endpoint	GET /admin; POST /admin/review/{id}
Preconditions	F2; S1 logged in; Pending S2 plan.
Test Data	Valid approve/reject/request_changes payloads as regular_user.
Test Steps	GET queue; POST each decision; query plan in fresh session.
Expected Result	Role guard denies access; no advisor queue/private data; no status/feedback/advisor/link changes; denial page currently HTTP 200 under H3 (dependency raises 401).

Current contract clarification: historical H3/HTTP-200 text in the preserved expectation above is superseded. Current unauthenticated HTML/JSON requests return 401; logged-in wrong-role requests guarded by the role dependency return 403. Current execution evidence is in Section 9.2; the separate HTML Student-mutation role gap remains F01.

IT-14 — Business-invalid submission inputs

Field	Specification
Test ID	IT-14
Objective	Business-invalid submission inputs.
Source Function or Endpoint	POST /app/plan
Preconditions	F2; S1 logged in; no plan for target semester; reset DB per variant.
Test Data	Omitted/empty course_ids; [C3,C3]; [999999]; [C1] completed; [C2] in progress; unmet prerequisite C5; invalid/blank semester; valid C3 plus invalid item.
Test Steps	POST form per variant; capture redirect; GET warning; compare SemesterPlan/PlanCourse counts.
Expected Result	303 warning for each business rejection, no partial plan/links. Messages identify empty/duplicate/missing/ineligible selections or invalid semester. Whitespace around a valid semester is accepted and stored trimmed (positive control).

IT-15 — Duplicate active semester plan

Field	Specification
Test ID	IT-15
Objective	Duplicate active semester plan.
Source Function or Endpoint	POST /app/plan
Preconditions	F2; S1 logged in; owned T1 plan separately Pending or Approved.
Test Data	Same-semester [C4]; second positive control uses T2.
Test Steps	POST duplicate; snapshot row/links; then submit to T2.
Expected Result	Pending/Approved duplicate ->303 warning already under review or approved; original plan/links unchanged, no extra T1 plan; independent T2 submission accepted.

IT-16 — Invalid review payload shapes and syntax

Field	Specification
Test ID	IT-16
Objective	Invalid review payload shapes and syntax.
Source Function or Endpoint	POST /admin/review/{id}
Preconditions	F2; A1 logged in, Pending plan; reset per variant.
Test Data	Malformed JSON '{'; JSON [] or null; {}; decision='' or integer; feedback=null/list/integer; noninteger plan path ID; valid form control.
Test Steps	POST with accurate Content-Type per variant; capture raw response; compare persisted snapshot.
Expected Result	Malformed JSON ->400; non-object/missing-or-invalid decision/non-string feedback/noninteger path ->422; no state changes; valid form decision with string feedback ->303 and normal review. JSON syntax checks must send raw malformed bytes, not client JSON serialization.

IT-17 — Review business validation and feedback boundary

Field	Specification
Test ID	IT-17
Objective	Review business validation and feedback boundary.
Source Function or Endpoint	POST /admin/review/{id}
Preconditions	F2; A1 authenticated; independent Pending/non-Pending/missing plan cases.
Test Data	Unknown decision; feedback lengths 0,499,500,501; 501 spaces; non-Pending statuses; missing integer ID.
Test Steps	POST JSON and form boundary variants; follow redirect for flash; query fresh state.
Expected Result	Business-invalid cases ->303 /admin warning, unchanged DB; valid 0/499/500 feedback accepted and trimmed; 501 rejected before trimming. Missing plan says Plan not found; non-Pending says Only pending plans can be reviewed. No invented mandatory-feedback rule.

IT-18 — Invalid form types and path identifiers

Field	Specification
Test ID	IT-18
Objective	Invalid form types and path identifiers.
Source Function or Endpoint	POST /app/plan; POST /app/plan/{id}/cancel; POST /app/plan/{id}/courses/{course_id}/remove
Preconditions	F2; S1 logged in, owned Pending [C3,C4].
Test Data	course_ids='abc' or mixed [C3,'abc']; plan_id='abc'; course_id='abc'; missing integer plan ID 999999; missing course membership.
Test Steps	Send repeated form fields or path variants; capture validation response and DB snapshot.
Expected Result	Noninteger form/path IDs ->422 before service mutation; nonexistent integer plan or absent plan course ->303 warning; plan and links unchanged. Student submission endpoint expects form data; do not assume a JSON REST contract.

IT-19 — Database uniqueness constraints

Field	Specification
Test ID	IT-19
Objective	Database uniqueness constraints.
Source Function or Endpoint	PlanCourse and StudentCourse SQLModel tables; AdvisingRepository.create_plan
Preconditions	F2 fresh engine with foreign keys enabled; seed referenced users/courses and valid Pending plan.
Test Data	Duplicate (plan_id,C3); duplicate (S1,C1); invalid foreign-key link.
Test Steps	Use isolated repository/session boundary to insert each invalid row; catch IntegrityError; rollback; open fresh session.
Expected Result	Real database rejects duplicate pairs and dangling foreign key; committed valid rows remain intact. This is a DB integration case rather than HTTP; UT-10 covers the earlier service guard.

IT-20 — Create/resubmit transaction rollback

Field	Specification
Test ID	IT-20
Objective	Create/resubmit transaction rollback.
Source Function or Endpoint	AdvisingRepository.create_plan; resubmit_plan
Preconditions	F2 real isolated DB; existing Rejected plan [C3] for resubmit case.
Test Data	Create with [C4,C4]; resubmit existing plan with [C4,C4] to trigger real unique-constraint failure.
Test Steps	Invoke repository directly to reach DB failure past service duplicate guard; catch IntegrityError; inspect a fresh session after rollback.
Expected Result	Failed create leaves no new plan or links; failed resubmit preserves original status/feedback/advisor/timestamp/C3 links; session usable after rollback. Keep actual DB/CRUD real; do not mock commit. Normal route persistence remains covered by IT-03/04/06.

8. Test implementation

8.1 Environment and CLI

Retained marked-run screenshots show Windows/win32, Python 3.14.0, pytest 9.1.1, pluggy 1.6.0, anyio 4.15.1 and pytest-asyncio 1.4.0 in the repository .venv. Historical unit metadata records FastAPI 0.141.1, SQLModel 0.0.42, PyJWT 2.14.0 and pwdlib 0.3.1; those versions are not independently visible in the later marked screenshots. CLI marker selection is configured in pyproject.toml.

The full audit run retains its command, exit code 0 and source hashes. Marked runs use a writable --basetemp under LOCALAPPDATA after an earlier Windows temp-directory issue. This PDF task does not rerun pytest, Postman or browser cases.

8.2 Unit implementation and independent oracles

tests/unit/conftest.py supplies in-memory model objects, autospecced repository doubles and guards against accidental connection access. UT-01–19 exercise the real AdvisingService while observing repository delegation; UT-20–23 cover authentication orchestration, database-resolved roles, hashing and JWT contracts. Real cryptography in UT-23 remains isolated from persistence. Literal arithmetic expected values prevent a test from merely repeating the service formula.

```
def test_UT_01_degree_totals_and_category(advising):
    result = advising.service.get_student_dashboard(1, T1)
    assert tuple(result[k] for k in ("completed_credits", "in_progress_credits", "remaining_credits", "total_credits", "progress_percent"))
== (3, 3, 87, 93, 3)
    assert result["category_progress"] == [dict(category="Core Courses", earned=3, required=30, remaining=27, percent=10)]
```

Source: tests/unit/test_advising.py: test_UT_01_degree_totals_and_category (verbatim).

8.3 Integration implementation and database isolation

The F2 fixture creates a fresh sqlite:// engine with StaticPool and check_same_thread=False, imports all models, creates metadata and explicitly enables foreign keys. Requests use real authentication, handlers, services, repositories and fresh Sessions. Separate TestClient cookie jars model Student S1, Student S2 and Advisor A1. Startup, recovery engine and dependency overrides are isolated; teardown restores them and disposes the engine. Snapshot comparisons in fresh identity maps verify committed state and unchanged state after denials. IT-19/20 reach real constraints and rollback below service validation.

```
def snapshot(self):
    """Read all tables through a fresh identity map, including auth/seed data."""
    with Session(self.engine) as session:
        return {
            model.__tablename__: [
                row.model_dump()
                for row in session.exec(select(model).order_by(model.id)).all()
            ]
            for model in self.models
        }
```

Source: tests/integration/conftest.py: StudentEnvironment.snapshot (verbatim).

F07 limits fidelity: the fixture turns foreign keys on, whereas the actual application SQLite engine was reproduced with PRAGMA 0. SQLite test success does not establish PostgreSQL behavior, concurrent submission protection, production settings or UI usability.

8.4 Authentication, authorization and boundaries

Tests cover valid and invalid login, malformed/expired/missing tokens, current DB roles, separate identities, foreign-plan concealment, read-back after forbidden writes and Student denial from Advisor operations. JSON APIs have 401/403/404/409/422 contracts and response allowlists. Positive cases cover valid eligible courses and 500-character feedback; negative/boundary cases include empty/duplicate/missing courses, unfinished prerequisites, zero denominators, over-credit caps, wrong semesters, invalid fields, 501-character feedback, non-Pending review, last-course removal and forbidden cancellation states. The suite misses the HTML Advisor submission/self-approval path reproduced in F01.

8.5 Postman Collection Runner strategy

The genuine v2.1 JSON collection has 90 sequential requests across readiness/authentication (12), role/progress/catalogue (9), Student lifecycle/validation (29), Advisor decisions/resubmission (26) and cancellation/history/logout (14). Student, Advisor and independent Student identity variables are kept separate; dynamic course and plan IDs support read-back assertions. Run one iteration in original order against the disposable instance. Exported password and token variables are blank. A static validator compiles 93 script blocks and covers all 15 JSON operations; compilation is distinct from the recorded Runner execution.

Source: docs/assignment2/postman/MyAdvisor-Assignment2.postman_collection.json; postman/validate_collection.cjs.

8.6 Test-script example from the actual collection

49 - Cannot remove final course

```
try {
let assertionFailed = false;
function check(name, fn) {
  pm.test(name, function() {
    try { fn(); } catch (error) { assertionFailed = true; throw error; }
  });
}
check("HTTP 409", () => pm.response.to.have.status(409));
check("JSON response", () => pm.expect(pm.response.headers.get("Content-Type")).to.include("application/json"));
const data = pm.response.json();
check("Error detail returned", () => pm.expect(data).to.have.property("detail"));
```

Verbatim opening excerpt from request 49; the full script remains in the genuine exported collection. No synthetic example or execution result is introduced.

9. Test reporting

Recorded run	Actual outcome	Provenance
Unit selection	113 passed; 602 deselected; 1 warning	26 Oct; 6.23s; 3 terminal captures
Integration selection	194 passed; 521 deselected; 1 warning	26 Oct; 30.52s; 3 terminal captures
Full pytest suite	715 passed; no failures/errors; 1 warning	27 Oct independent audit; 76.25s; exit 0; text/XML/hashes
Postman Runner	90 requests; 470 assertions passed; 0 assertion failures / execution errors; 1 iteration	27 Oct; 10s 697ms; 6 genuine captures
Manual UAT	1 Pass; 0 Fail; 19 Not Run	27 Oct; 18 partial cases + 1 unverified; 19 captures

The full pytest total includes the marked selections: 113 unit + 194 integration + 408 other collected nodes = 715. These overlapping results must not be added across runs. Zero observed automated failures does not negate the independent audit defects or incomplete UAT.

9.1 Unit Testing Results

Prompt 15 audit addendum ? 27 October 2026: an independent **complete-suite** rerun now records **715 passed, 0 failures/errors/skips, 1 warning in 76.25s; exit 0**. Its text output, JUnit XML and run metadata/source hashes are retained. The XML includes all 113 unit and 194 integration nodes (plus 408 other nodes); this is not a separate marked-unit/integration run or a reconstruction of the older marked outputs. The historical provenance statements below describe those original runs. Final audit records newly reproduced application defects that the passing suite misses. No application/test source was changed.

Prepared: 26 October 2026; evidence rechecked 27 October 2026. Evidence review only; no tests were rerun to prepare this report.

Scope and objectives

Verify isolated degree-progress calculations, completed-only prerequisites, semester and selection validation, plan-state rules, advisor decisions and feedback boundaries, cancellation/removal guards, authentication orchestration, role resolution, password hashing and JWT contracts. The unit plan defines UT-01-UT-23 and local requirements R01-R14. Database persistence, HTTP cookie delivery and browser acceptance require separate evidence.

Actual results and provenance

Latest verified result supplied by the student for this reporting task

Field	Recorded detail
Passed	113
Failed	0
Errors	0
Deselected	602
Warnings	1

Retained earlier unit output, 26 October 2026

Field	Recorded detail
Passed	113
Failed	0
Errors	0
Deselected	408
Warnings	1

Full-suite result supplied by the student, for context

Field	Recorded detail
Passed	715
Failed	0
Errors	0
Deselected	Not applicable
Warnings	1

The latest unit selection totals 715 collected items (113 + 602). The retained verbose output, JUnit XML and run record describe an earlier 521-item collection, completed in 1.98 seconds with exit code 0. JUnit independently records 113 tests, zero failures/errors/skips. Preserve that historical result; it is not the raw output of the latest run. The latest screenshot confirms **6.23 seconds** and the execution command. A complete latest text/JUnit export and numeric exit-code record are **missing**. Full-suite and marked totals overlap; do not add them together.

Executed tests and UT traceability

The following per-group pass counts come from the retained unit execution and agree with its node IDs/JUnit. They total 112 variants. The retained tests/test_login_routing.py::test_admin_login_redirects_to_home is the additional passed UT-20 companion, yielding 113. Latest screenshots show portions of the passed node list and the final total; the complete latest per-node export is missing, so these detailed group counts are attributed to the retained run.

Functions UT-01-UT-19 are in test_advising.py; UT-20-UT-23 are in test_auth.py. Parameterized node suffixes and every actual result are retained in the verbose output.

Case	Planned objective	Executed function(s)	Retained variants
UT-01	Degree totals and category calculation	test_UT_01_degree_totals_and_category	1 passed
UT-02	Empty academic record and missing program fallback	test_UT_02_empty_record	2 passed
UT-03	Progress limits, zero denominators and duplicate records	test_UT_03_progress_boundaries	4 passed
UT-04	Prerequisites satisfied or absent	test_UT_04_satisfied_prerequisites	4 passed
UT-05	Missing and in-progress prerequisites	test_UT_05_missing_prerequisites	3 passed
UT-06	Tracked or unknown target courses	test_UT_06_tracked_and_unknown_courses	3 passed
UT-07	Valid semester names and trimming	test_UT_07_semesters	10 passed
UT-08	Invalid semester names	test_UT_08_invalid_semesters	7 passed
UT-09	Empty selection	test_UT_09_empty_selection	1 passed
UT-10	Duplicate selected course IDs	test_UT_10_duplicate_selection	2 passed
UT-11	Missing or ineligible selection aborts submission	test_UT_11_selection_aborts_without_writes	4 passed
UT-12	Submission status decision table	test_UT_12_submission_status_table	6 passed
UT-13	Dashboard edit and cancelled-selection rules	test_UT_13_dashboard_selection_and_edit	6 passed
UT-14	Advisor decisions and feedback normalization	test_UT_14_advisor_decisions	6 passed

Case	Planned objective	Executed function(s)	Retained variants
UT-15	Invalid advisor decision	test_UT_15_invalid_decision	4 passed
UT-16	Feedback length boundary before trimming	test_UT_16_feedback_raw_length	18 passed
UT-17	Review only existing Pending plans	test_UT_17_review_requires_pending_plan	5 passed
UT-18	Cancellation status and ownership matrix	test_UT_18_cancellation_matrix	7 passed
UT-19	Remove course guard matrix	test_UT_19_remove_course_matrix	9 passed
UT-20	Authentication success	test_UT_20_authentication_success	1 passed
UT-21	Authentication failures and short circuit	test_UT_21_authentication_failure	2 passed
UT-22	Cookie token resolution and admin role	test_UT_22_identity_failures, test_UT_22_valid_identity_and_admin_guard	5 passed
UT-23	Password hash and JWT utility contracts	test_UT_23_password_hash_contract, test_UT_23_jwt_claims_expiry_and_input_unchanged	2 passed

The requirements matrix in the linked plan now includes report references and manual/API execution dispositions.

Execution command and environment

The latest start screenshot records this PowerShell setup and command from the repository containing pyproject.toml:

```
$base = Join-Path $env:LOCALAPPDATA "MyAdvisor-A2-Pytest"
.\.venv\Scripts\python.exe -m pytest tests -m unit -v --basetemp="$base\unit"
```

The screenshot confirms Windows/win32, the repository .venv interpreter, Python **3.14.0**, pytest **9.1.1**, pluggy **1.6.0**, anyio **4.15.1** and pytest-asyncio **1.4.0**, with pyproject.toml configuration. It shows **715 collected / 602 deselected / 113 selected**. The final screenshot records **113 passed, 602 deselected, 1 warning in 6.23s** and a returned PowerShell prompt. A numeric exit code is not shown.

For comparison, the retained earlier command was:

```
python -m pytest -m unit -v -p no:cacheprovider --junitxml=docs/assignment2/evidence/unit/results.xml
```

The earlier run started at 22:38:26 UTC-04:00 on 26 October 2026 and recorded FastAPI 0.141.1, SQLModel 0.0.42, PyJWT 2.14.0 and pwdlib 0.3.1; these dependency versions are not shown in the latest screenshots. Its baseline commit 679b9250c13f38a2f97ed4e6d08a325fe48219d3 had local changes and does not identify the latest tested snapshot. Latest build/commit and named tester are **missing**. Screenshot capture/execution date is **26 October 2026**, as confirmed by the student and filenames; the terminal has no independent execution timestamp/time-zone record.

Fixtures use in-memory objects, autospecced repositories and connection guards. Hash/signing checks use real local cryptography. The writable temporary base resolves the earlier Windows temporary-directory issue. The earlier cache-disabled command and latest command are distinct executions.

Screenshot evidence

All three supplied Unit images were inspected visually. Capture date: **26 October 2026**.

Evidence	What is visible
231703: start and early tests	$base definition, actual command, environment, collection totals, UT-01 through part of UT-08 and retained login companion passing
231742: later tests and warning	Later UT-16 through UT-23 variants passing, warning text
231747: completed summary	Final pass/deselection/warning counts, duration and returned prompt

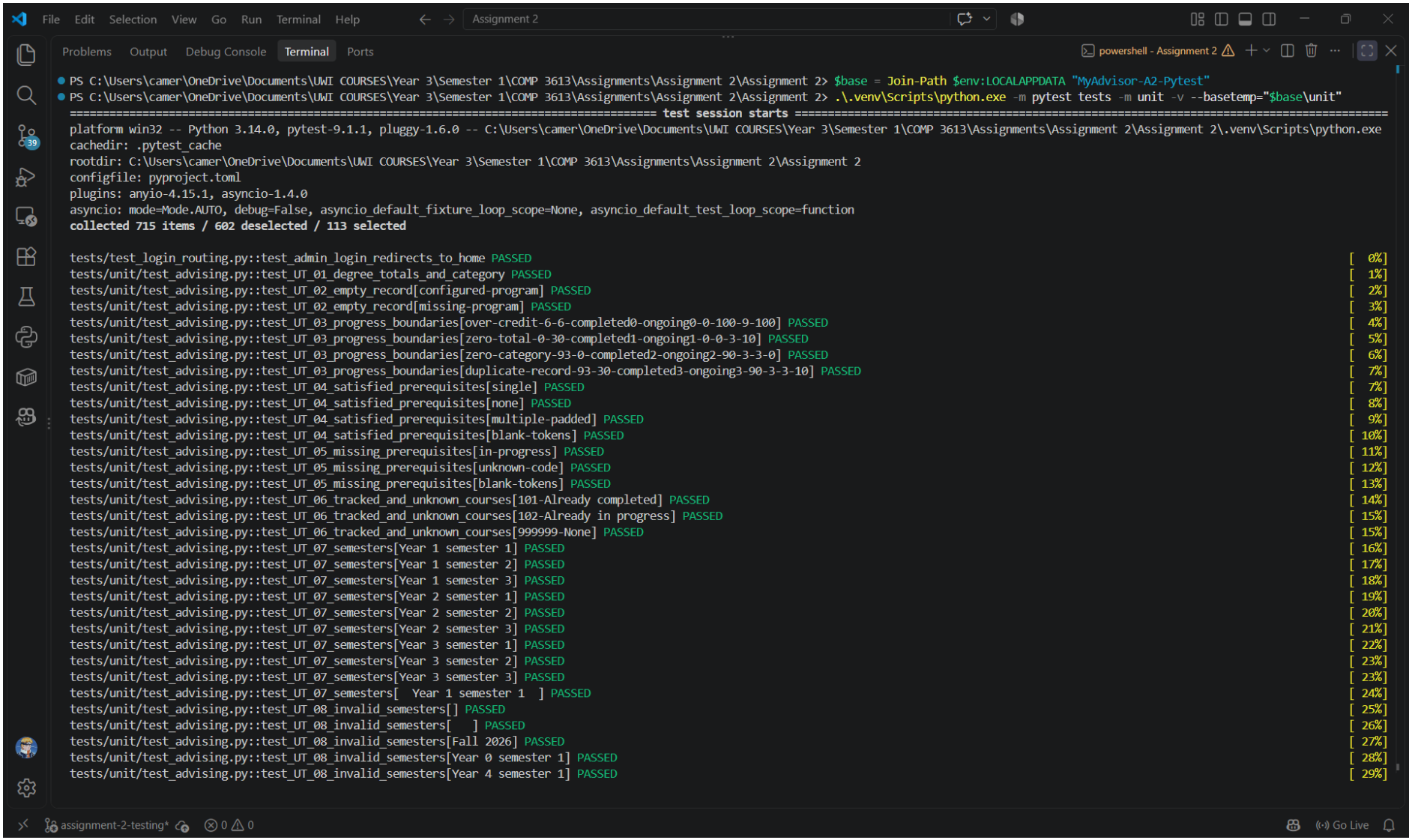
The screenshots cover portions of the node list rather than a complete scrollback export. The earlier text/XML remains the complete retained per-variant evidence.

Issues, warnings and conclusion

Both the latest screenshots and retained log identify StarletteDeprecationWarning: using httpx with starlette.testclient is deprecated; the warning suggests httpx2. This is a dependency warning, not a failed assertion. No application defect or unit failure is recorded. Temporary-directory and cache access issues concern execution setup, not failed business assertions.

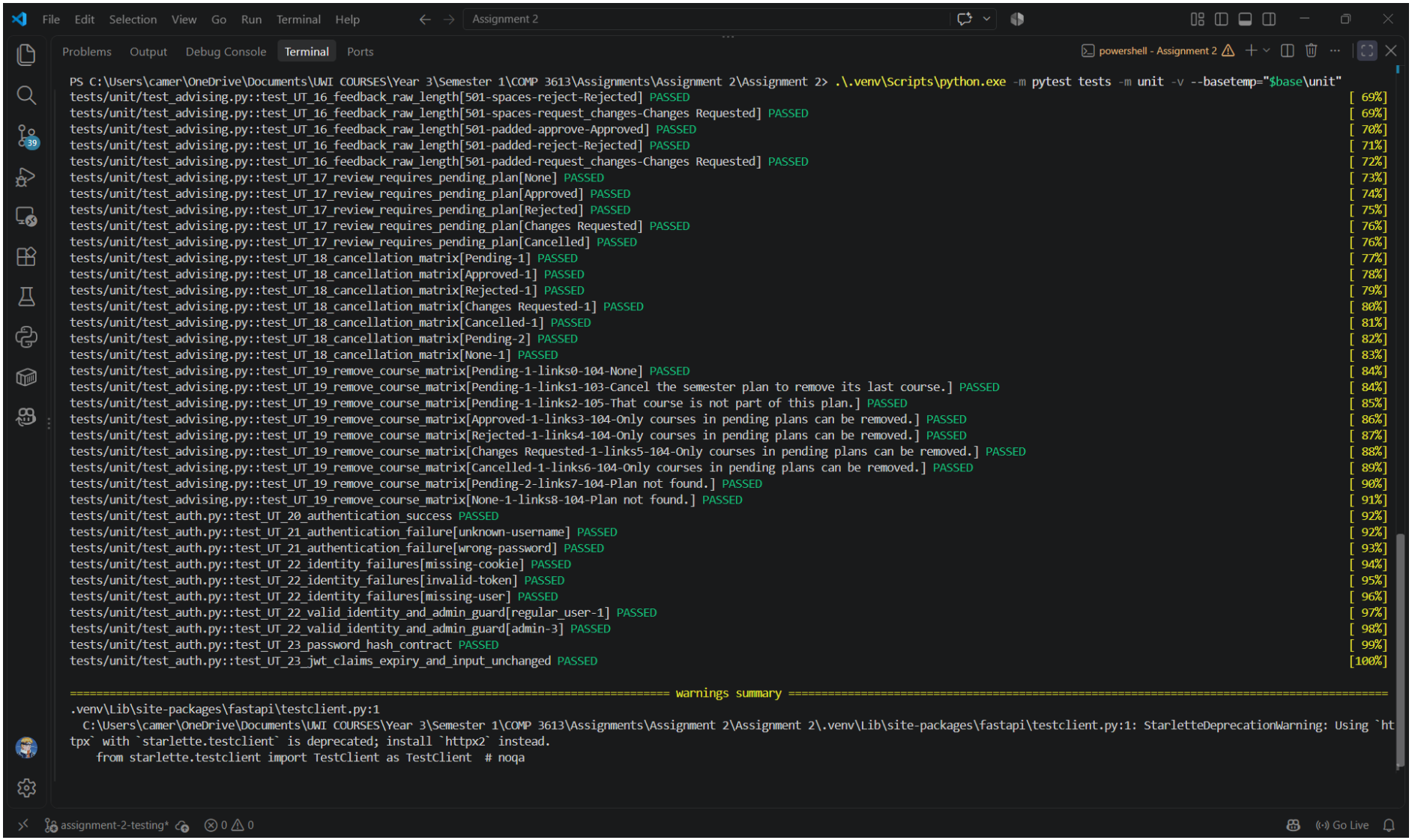
All 113 unit tests passed in the student's latest verified result; the older machine-readable evidence supports all 23 planned UT groups. The latest summary, command and warning are corroborated by screenshots; complete latest raw-output and build/tester metadata remain missing. Passing these isolated tests does not establish integration or manual acceptance.

Figure 9.1.1. Unit terminal evidence



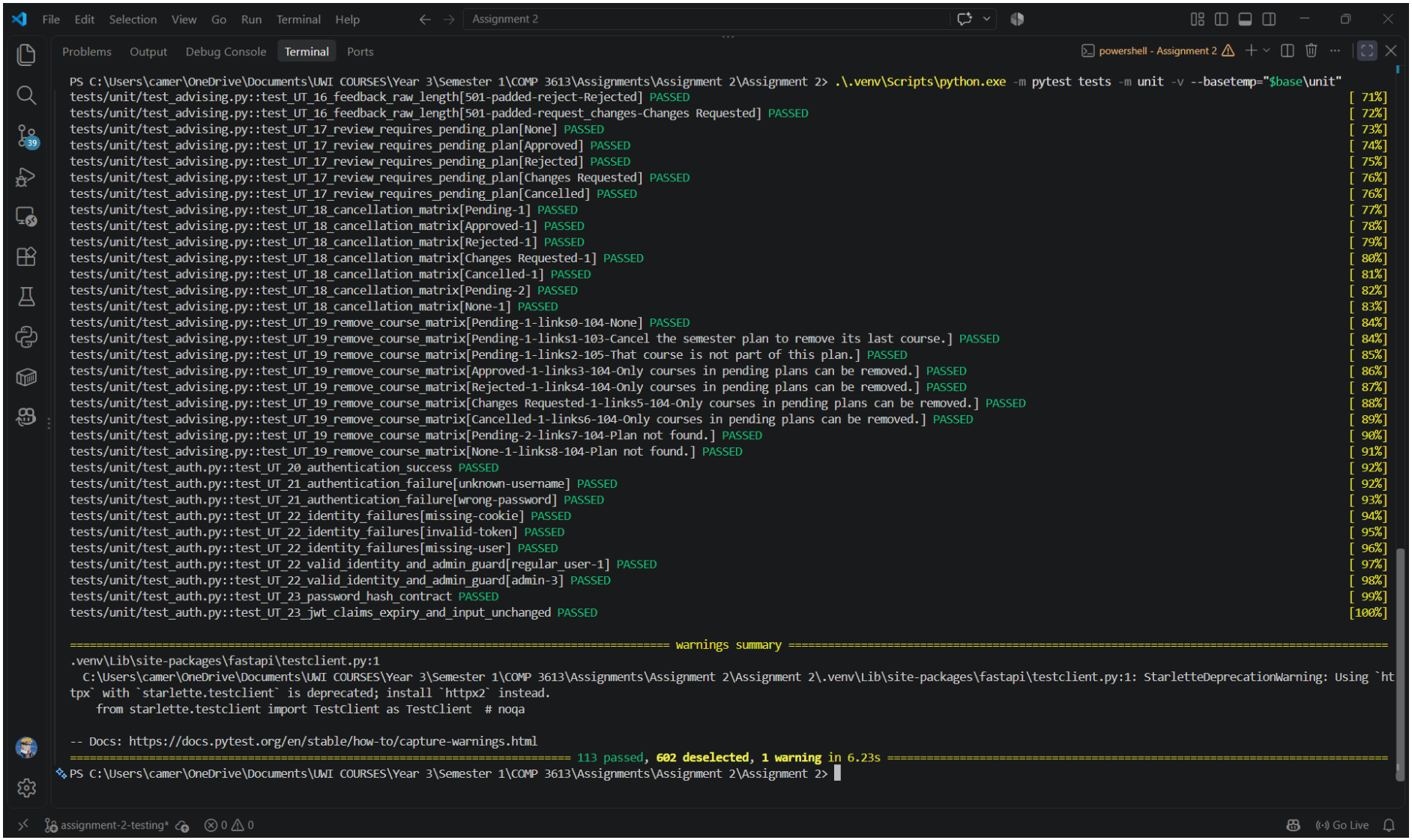
Original terminal screenshot: Screenshot 2026-10-26 231703.png. Capture date 26 October 2026. The reported warning and selected/deselected counts are retained.

Figure 9.1.2. Unit terminal evidence



Original terminal screenshot: Screenshot 2026-10-26 231742.png. Capture date 26 October 2026. The reported warning and selected/deselected counts are retained.

Figure 9.1.3. Unit terminal evidence



Original terminal screenshot: Screenshot 2026-10-26 231747.png. Capture date 26 October 2026. The reported warning and selected/deselected counts are retained.

9.2 Integration Testing Results

Prompt 15 audit addendum ? 27 October 2026: an independent **complete-suite** rerun now records **715 passed, 0 failures/errors/skips, 1 warning in 76.25s; exit 0**. Its text output, JUnit XML and run metadata/source hashes are retained. The XML includes all 113 unit and 194 integration nodes (plus 408 other nodes); this is not a separate marked-unit/integration run or a reconstruction of the older marked outputs. The historical provenance statements below describe those original runs. Final audit records newly reproduced application defects that the passing suite misses. No application/test source was changed.

Prepared: 26 October 2026; evidence rechecked 27 October 2026. Evidence review only; no tests were rerun to prepare this report.

Scope, objectives and components tested together

The integration plan covers IT-01-IT-20: real login and signed-cookie authentication; Student degree progress and course selection; submission and committed plan/course links; Advisor queue, approval, rejection and requests for changes; Student feedback, same-ID resubmission, cancellation and replacement; ownership and role restrictions; invalid requests; database constraints and rollback.

HTTP cases join FastAPI TestClient, routing/dependencies, cookie/JWT and current roles, request parsing, services, repositories, SQLModel and isolated SQLite. IT-19/20 exercise real database constraints and transaction rollback directly. Startup/demo seeding is isolated; no production/PostgreSQL, concurrency, or manual-browser acceptance result is claimed.

Actual counts and evidence provenance

Latest marked integration run verified by the student

Field	Recorded detail
Passed	194
Failed	0
Errors	0
Deselected	521
Warnings	1

Retained Student-only integration run, 26 October 2026

Field	Recorded detail
Passed	59
Failed	0
Errors	0
Deselected	Not reported
Warnings	1

Latest full suite verified by the student, for context

Field	Recorded detail
Passed	715
Failed	0
Errors	0
Deselected	Not applicable
Warnings	1

The latest integration selection totals 715 items (194 + 521). The latest final screenshot confirms **30.52 seconds** and the actual command. Its complete raw verbose/JUnit export and numeric exit-code record are **missing**. The retained Student execution record and verbose output establish 59 passed in 4.62 seconds, exit code 0. They cover Student portions only, not all 194 latest cases. Do not add these overlapping runs to the latest totals. The 715 full-suite result is broader than the 113 unit and 194 marked integration selections; remaining tests are not assigned to either category here.

IT traceability

The table maps every planned IT group to current source functions via pytest.mark.it, including parameter-level marks. It is implementation traceability for the latest aggregate result, not a fabricated per-variant execution log. Multiple IT IDs can share one test; table rows are not test counts. The latest screenshots show passed Advisor and Student variants and the final aggregate, but do not cover the complete scrollback. Complete latest per-variant output and per-IT executed counts are **missing**; the retained Student text log provides its historical per-node outcomes.

Case	Planned workflow/boundary	Current implementing function(s)
IT-01	Real login, cookie and logout	test_student_login_cookie_and_logout; test_bad_credentials_never_authenticate
IT-02	Student degree progress through HTTP	test_degree_progress_through_authenticated_html_and_json
IT-03	Eligible selection, submission and advisor queue	test_advisor_queue_receives_real_student_submission_and_only_pending_plans; test_available_courses_use_completed_prerequisites_and_tracked_status; test_valid_submission_persists_courses_and_can_be_retrieved
IT-04	Persistence across requests and sessions	test_advisor_queue_receives_real_student_submission_and_only_pending_plans; test_student_resubmits_reviewed_plan_and_advisor_reviews_replacement; test_valid_submission_persists_courses_and_can_be_retrieved
IT-05	Advisor approval and rejection	test_advisor_decision_persists_and_student_receives_only_own_feedback
IT-06	Request changes and resubmit	test_advisor_decision_persists_and_student_receives_only_own_feedback; test_student_resubmits_reviewed_plan_and_advisor_reviews_replacement; test_resubmission_reuses_plan_and_replaces_old_selection

Case	Planned workflow/boundary	Current implementing function(s)
IT-07	Rejected plan resubmission without cancellation	test_student_resubmits_reviewed_plan_and_advisor_reviews_replacement; test_resubmission_reuses_plan_and_replaces_old_selection
IT-08	Allowed cancellation and replacement	test_cancelled_plan_leaves_advisor_queue_and_replacement_returns; test_allowed_cancellation_retains_history_and_replacement_gets_new_id
IT-09	Forbidden cancellation statuses	test_cancellation_rejected_for_forbidden_status
IT-10	Submitted course removal and last-course rule	test_pending_plan_can_remove_course_but_keeps_last_course; test_course_removal_rejected_for_non_pending_plan
IT-11	Cross-student ownership and data visibility	test_advisor_decision_persists_and_student_receives_only_own_feedback; test_other_students_plan_and_private_data_are_inaccessible
IT-12	Unauthenticated and invalid-token access	test_advisor_endpoints_deny_unauthenticated_and_invalid_tokens; test_protected_student_requests_deny_invalid_auth_without_writes
IT-13	Student cannot use advisor operations	test_students_cannot_access_advisor_queue_or_decisions
IT-14	Business-invalid submission inputs	test_invalid_semester_rejects_without_writes; test_duplicate_missing_or_ineligible_courses_are_atomic_rejections; test_invalid_resubmission_preserves_review_and_old_courses
IT-15	Duplicate active semester plan	test_active_plan_blocks_duplicate_but_allows_different_semester
IT-16	Invalid review payload shapes and syntax	test_invalid_review_json_is_rejected_before_writes; test_noninteger_review_path_is_rejected_without_writes
IT-17	Review business validation and feedback boundary	test_unknown_decision_is_rejected_without_writes; test_feedback_at_limit_is_accepted; test_feedback_over_limit_is_rejected_before_trimming; test_already_reviewed_plan_cannot_be_reviewed_again; test_nonpending_status_rejects_review; test_missing_plan_rejects_review
IT-18	Invalid form types and path identifiers	test_invalid_form_or_path_types_return_422_without_writes; test_missing_plan_or_course_warns_without_writes
IT-19	Database uniqueness constraints	test_database_constraints_preserve_committed_records

Case	Planned workflow/boundary	Current implementing function(s)
IT-20	Create/resubmit transaction rollback	test_failed_plan_transaction_rolls_back_all_changes_and_session_is_reusable

The retained Student IT traceability describes actual expected-versus-observed HTTP and fresh-session state assertions for IT-01-04, Student portions of IT-06-12, IT-14-15 and IT-18. Current Advisor and database test source supplies the additional planned boundaries, and latest screenshots show Advisor lifecycle/boundary tests passing. No complete latest detailed text/JUnit export is available. Plan body fields saying Not Run and H3/earlier-scope comments are historical: the reporting addendum records their current evidence disposition without rewriting old executions.

Execution command and environment

The latest integration screenshots show this actual PowerShell command:

```
.\.venv\Scripts\python.exe -m pytest tests -m integration -v --basetemp="$base\integration"
```

The Unit start screenshot shows $base = Join-Path $env:LOCALAPPDATA "MyAdvisor-A2-Pytest" in the same repository terminal. The integration start confirms Windows/win32, the repository .venv interpreter, Python **3.14.0**, pytest **9.1.1**, pluggy **1.6.0**, anyio **4.15.1**, pytest-asyncio **1.4.0** and pyproject.toml configuration. It records **715 collected / 521 deselected / 194 selected**. The final screenshot shows **194 passed, 521 deselected, 1 warning in 30.52s** and the returned prompt. A numeric exit code is not displayed.

Recorded earlier Student-only command:

```
python -m pytest tests/integration/test_student_workflow.py -m integration -v -p no:cacheprovider
```

The retained Student run records FastAPI 0.141.1 and SQLModel 0.0.42; these versions are not visible in the latest screenshots. Current fixtures use per-case in-memory SQLite, foreign keys, real request Sessions, separate actor clients and external-connection guards. Latest exact dependency inventory, build/commit and named tester are **missing**. Capture/execution date is **26 October 2026**, confirmed by the student and filenames; no separate terminal execution timestamp/time-zone record is shown. The older baseline commit with local changes must not identify the latest build.

For full-suite context PROGRESS retains the student's **715 passed, 1 warning in 56.83s**, using verbose execution and --basetemp="$base\all"; the full-suite screenshot/raw output is **missing**. These six screenshots evidence the marked runs only.

Screenshot evidence

All three Integration images were inspected visually. Capture date: **26 October 2026**.

Evidence	What is visible
231956: start and Advisor lifecycle	Command/environment/collection totals, queue receipt, approval/rejection/changes requested, Student resubmission and cancellation variants passing
232025: Advisor validation	Oversized feedback, repeated review and non-Pending status variants passing
232058: completed summary	Student validation/resubmission/cancellation/security variants, exact warning, final totals and duration

Intermediate nodes, including the entire database-test output, are not all visible in these captures; no screenshot is claimed to prove every individual parameter variant.

Defects, retests and warnings

The retained Student run passed initially (59 in 4.56s) and finally (59 in 4.62s), with no application fix claimed. Historical H3 (HTML denial returned HTTP 200) is superseded by retained executed Student 401 checks. The separate Workflow 1 API record also records current HTML/JSON 401 behavior.

The Workflow 3 API record documents six initially failing expiry cases caused by the test fixture supplying an overwritten exp claim. The fixture was corrected to use negative expires_delta; the recorded final broader run

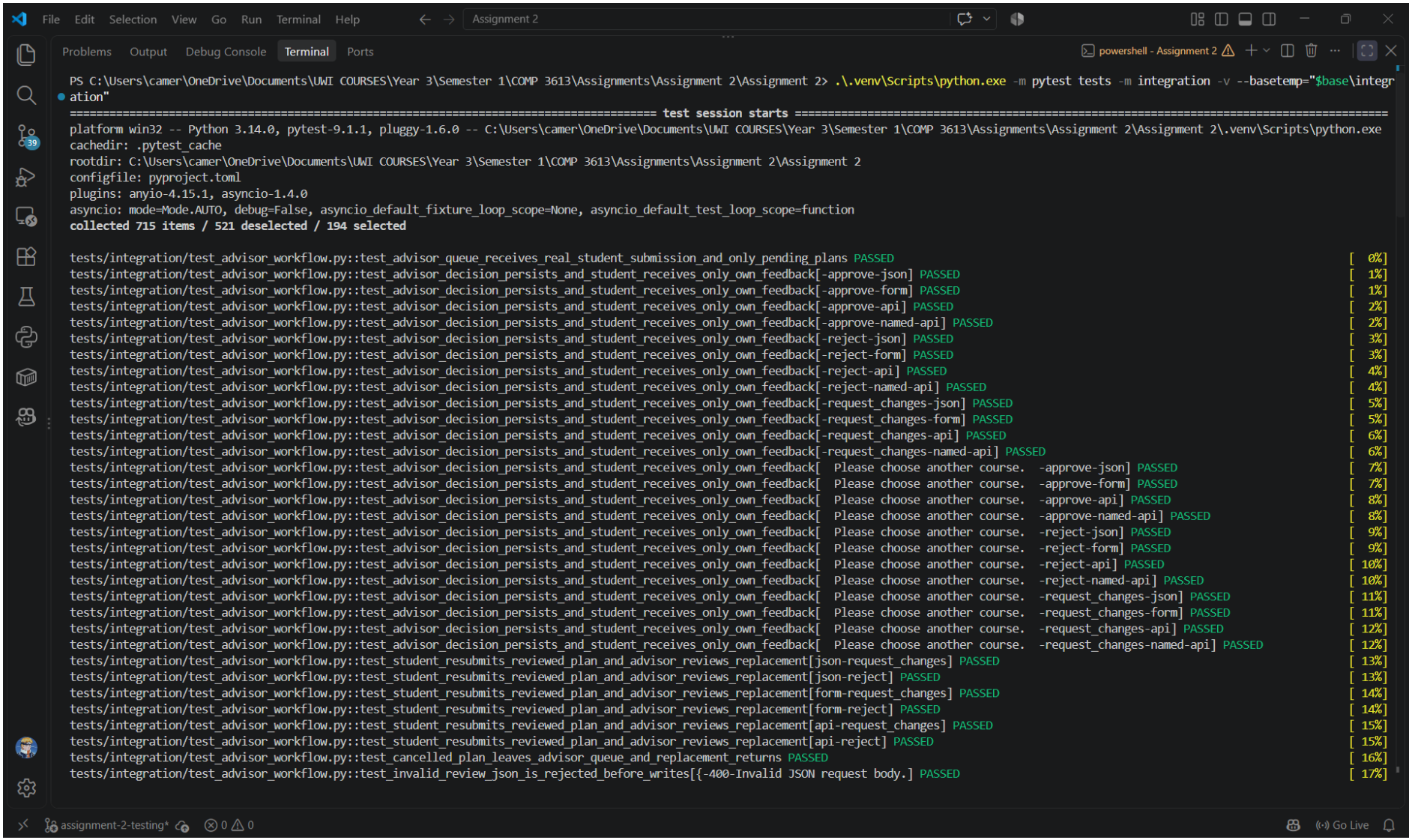
had 409 passed. This is historical automated fixture/retest evidence, not a Postman result or a defect discovered in the latest 194-test run. No dated per-defect retest export for that latest run is available.

The latest completed screenshot explicitly identifies the same StarletteDeprecationWarning as the earlier logs: httpx use with Starlette TestClient is deprecated, with httpx2 suggested. It is a dependency warning rather than an assertion failure. The Windows temporary-directory setup issue is recorded as resolved in PROGRESS.

Conclusion

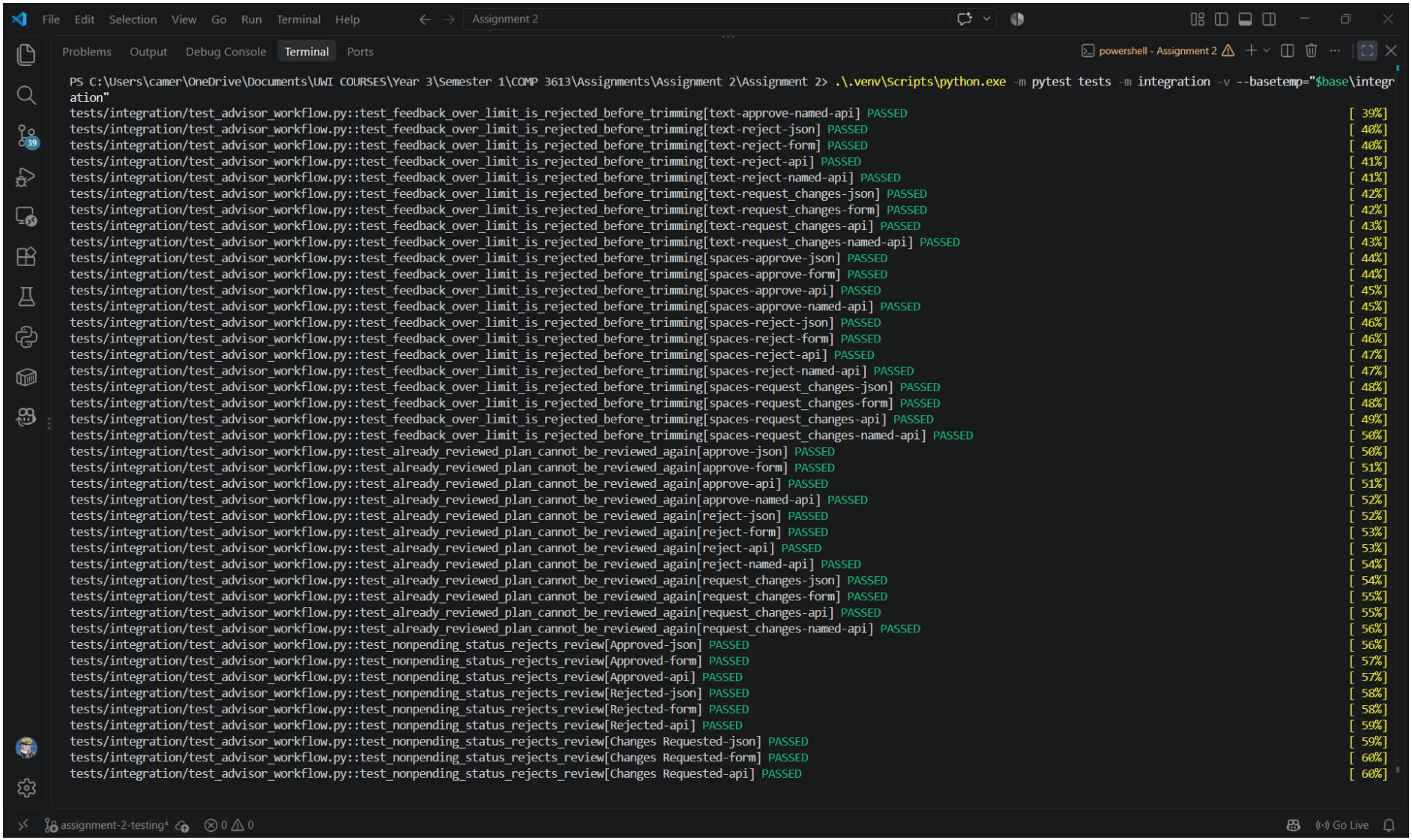
The student's verified latest marked integration result is **194 passed, 521 deselected, 1 warning**. Current source maps all 20 planned IT groups; older retained execution proves the documented Student subset. Latest screenshots corroborate the command, totals, duration, warning and visible variants. A complete latest per-node export remains missing, limiting independent verification of every planned variant. No manual acceptance or Collection Runner success follows from this automated result.

Figure 9.2.1. Integration terminal evidence



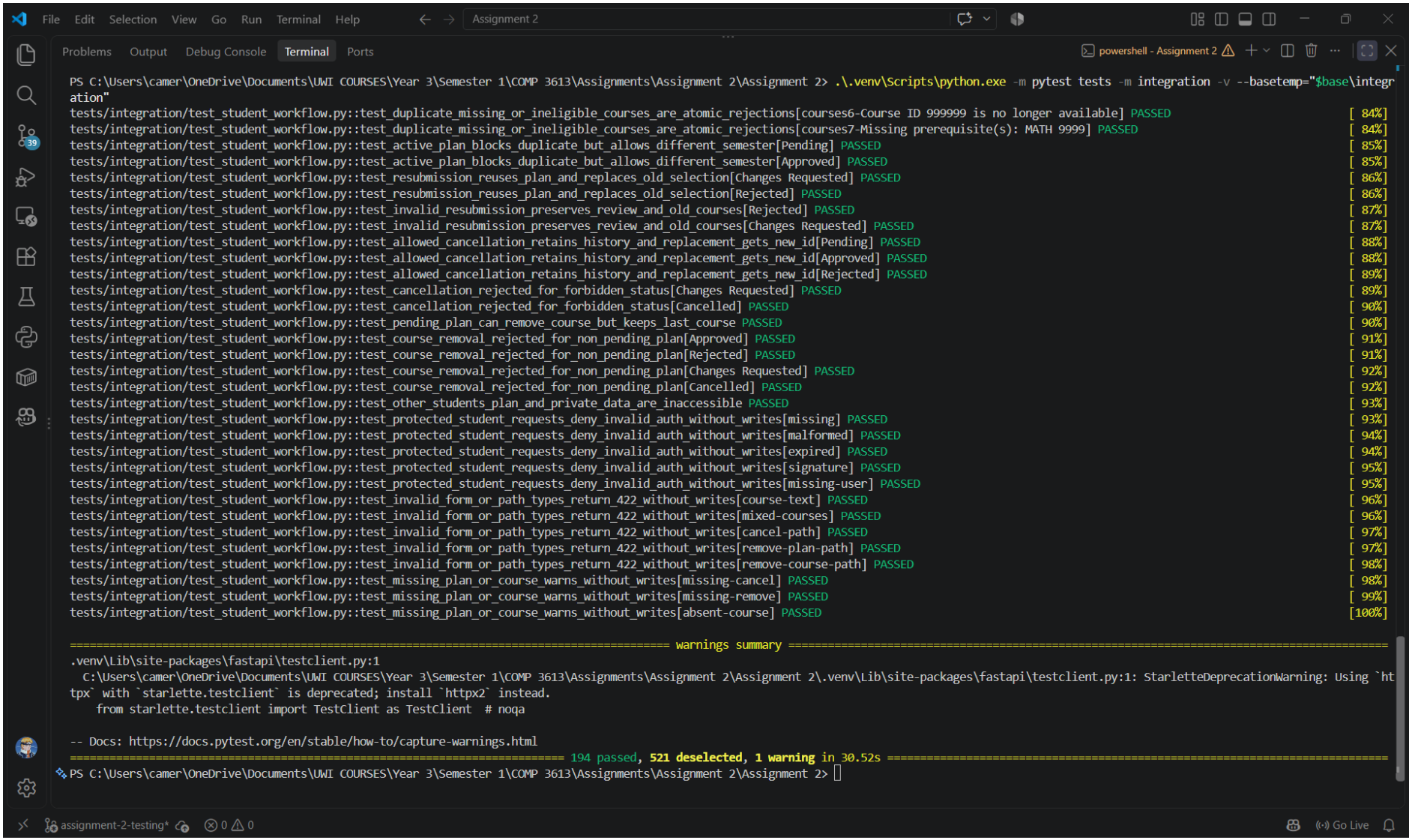
Original terminal screenshot: Screenshot 2026-10-26 231956.png. Capture date 26 October 2026. The reported warning and selected/deselected counts are retained.

Figure 9.2.2. Integration terminal evidence



Original terminal screenshot: Screenshot 2026-10-26 232025.png. Capture date 26 October 2026. The reported warning and selected/deselected counts are retained.

Figure 9.2.3. Integration terminal evidence



Original terminal screenshot: Screenshot 2026-10-26 232058.png. Capture date 26 October 2026. The reported warning and selected/deselected counts are retained.

9.3 Complete Automated Test Suite

Prompt 15 independently executed the existing complete suite on 27 October 2026: **715 passed, 0 failures, 0 errors, 0 skips, 1 warning in 76.25 seconds; exit code 0**. JUnit contains 715 testcase nodes. The recorded source-hash inventory identifies the audited local code; historical screenshots do not acquire this provenance retroactively. All 113 marked unit and 194 marked integration nodes occur within the audit XML.

Source: docs/assignment2/evidence/audit/pytest-output.txt; pytest-results.xml; pytest-run.json.

Verbatim retained terminal text (the audit did not capture a full-suite screenshot):

```
..................................................................... [ 10%]
..................................................................... [ 20%]
..................................................................... [ 30%]
..................................................................... [ 40%]
..................................................................... [ 50%]
..................................................................... [ 60%]
..................................................................... [ 70%]
..................................................................... [ 80%]
..................................................................... [ 90%]
.....................................................................  [100%]
============================== warnings summary ===============================
.venv\Lib\site-packages\fastapi\testclient.py:1
  [repository root]\.venv\Lib\site-packages\fastapi\testclient.py:1: StarletteDeprecationWarning: Using `httpx` with `starlette.testclient
` is deprecated; install `httpx2` instead.
    from starlette.testclient import TestClient as TestClient  # noqa

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
715 passed, 1 warning in 76.25s (0:01:16)
```

The warning is Starlette’s deprecated httpx TestClient transport. Retained earlier user-reported 715 passes in 56.83s and historical 113-pass/408-deselected and 59-Student-integration logs remain separate executions. No line/branch coverage percentage was measured.

9.4 Postman API Results

Prepared: 26 October 2026; finalized from supplied execution evidence on 27 October 2026. Evidence review only; no tests were rerun to prepare this report.

Collection version and API scope

Reviewed MyAdvisor-Assignment2.postman_collection.json and run instructions. Name: **MyAdvisor - COMP 3613 Assignment 2**. Format: **Postman Collection v2.1.0**. No separate application collection revision/version is declared; do not confuse schema format with a release version. SHA-256 of the reviewed file: 409cffa7540cd73880d74139f77346cdb47d06a18d9ae8e1b11cf70ef2c44b6d.

There are **90 ordered requests** across five folders, covering all 15 implemented JSON API operations and supporting health, login, registration and logout. Scope follows WF1 Track Degree Progress, WF2 Plan Semester Course Selections and WF3 Review and Approve Course Plans. Cookie authentication uses separate Student/Advisor/independent-Student tokens. Dynamic IDs and read-back checkpoints verify API-visible persistence. The prescribed run uses one iteration on an isolated local SQLite-backed server, no environment/data file, redirects/cookie jar disabled, and original request order.

Folder	Executed requests
01 - Local readiness and real cookie authentication	12
02 - Role boundaries, progress and catalogue	9
03 - Submission validation and student plan lifecycle	29
04 - Advisor review, feedback validation and student resubmission	26
05 - Cancellation, retained history and logout	14

Requests and assertions

All **90 requests (01–90)** in the inventory below were executed in the user-verified run; the five scroll captures show their green assertion results with zero failures. Request numbers are the collection’s actual identifiers, not newly invented test IDs. The request table records scope; it does not manufacture individual response bodies or timings. Assertions inspect expected statuses, JSON error codes/details and validation fields, public response shapes/privacy, authenticated ownership/reviewer attribution, progress calculations and eligibility, exact courses/status/feedback, read-back after denied writes, queue membership, same-ID resubmission and new-ID replacement after cancellation. Positive boundaries include 500-character feedback; negative cases cover 501 characters, malformed/invalid input, missing authentication, wrong roles and foreign-resource access. Login/logout assertions check redirects and cookies. Expected denials must match their specific 401/403/404/409/422 contract.

Request	Method	URL template
01 - Server health and initialize run variables	GET	{{base_url}}/health
02 - Unauthenticated /api/degree-progress	GET	{{base_url}}/api/degree-progress
03 - Unauthenticated /api/courses	GET	{{base_url}}/api/courses
04 - Unauthenticated /api/semester-plans	GET	{{base_url}}/api/semester-plans

Request	Method	URL template
05 - Unauthenticated /api/advisor/plans	GET	{{base_url}}/api/advisor/plans
06 - Unauthenticated /api/users	GET	{{base_url}}/api/users
07 - Malformed cookie denied	GET	{{base_url}}/api/degree-progress
08 - Incorrect login rejected	POST	{{base_url}}/login
09 - Student login and capture cookie	POST	{{base_url}}/login
10 - Advisor login and capture cookie	POST	{{base_url}}/login
11 - Register independent student for ownership checks	POST	{{base_url}}/register
12 - Other login and capture cookie	POST	{{base_url}}/login
13 - Student forbidden from /api/advisor/plans	GET	{{base_url}}/api/advisor/plans
14 - Student forbidden from /api/users	GET	{{base_url}}/api/users
15 - Advisor forbidden from /api/degree-progress	GET	{{base_url}}/api/degree-progress
16 - Advisor forbidden from /api/courses	GET	{{base_url}}/api/courses
17 - Advisor forbidden from /api/semester-plans	GET	{{base_url}}/api/semester-plans
18 - Advisor users list excludes passwords	GET	{{base_url}}/api/users
19 - Student degree progress calculations and privacy	GET	{{base_url}}/api/degree-progress
20 - Discover eligible and blocked dynamic course IDs	GET	{{base_url}}/api/courses
21 - Choose a semester without an existing active plan	GET	{{base_url}}/api/semester-plans
22 - Empty selection rejected	POST	{{base_url}}/api/semester-plans
23 - Duplicate selection rejected	POST	{{base_url}}/api/semester-plans
24 - Unknown course rejected	POST	{{base_url}}/api/semester-plans
25 - Missing prerequisites rejected explicitly	POST	{{base_url}}/api/semester-plans
26 - Invalid semester rejected	POST	{{base_url}}/api/semester-plans
27 - String course ID rejected	POST	{{base_url}}/api/semester-plans
28 - Spoof student ID rejected	POST	{{base_url}}/api/semester-plans
29 - Malformed JSON rejected	POST	{{base_url}}/api/semester-plans
30 - Verify invalid submissions created no records	GET	{{base_url}}/api/semester-plans

Request	Method	URL template
31 - Submit two eligible courses and capture plan ID	POST	{{base_url}}/api/semester-plans
32 - Retrieve persisted pending plan	GET	{{base_url}}/api/semester-plans/{{plan_id}}
33 - Other student cannot read foreign plan	GET	{{base_url}}/api/semester-plans/{{plan_id}}
34 - Missing plan conceals ownership identically	GET	{{base_url}}/api/semester-plans/2147483647
35 - Other student cannot revise foreign plan	PUT	{{base_url}}/api/semester-plans/{{plan_id}}/courses
36 - Other student cannot remove foreign course	DELETE	{{base_url}}/api/semester-plans/{{plan_id}}/courses/{{course_a_id}}
37 - Other student cannot cancel foreign plan	DELETE	{{base_url}}/api/semester-plans/{{plan_id}}
38 - Other student list excludes foreign plan	GET	{{base_url}}/api/semester-plans
39 - Duplicate pending submission conflicts	POST	{{base_url}}/api/semester-plans
40 - Pending plan cannot be revised	PUT	{{base_url}}/api/semester-plans/{{plan_id}}/courses
41 - Invalid path ID rejected	GET	{{base_url}}/api/semester-plans/0
42 - Student denied advisor action detail	GET	{{base_url}}/api/advisor/plans/{{plan_id}}
43 - Student denied advisor action /review	POST	{{base_url}}/api/advisor/plans/{{plan_id}}/review
44 - Student denied advisor action /approve	POST	{{base_url}}/api/advisor/plans/{{plan_id}}/approve
45 - Student denied advisor action /reject	POST	{{base_url}}/api/advisor/plans/{{plan_id}}/reject
46 - Student denied advisor action /request-changes	POST	{{base_url}}/api/advisor/plans/{{plan_id}}/request-changes
47 - Verify ownership and role denials left pending plan intact	GET	{{base_url}}/api/semester-plans/{{plan_id}}
48 - Remove selected course from pending plan	DELETE	{{base_url}}/api/semester-plans/{{plan_id}}/courses/{{course_b_id}}
49 - Cannot remove final course	DELETE	{{base_url}}/api/semester-plans/{{plan_id}}/courses/{{course_a_id}}
50 - Absent selected course returns not found	DELETE	{{base_url}}/api/semester-plans/{{plan_id}}/courses/{{course_b_id}}
51 - Advisor pending queue includes submitted plan	GET	{{base_url}}/api/advisor/plans
52 - Advisor sees student plan detail	GET	{{base_url}}/api/advisor/plans/{{plan_id}}

Request	Method	URL template
53 - Advisor missing plan returns not found	GET	{{base_url}}/api/advisor/plans/2147483647
54 - Unknown decision rejected	POST	{{base_url}}/api/advisor/plans/{{plan_id}}/review
55 - Missing decision rejected	POST	{{base_url}}/api/advisor/plans/{{plan_id}}/review
56 - Oversized feedback rejected	POST	{{base_url}}/api/advisor/plans/{{plan_id}}/approve
57 - Nonstring feedback rejected	POST	{{base_url}}/api/advisor/plans/{{plan_id}}/reject
58 - Spoof advisor identity rejected	POST	{{base_url}}/api/advisor/plans/{{plan_id}}/review
59 - Action cannot override decision rejected	POST	{{base_url}}/api/advisor/plans/{{plan_id}}/approve
60 - Verify invalid reviews left pending plan unassigned	GET	{{base_url}}/api/advisor/plans/{{plan_id}}
61 - Generic review requests changes and trims feedback	POST	{{base_url}}/api/advisor/plans/{{plan_id}}/review
62 - Reviewed plan leaves pending queue	GET	{{base_url}}/api/advisor/plans
63 - Student sees changes and feedback	GET	{{base_url}}/api/semester-plans/{{plan_id}}
64 - Changes requested cannot be cancelled	DELETE	{{base_url}}/api/semester-plans/{{plan_id}}
65 - Student revises and resubmits with same ID	PUT	{{base_url}}/api/semester-plans/{{plan_id}}/courses
66 - Advisor assignment cleared after resubmission	GET	{{base_url}}/api/advisor/plans/{{plan_id}}
67 - Dedicated request-changes action	POST	{{base_url}}/api/advisor/plans/{{plan_id}}/request-changes
68 - POST resubmits existing changes-requested plan	POST	{{base_url}}/api/semester-plans
69 - Dedicated reject action accepts 500-character boundary	POST	{{base_url}}/api/advisor/plans/{{plan_id}}/reject
70 - Student sees rejection	GET	{{base_url}}/api/semester-plans/{{plan_id}}
71 - Repeated review conflicts	POST	{{base_url}}/api/advisor/plans/{{plan_id}}/approve
72 - POST resubmits rejected plan with same ID	POST	{{base_url}}/api/semester-plans
73 - Dedicated approve action	POST	{{base_url}}/api/advisor/plans/{{plan_id}}/approve
74 - Student sees approval and feedback	GET	{{base_url}}/api/semester-plans/{{plan_id}}

Request	Method	URL template
75 - Approved plan cannot be edited	PUT	{{base_url}}/api/semester-plans/{{plan_id}}/courses
76 - Approved plan cannot be reviewed again	POST	{{base_url}}/api/advisor/plans/{{plan_id}}/reject
77 - Verify forbidden edits and reviews preserved approval	GET	{{base_url}}/api/semester-plans/{{plan_id}}
78 - Cancel approved plan and retain selections	DELETE	{{base_url}}/api/semester-plans/{{plan_id}}
79 - Repeated cancellation conflicts	DELETE	{{base_url}}/api/semester-plans/{{plan_id}}
80 - Retrieve cancelled history	GET	{{base_url}}/api/semester-plans/{{plan_id}}
81 - Cancelled plan remains visible to advisor	GET	{{base_url}}/api/advisor/plans/{{plan_id}}
82 - Cancelled plan cannot be reviewed	POST	{{base_url}}/api/advisor/plans/{{plan_id}}/approve
83 - Cancelled plan allows new submission with new ID	POST	{{base_url}}/api/semester-plans
84 - Cancel replacement pending plan for repeatable run	DELETE	{{base_url}}/api/semester-plans/{{plan_id}}
85 - Student list retains both cancelled records	GET	{{base_url}}/api/semester-plans
86 - Final advisor queue excludes collection plans	GET	{{base_url}}/api/advisor/plans
87 - Student logout clears browser cookie	GET	{{base_url}}/logout
88 - Advisor logout clears browser cookie	GET	{{base_url}}/logout
89 - Other logout clears browser cookie	GET	{{base_url}}/logout
90 - No cookie after logout is denied	GET	{{base_url}}/api/degree-progress

Assertions are conditional and can execute in loops. Neither 90 requests nor 93 script blocks is an actual assertion count. The existing static validator record reports 93 scripts compiled and route/structure checks; it does not run requests/assertions.

Actual Collection Runner outcome

Execution measure	Evidence-based value
Requests executed	**90**, user-verified; numbered rows 01–90 across the captures
Assertions passed / failed	**470 / 0**, user-verified and corroborated by green row counts and summary
Execution errors	**0**, summary Errors field and user confirmation
Iterations completed	**1**
Execution/capture date	**27 October 2026**, user-confirmed
Start time shown / duration	**01:25:19 AM / 10s 697ms**; screenshot clock time zone not shown
Average response time	**29 ms**, descriptive for this local run only
Source / environment selection	**Runner / none**, visible summary
Actual server	**http://127.0.0.1:5001**, disposable local test server, user-confirmed
Tester name / build / exact database	Not recorded; do not infer tester from application identities
Machine-readable Runner/Newman result export	**Not found**; collection JSON/YAML are definitions, not result exports

The 470 assertions are runtime results, distinct from 90 HTTP requests and the collection’s 93 script blocks. A correctly denied negative request counts as a successful test when its assertions pass. Zero assertion failures and execution errors are observed for this one completed run; no full raw response archive or independent build identifier is available.

Screenshot evidence

All six local Postman PNGs were opened and visually inspected; all are readable, genuine Runner views. Their headers show the same run time, duration, iteration count, assertion count and zero errors. These are overlapping scroll captures of **one run**, not six runs. Paths and hashes are recorded in the evidence inventory.

Screenshot	Readable request coverage
012608	01–19 fully visible; 20 at lower edge; authentication, role boundaries and progress
012643	21–39 fully visible; 40 at lower edge; invalid submissions, persisted plan and ownership
012658	41–59 fully visible; 60 at lower edge; role denials, removal and review validation
012739	61–79 fully visible; 80 at lower edge; changes requested, feedback, resubmission, rejection and approval
012839	72–90; approved-state guards, cancellation/history, replacement and logout
postman-results.png	Additional view of 72–90 and same completed summary; capture date not independently embedded

Authentication, workflows and negative testing

Requests 02–12 and 87–90 exercise missing/malformed authentication, incorrect login, real role-specific cookie login, independent Student registration and cookie-clearing logout. Requests 13–18, 28, 33–47 and 58–59 verify role/ownership restrictions, privacy and spoofing rejection, including unchanged state after denials.

WF1 uses request 19 for degree calculations/privacy. WF2 uses catalogue/eligibility (20), semester choice and invalid input (21–30), persisted submission (31–32), duplicate/state/ownership guards (33–47), removal/last-course checks (48–50), revision and returned status (63–72), and cancellation/history/replacement (78–86). WF3 covers queue/detail (51–53), malformed or invalid review/feedback/identity (54–60), requests for changes and feedback (61–68), rejection at the 500-character boundary (69–72), approval and state guards (73–77). Read-back checkpoints verify that invalid or forbidden operations do not change the protected plan. These groups overlap and do not represent additional runs.

Limitations, defects and conclusion

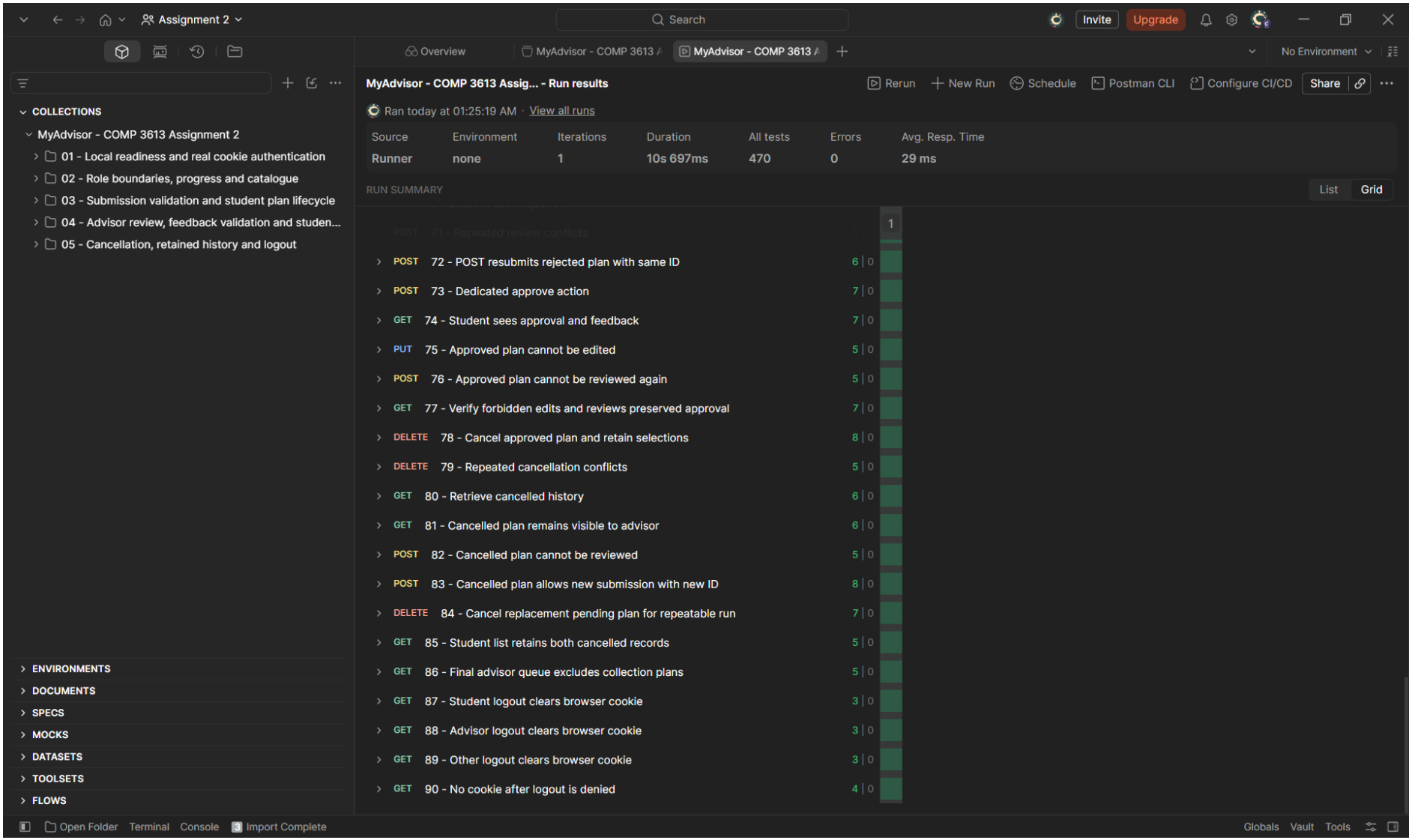
The local sequential SQLite-backed run excludes browser usability, production/PostgreSQL behavior, database outages, concurrency, load and races. API read-back demonstrates visible persistence but does not replace direct rollback/constraint tests (IT-19/20). Logout clears cookies; this does not establish revocation of a previously issued JWT. Capture headers do not independently establish the actual URL, database, tester or tested commit; URL/date are attributed to the user’s verified record. The reviewed collection hash identifies today’s definition, not a cryptographically attested executed build.

No assertion failure, execution error or Runner defect/retest is recorded for this run. **All 90 requests completed with 470 successful assertions, zero assertion failures and zero execution errors in one iteration.** API checks succeeded within that scope. Browser acceptance remains pending in the UAT report.

Prompt 14 final reconciliation ? 27 October 2026

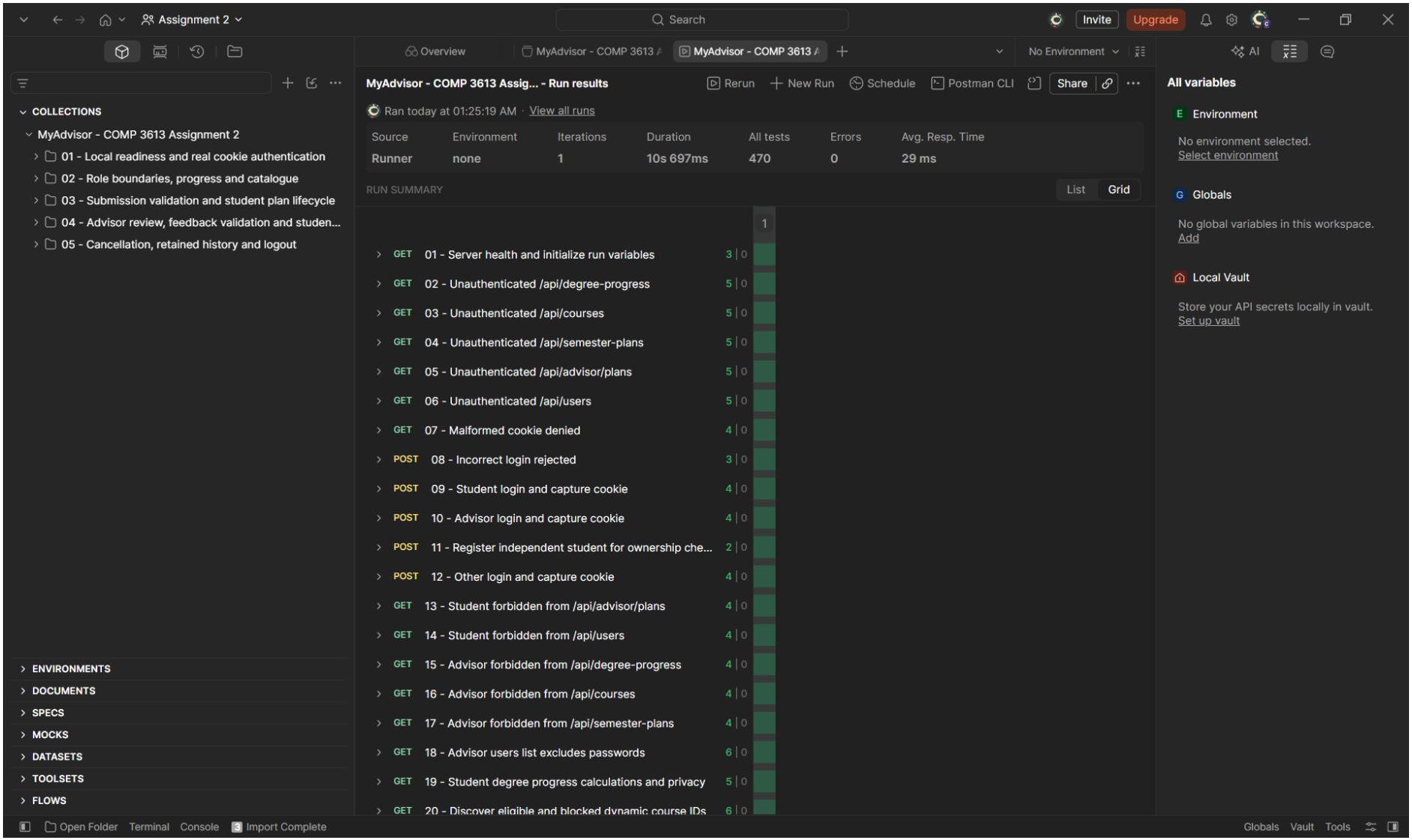
The user reconfirms 90 requests successfully executed, 470 assertions passed, zero assertion failures, zero execution errors and one iteration. Existing summary image visually rechecked; no rerun or new result export. UAT tester/date confirmation applies to the manual captures, not automatically to this API run. Manual disposition is now **1 Pass / 0 Fail / 19 Not Run**, including 18 partially verified cases; UAT report records remaining checks.

Figure 9.4.1. Genuine Postman Runner evidence



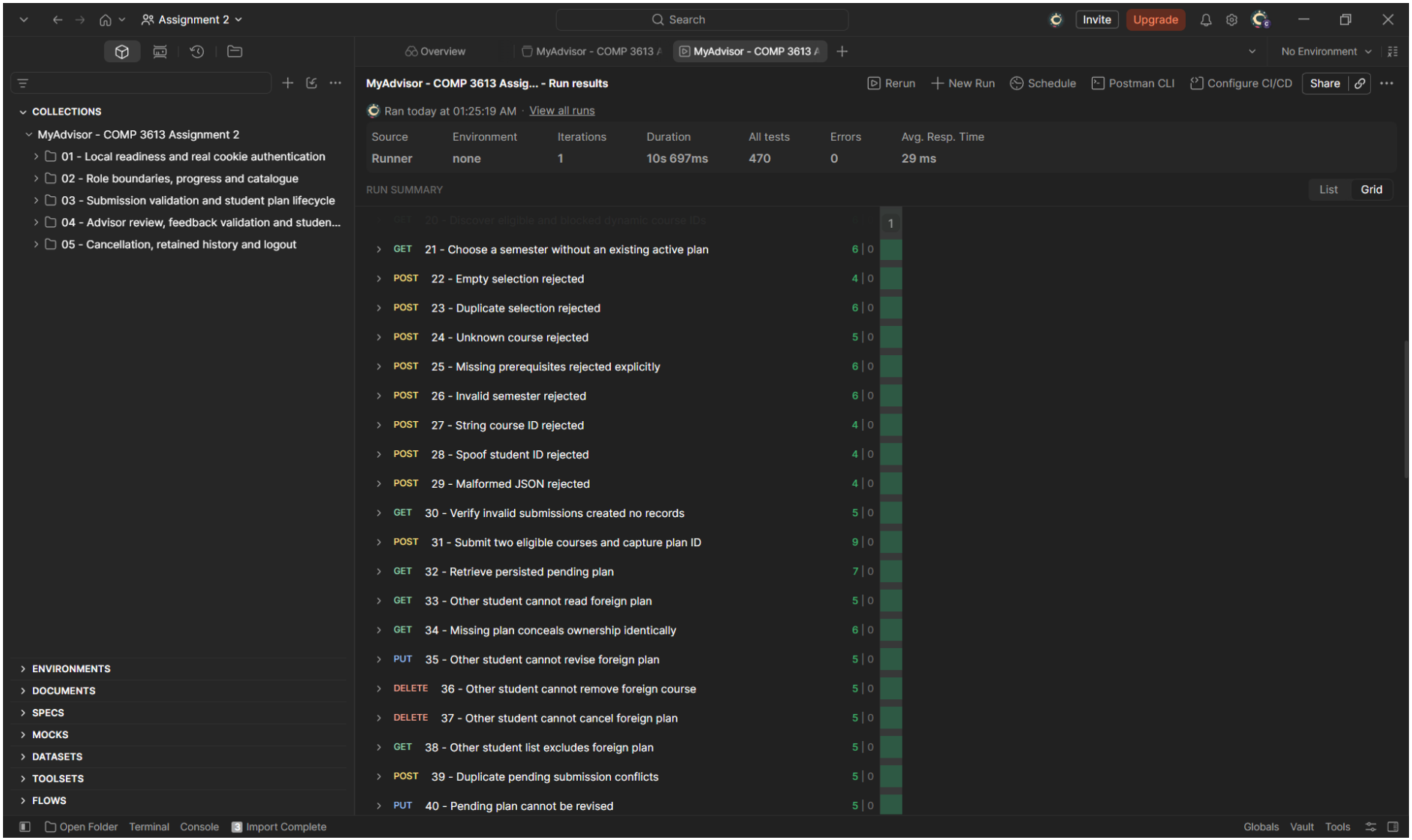
Original postman-results.png. Completed 27 October 2026: 470 passing assertions, 0 errors, 1 iteration. Scroll captures collectively cover requests 01–90; partly clipped edge rows are not reconstructed.

Figure 9.4.2. Genuine Postman Runner evidence



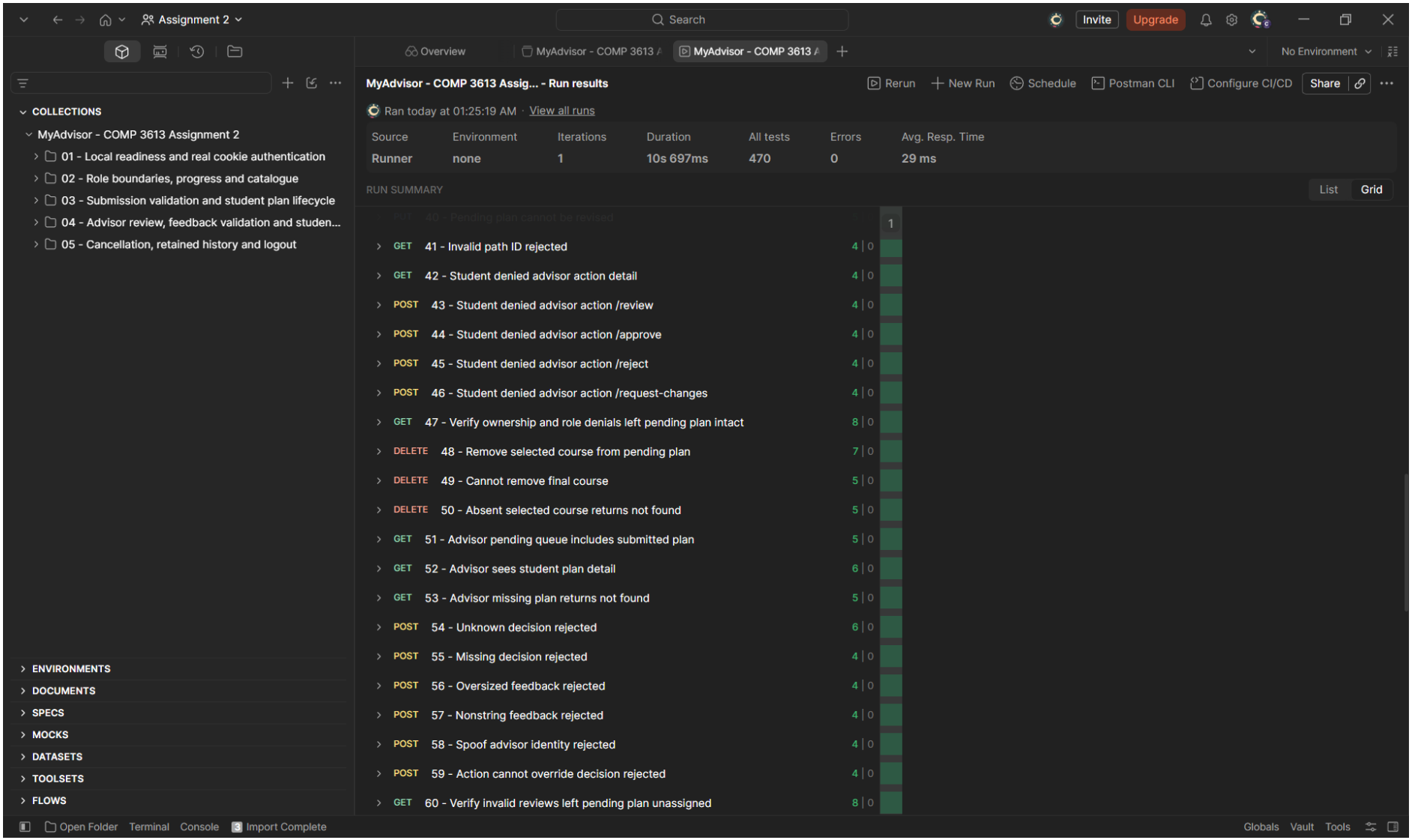
Original Screenshot 2026-10-27 012608.png. Completed 27 October 2026: 470 passing assertions, 0 errors, 1 iteration. Scroll captures collectively cover requests 01–90; partly clipped edge rows are not reconstructed.

Figure 9.4.3. Genuine Postman Runner evidence



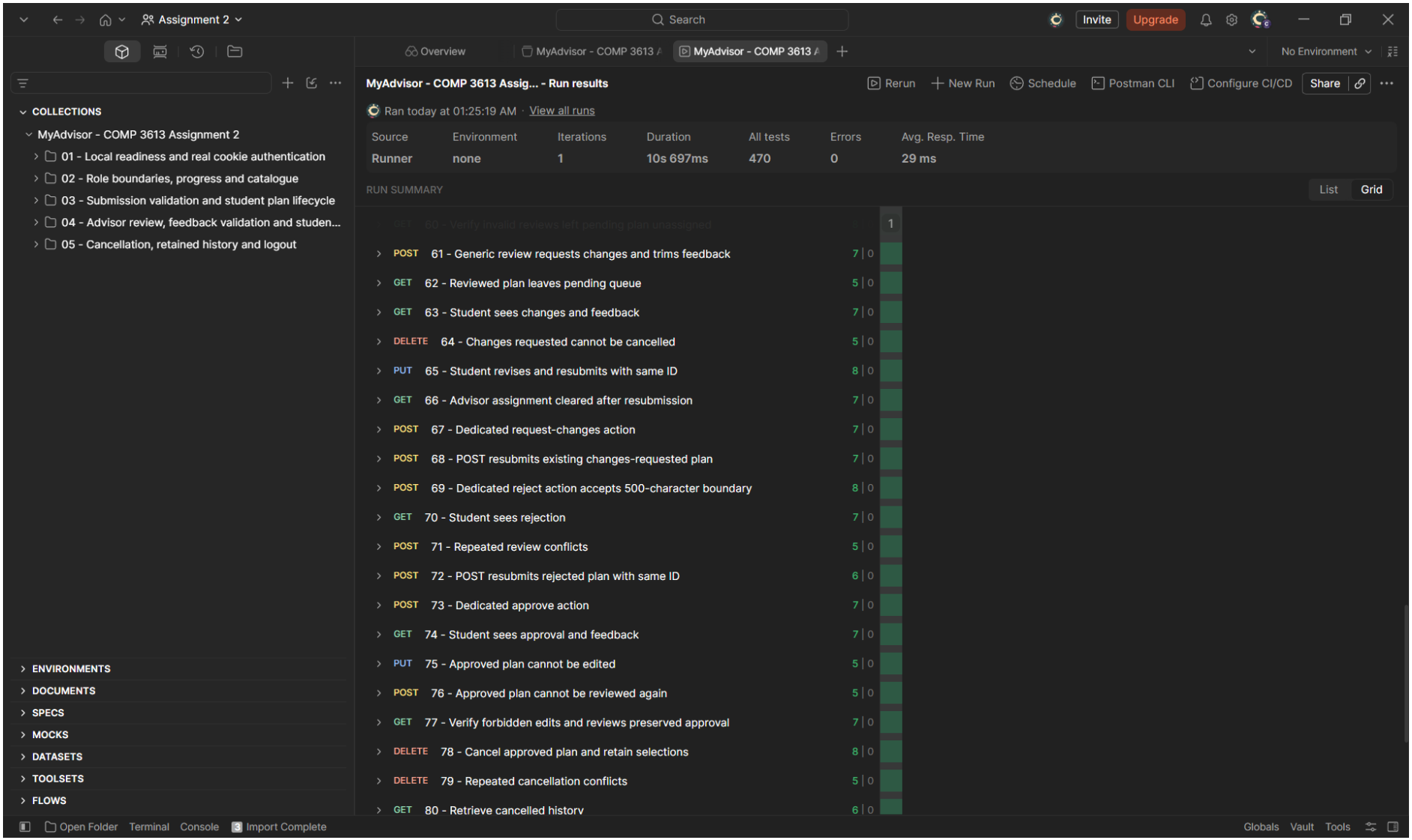
Original Screenshot 2026-10-27 012643.png. Completed 27 October 2026: 470 passing assertions, 0 errors, 1 iteration. Scroll captures collectively cover requests 01–90; partly clipped edge rows are not reconstructed.

Figure 9.4.4. Genuine Postman Runner evidence



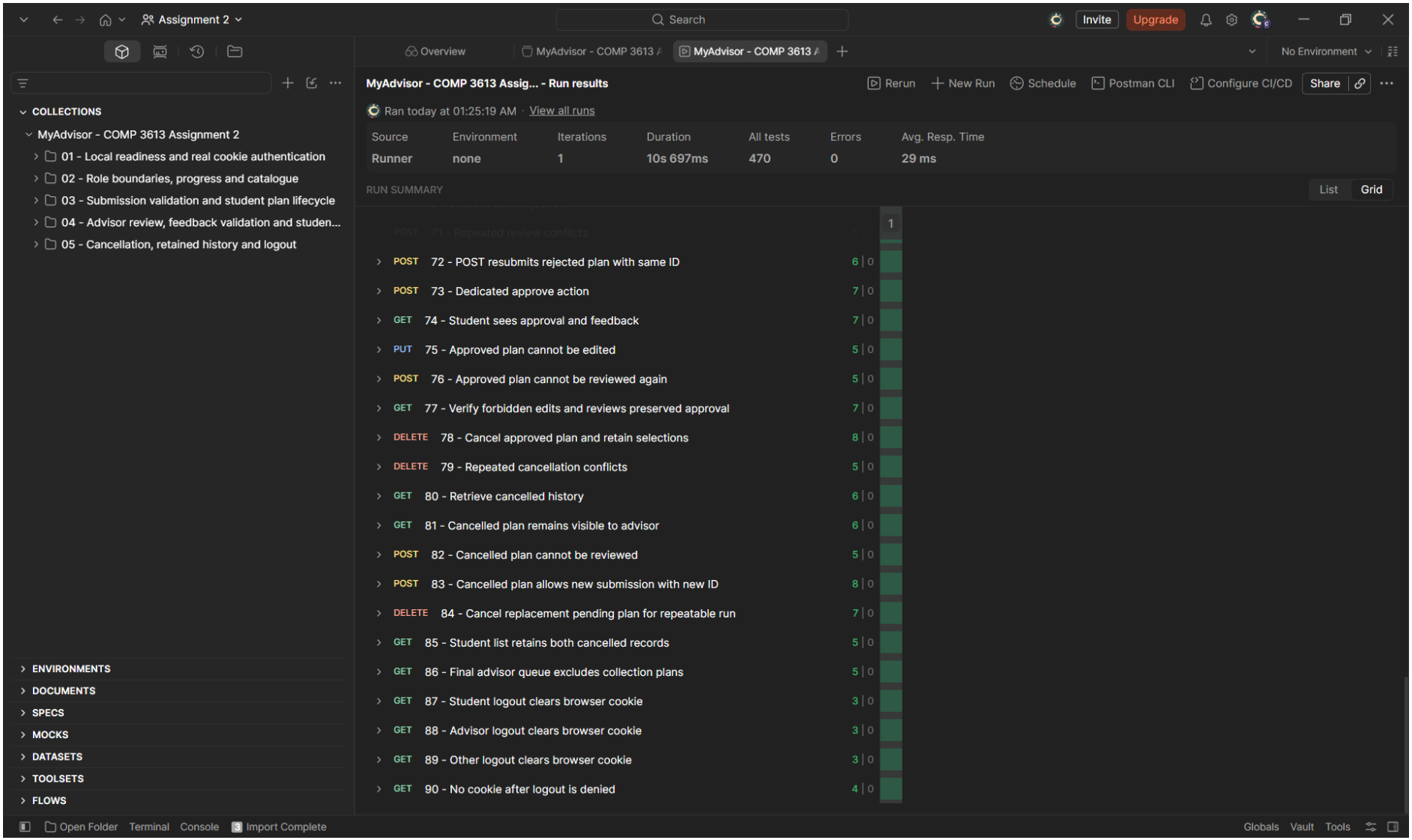
Original Screenshot 2026-10-27 012658.png. Completed 27 October 2026: 470 passing assertions, 0 errors, 1 iteration. Scroll captures collectively cover requests 01–90; partly clipped edge rows are not reconstructed.

Figure 9.4.5. Genuine Postman Runner evidence



Original Screenshot 2026-10-27 012739.png. Completed 27 October 2026: 470 passing assertions, 0 errors, 1 iteration. Scroll captures collectively cover requests 01–90; partly clipped edge rows are not reconstructed.

Figure 9.4.6. Genuine Postman Runner evidence



Original Screenshot 2026-10-27 012839.png. Completed 27 October 2026: 470 passing assertions, 0 errors, 1 iteration. Scroll captures collectively cover requests 01–90; partly clipped edge rows are not reconstructed.

9.5 Manual UAT Results

Tester Cameron Ramesar explicitly confirmed execution and all 19 captures on 27 October 2026. The finalized execution sheet is authoritative: **1 Pass / 0 Fail / 19 Not Run**. Not Run means complete-case result remains unverified: 18 cases have observed partial results and UAT-05 lacks its required Pending/relogin workflow evidence. No expected outcome is counted as observed. UAT-06 passes only the recorded review objective with substituted data, not UI usability acceptance.

Actual initial plan A: MATH 1300 + MATH 2200 + MATH 2250, 3 courses / 9 credits. Revised A: MATH 1300 + MATH 2200 + ENGL 1000, still 3 / 9. The planned A was COMP 1201 + MATH 2250, 2 / 6; revised planned A was COMP 1201 + ENGL 1000. Initial Mathematics choices have no prerequisites and therefore do not establish the intended nonempty-prerequisite positive path. This is a documented data deviation, not an invented failure.

Prompt 16 visual correction: the filenames of the two cancellation captures are reversed relative to their content. UAT-16-pending-cancelled.png shows semester B Cancelled with rejection feedback and supports case UAT-15. UAT-15-rejected-cancelled.png shows semester C Cancelled and supports case UAT-16. The image bytes/names are unchanged; references are corrected in the current sheet/report/matrix. No chronology or remaining substep is inferred from filenames, and totals are unchanged.

Case	Recorded status	Evidence disposition
UAT-01	Not Run	Partially verified; required substeps remain
UAT-02	Not Run	Partially verified; required substeps remain
UAT-03	Not Run	Partially verified; required substeps remain
UAT-04	Not Run	Partially verified; required substeps remain
UAT-05	Not Run	Required workflow unverified
UAT-06	Pass	Review objective verified with substituted data
UAT-07	Not Run	Partially verified; required substeps remain
UAT-08	Not Run	Partially verified; required substeps remain
UAT-09	Not Run	Partially verified; required substeps remain
UAT-10	Not Run	Partially verified; required substeps remain
UAT-11	Not Run	Partially verified; required substeps remain
UAT-12	Not Run	Partially verified; required substeps remain
UAT-13	Not Run	Partially verified; required substeps remain
UAT-14	Not Run	Partially verified; required substeps remain
UAT-15	Not Run	Partially verified; required substeps remain
UAT-16	Not Run	Partially verified; required substeps remain
UAT-17	Not Run	Partially verified; required substeps remain
UAT-18	Not Run	Partially verified; required substeps remain
UAT-19	Not Run	Partially verified; required substeps remain
UAT-20	Not Run	Partially verified; required substeps remain

Source: docs/assignment2/reports/uat-execution.md; uat-final-report.md; uat-supplemental-evidence.md.

UAT-01 — View degree progress

Execution field	Value
Actual result	Degree Progress shows 6 completed, 3 in progress, 84 remaining, 93 required and 6/93 (6%); category denominators total 120; recent academic activity is visible. Data deviation: No demonstrated course-data deviation for this case; unobserved inputs are not claimed executed. Remaining: Confirm totals/activity after refresh; reviewer disposition of DATA-01 (120 category credits versus 93 degree credits).
Defect ID if applicable	DATA-01
Pass/Fail/Not Run	Not Run
Proposed review disposition	Partially verified; complete case remains unconfirmed
Tester	Cameron Ramesar (explicit user confirmation)
Execution date	27 October 2026 (user-confirmed)
Screenshot evidence reference	UAT-02.png

UAT-02 — Browse, search, and filter available courses

Execution field	Value
Actual result	Year 1 semester 1, math search and All departments are visible; MATH 1300, MATH 2200 and MATH 2250 are Eligible, each 3 credits; MATH 1200 is already completed. Data deviation: Actual A initial: MATH 1300/2200/2250, 3 courses/9 credits; planned COMP 1201 + MATH 2250, 2 courses/6 credits. Remaining: Confirm semester switching, code and title searches, department and combined filters, no-match search and Clear Filters outcomes.
Defect ID if applicable	No confirmed functional mismatch/retest; see issue log
Pass/Fail/Not Run	Not Run
Proposed review disposition	Partially verified; complete case remains unconfirmed
Tester	Cameron Ramesar (explicit user confirmation)
Execution date	27 October 2026 (user-confirmed)
Screenshot evidence reference	UAT-03.png

UAT-03 — Select only eligible courses and review the selection

Execution field	Value
Actual result	Three eligible Mathematics courses are selected; selection count is 3. Completed-course restriction is visible. Post-submission image also shows in-progress and missing-prerequisite reasons, but that form is locked. Data deviation: Actual A initial: MATH 1300/2200/2250, 3 courses/9 credits; planned COMP 1201 + MATH 2250, 2 courses/6 credits. Remaining: Confirm empty-selection prevention, pre-submission ineligible controls, actual count changes, review total and Back preserving choices; exact planned 0?2?1?2 sequence was not established.
Defect ID if applicable	No confirmed functional mismatch/retest; see issue log
Pass/Fail/Not Run	Not Run
Proposed review disposition	Partially verified; complete case remains unconfirmed
Tester	Cameron Ramesar (explicit user confirmation)
Execution date	27 October 2026 (user-confirmed)
Screenshot evidence reference	UAT-03.png; UAT-04.png

UAT-04 — Submit semester plan A

Execution field	Value
Actual result	Submission success message; semester 1 Pending with MATH 1300, MATH 2200 and MATH 2250 (9 credits). Standard selection is locked; Remove and Cancel controls are visible. Data deviation: Actual A initial: MATH 1300/2200/2250, 3 courses/9 credits; planned COMP 1201 + MATH 2250, 2 courses/6 credits. Remaining: Confirm refresh persistence and repeat submission prevention through the Pending form.
Defect ID if applicable	No confirmed functional mismatch/retest; see issue log
Pass/Fail/Not Run	Not Run
Proposed review disposition	Partially verified; complete case remains unconfirmed
Tester	Cameron Ramesar (explicit user confirmation)
Execution date	27 October 2026 (user-confirmed)
Screenshot evidence reference	UAT-04.png

UAT-05 — View submission status

Execution field	Value
Actual result	No screenshot or supplied observation establishes the originally required Pending Home/Messages and sign-out/sign-in workflow. Later decision Messages are corroborating navigation evidence only. Data deviation: Actual A initial: MATH 1300/2200/2250, 3 courses/9 credits; planned COMP 1201 + MATH 2250, 2 courses/6 credits. Remaining: Confirm Pending Home/Messages summary (actual 3 courses/9 credits) and selection retained after sign-out/sign-in.
Defect ID if applicable	No confirmed functional mismatch/retest; see issue log
Pass/Fail/Not Run	Not Run
Proposed review disposition	Unverified required workflow
Tester	Cameron Ramesar (explicit user confirmation)
Execution date	27 October 2026 (user-confirmed)
Screenshot evidence reference	None for the required workflow

UAT-06 — Advisor reviews the submitted plan

Execution field	Value
Actual result	Advisor Review shows the correct Student identity/ID/program, October 27 submission date, semester 1 Pending, MATH 1300/2200/2250, 3 courses and 9 credits, comments counter and all three decision controls. Reading the card leaves it Pending. Actual data satisfies the review objective with the documented substitution. Data deviation: Actual A initial: MATH 1300/2200/2250, 3 courses/9 credits; planned COMP 1201 + MATH 2250, 2 courses/6 credits. Remaining: No remaining outcome check for this review objective with substituted data. Separate-profile setup/login attempts remain unverified under UAT-18/19; table contrast issue UI-01 remains open.
Defect ID if applicable	UI-01; impact/retest unconfirmed
Pass/Fail/Not Run	Pass
Proposed review disposition	Verified complete review objective with substituted data
Tester	Cameron Ramesar (explicit user confirmation)
Execution date	27 October 2026 (user-confirmed)
Screenshot evidence reference	UAT-05.png

UAT-07 — Request changes and deliver feedback

Execution field	Value
Actual result	Advisor Messages and Student Plan Courses show Changes Requested for semester 1 with identical feedback: "Replace MATH 2250 with ENGL 1000, then resubmit." Original three Mathematics selections remain selected in the editable form; update/resubmit instruction is visible and Cancel is absent. Data deviation: Actual A initial: MATH 1300/2200/2250, 3 courses/9 credits; planned COMP 1201 + MATH 2250, 2 courses/6 credits. Remaining: Confirm Advisor action confirmation, removal from Pending queue, and Student Messages delivery (Student Plan Courses delivery is established).
Defect ID if applicable	No confirmed functional mismatch/retest; see issue log
Pass/Fail/Not Run	Not Run
Proposed review disposition	Partially verified; complete case remains unconfirmed
Tester	Cameron Ramesar (explicit user confirmation)
Execution date	27 October 2026 (user-confirmed)
Screenshot evidence reference	UAT-07-request-changes.png; UAT-07-student-feedback.png

UAT-08 — Modify and resubmit the returned plan

Execution field	Value
Actual result	Advisor Pending queue shows revised semester 1: ENGL 1000, MATH 1300 and MATH 2200; 3 courses, 9 credits. MATH 2250 was replaced. Empty Advisor comments input is visible; it does not independently prove stored feedback was cleared. Data deviation: Actual revised A: MATH 1300/2200 + ENGL 1000, 3 courses/9 credits; planned COMP 1201 + ENGL 1000, 2 courses/6 credits. B/C data otherwise matches the plan where shown. Exact executed feedback is recorded above; no feedback deviation is inferred. Remaining: Confirm Student Pending state and review total for the actual 3-course/9-credit revision, current Pending Messages with prior feedback cleared, and exactly one revised Advisor queue entry after refresh.
Defect ID if applicable	No confirmed functional mismatch/retest; see issue log
Pass/Fail/Not Run	Not Run
Proposed review disposition	Partially verified; complete case remains unconfirmed
Tester	Cameron Ramesar (explicit user confirmation)
Execution date	27 October 2026 (user-confirmed)
Screenshot evidence reference	UAT-08-resubmitted.png

UAT-09 — Approve the revised plan

Execution field	Value
Actual result	Student semester 1 is Approved; feedback: "Revised selection approved." ENGL 1000, MATH 1300 and MATH 2200 remain selected (9 credits); standard selection is locked, Remove absent and Cancel available. Student Messages in UAT-11 also shows this approval/feedback. Data deviation: Actual revised A: MATH 1300/2200 + ENGL 1000, 3 courses/9 credits; planned COMP 1201 + ENGL 1000, 2 courses/6 credits. B/C data otherwise matches the plan where shown. Exact executed feedback is recorded above; no feedback deviation is inferred. Remaining: Confirm Advisor action confirmation and Advisor Approved feedback/queue removal; Student refresh persistence.
Defect ID if applicable	No confirmed functional mismatch/retest; see issue log
Pass/Fail/Not Run	Not Run
Proposed review disposition	Partially verified; complete case remains unconfirmed
Tester	Cameron Ramesar (explicit user confirmation)
Execution date	27 October 2026 (user-confirmed)
Screenshot evidence reference	UAT-09-approved.png; UAT-11-rejected.png

UAT-10 — Submit a separate plan for rejection

Execution field	Value
Actual result	Student semester 2 submission success and Pending state with only MATH 2250, Industrial Statistics, 3 credits. Data deviation: Actual revised A: MATH 1300/2200 + ENGL 1000, 3 courses/9 credits; planned COMP 1201 + ENGL 1000, 2 courses/6 credits. B/C data otherwise matches the plan where shown. Exact executed feedback is recorded above; no feedback deviation is inferred. Remaining: Confirm switching between semesters preserves each separate selection/status; later UAT-11 supports retained approval, not every switch.
Defect ID if applicable	No confirmed functional mismatch/retest; see issue log
Pass/Fail/Not Run	Not Run
Proposed review disposition	Partially verified; complete case remains unconfirmed
Tester	Cameron Ramesar (explicit user confirmation)
Execution date	27 October 2026 (user-confirmed)
Screenshot evidence reference	UAT-10-second-plan.png

UAT-11 — Reject the separate plan

Execution field	Value
Actual result	Student Messages shows semester 2 Rejected with feedback: "Plan rejected for this UAT scenario; choose a different course mix." Semester 1 remains Approved with its approval feedback. Data deviation: Actual revised A: MATH 1300/2200 + ENGL 1000, 3 courses/9 credits; planned COMP 1201 + ENGL 1000, 2 courses/6 credits. B/C data otherwise matches the plan where shown. Exact executed feedback is recorded above; no feedback deviation is inferred. Remaining: Confirm Advisor rejection confirmation/feedback and queue removal; Student rejected plan editability and Cancel availability.
Defect ID if applicable	No confirmed functional mismatch/retest; see issue log
Pass/Fail/Not Run	Not Run
Proposed review disposition	Partially verified; complete case remains unconfirmed
Tester	Cameron Ramesar (explicit user confirmation)
Execution date	27 October 2026 (user-confirmed)
Screenshot evidence reference	UAT-11-rejected.png

UAT-12 — Submit a pending plan for cancellation checks

Execution field	Value
Actual result	Advisor Review shows semester 3 Pending with COMP 1201 and MATH 2250, 2 courses and 6 credits. Data deviation: Actual revised A: MATH 1300/2200 + ENGL 1000, 3 courses/9 credits; planned COMP 1201 + ENGL 1000, 2 courses/6 credits. B/C data otherwise matches the plan where shown. Exact executed feedback is recorded above; no feedback deviation is inferred. Remaining: Confirm Student Pending view and that A Approved/B Rejected were unchanged during this submission.
Defect ID if applicable	No confirmed functional mismatch/retest; see issue log
Pass/Fail/Not Run	Not Run
Proposed review disposition	Partially verified; complete case remains unconfirmed
Tester	Cameron Ramesar (explicit user confirmation)
Execution date	27 October 2026 (user-confirmed)
Screenshot evidence reference	UAT-12-pending-plan.png

UAT-13 — Remove a course from a Pending plan and protect the last course

Execution field	Value
Actual result	"Course removed from the submitted plan." appears. Semester 3 stays Pending with only COMP 1201, 3 credits. No Remove button for the final course; "Cancel the plan to remove its last course." is visible. Data deviation: No demonstrated course-data deviation for this case; unobserved inputs are not claimed executed. Remaining: Confirm dismissing Remove changed nothing and refreshed Advisor queue showed COMP 1201/3 credits. Final-course UI guard is established; no forged removal request is claimed.
Defect ID if applicable	No confirmed functional mismatch/retest; see issue log
Pass/Fail/Not Run	Not Run
Proposed review disposition	Partially verified; complete case remains unconfirmed
Tester	Cameron Ramesar (explicit user confirmation)
Execution date	27 October 2026 (user-confirmed)
Screenshot evidence reference	UAT-13-course-removed.png

UAT-14 — Cancel an Approved plan

Execution field	Value
Actual result	Cancellation success; semester 1 Cancelled; "Revised selection approved." retained as feedback. Zero courses selected; new-selection instruction and eligible unchecked choice are visible; Cancel absent. Data deviation: Actual revised A: MATH 1300/2200 + ENGL 1000, 3 courses/9 credits; planned COMP 1201 + ENGL 1000, 2 courses/6 credits. B/C data otherwise matches the plan where shown. Exact executed feedback is recorded above; no feedback deviation is inferred. Remaining: Confirm dismissing cancellation preserved approval, refresh persisted Cancelled, and Messages showed Cancelled.
Defect ID if applicable	No confirmed functional mismatch/retest; see issue log
Pass/Fail/Not Run	Not Run
Proposed review disposition	Partially verified; complete case remains unconfirmed
Tester	Cameron Ramesar (explicit user confirmation)
Execution date	27 October 2026 (user-confirmed)
Screenshot evidence reference	UAT-14-approved-cancelled.png

UAT-15 — Cancel a Rejected plan

Execution field	Value
Actual result	Cancellation success; semester 2 Cancelled; rejection feedback retained exactly. Zero courses selected, new selection possible and Cancel absent. Data deviation: Actual revised A: MATH 1300/2200 + ENGL 1000, 3 courses/9 credits; planned COMP 1201 + ENGL 1000, 2 courses/6 credits. B/C data otherwise matches the plan where shown. Exact executed feedback is recorded above; no feedback deviation is inferred. Remaining: Confirm refresh/Messages Cancelled and A Cancelled/C Pending unchanged.
Defect ID if applicable	No confirmed functional mismatch/retest; see issue log
Pass/Fail/Not Run	Not Run
Proposed review disposition	Partially verified; complete case remains unconfirmed
Tester	Cameron Ramesar (explicit user confirmation)
Execution date	27 October 2026 (user-confirmed)
Screenshot evidence reference	UAT-16-pending-cancelled.png

UAT-16 — Cancel a Pending plan and remove it from the review queue

Execution field	Value
Actual result	Cancellation success; semester 3 Cancelled; zero selected courses; new-selection instruction and eligible unchecked choice visible; repeat Cancel absent. Data deviation: No demonstrated course-data deviation for this case; unobserved inputs are not claimed executed. Remaining: Confirm Student Messages Cancelled, removal from refreshed Advisor Pending queue and preservation of other students if present.
Defect ID if applicable	No confirmed functional mismatch/retest; see issue log
Pass/Fail/Not Run	Not Run
Proposed review disposition	Partially verified; complete case remains unconfirmed
Tester	Cameron Ramesar (explicit user confirmation)
Execution date	27 October 2026 (user-confirmed)
Screenshot evidence reference	UAT-15-rejected-cancelled.png

UAT-17 — Start a new plan after cancellation

Execution field	Value
Actual result	After cancellation, semester 3 shows successful submission and a new Pending selection containing only ENGL 1000, Academic Writing, 3 credits. Old COMP 1201/MATH 2250 selection is absent. Data deviation: No demonstrated course-data deviation for this case; unobserved inputs are not claimed executed. Remaining: Confirm Student refresh persistence and exactly one active new Advisor queue entry after refresh with ENGL 1000/3 credits. Historical Cancelled Messages entries are permitted, not a required additional check.
Defect ID if applicable	No confirmed functional mismatch/retest; see issue log
Pass/Fail/Not Run	Not Run
Proposed review disposition	Partially verified; complete case remains unconfirmed
Tester	Cameron Ramesar (explicit user confirmation)
Execution date	27 October 2026 (user-confirmed)
Screenshot evidence reference	UAT-17-new-plan.png

UAT-18 — Student login and clean-run baseline (execute first)

Execution field	Value
Actual result	Student Home shows the seeded identity and Student navigation, no submitted semester plan and 0 updates. This is authenticated-state evidence, not a capture of the login attempts. Data deviation: No demonstrated course-data deviation for this case; unobserved inputs are not claimed executed. Remaining: Confirm incorrect-login denial, valid-login destination, initial empty Messages/all A/B/C semesters and refresh retaining session.
Pass/Fail/Not Run	Not Run
Proposed review disposition	Partially verified; complete case remains unconfirmed
Tester	Cameron Ramesar (explicit user confirmation)
Execution date	27 October 2026 (user-confirmed)
Screenshot evidence reference	UAT-01.png
Defect ID if applicable	No confirmed functional mismatch/retest; see issue log

UAT-19 — Advisor login (execute after UAT-05, before UAT-06)

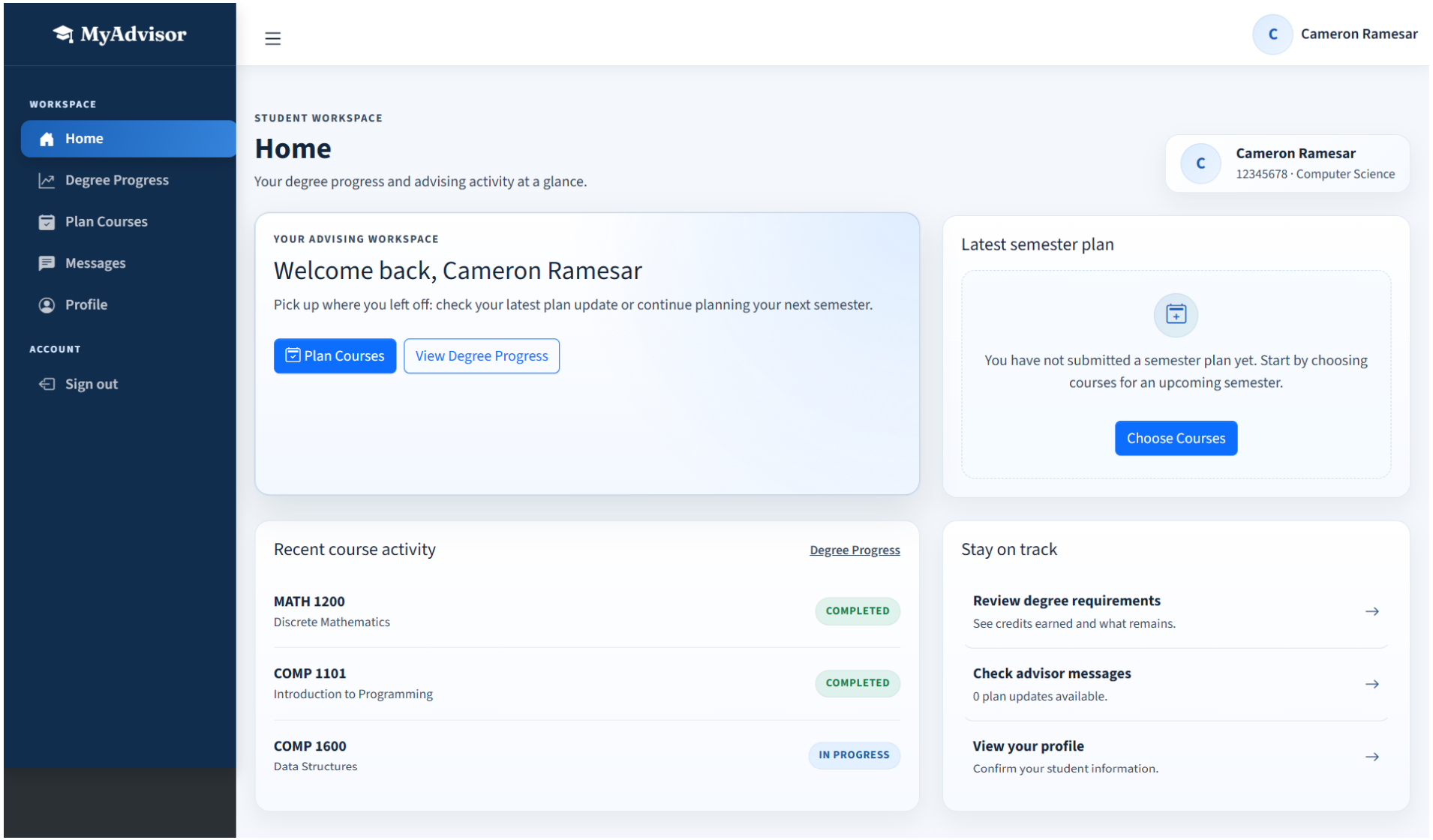
Execution field	Value
Actual result	Advisor Review shows Dr. Williams and Advisor navigation, with no Student planning/progress links. This is authenticated-state evidence, not a capture of login attempts. Data deviation: No demonstrated course-data deviation for this case; unobserved inputs are not claimed executed. Remaining: Confirm incorrect-login denial, valid Advisor Home destination and refresh preserving independent Student/Advisor sessions.
Pass/Fail/Not Run	Not Run
Proposed review disposition	Partially verified; complete case remains unconfirmed
Tester	Cameron Ramesar (explicit user confirmation)
Execution date	27 October 2026 (user-confirmed)
Screenshot evidence reference	UAT-05.png
Defect ID if applicable	No confirmed functional mismatch/retest; see issue log

UAT-20 — Browser-visible role and signed-out restrictions (execute last)

Execution field	Value
Actual result	401 Unauthorized / Session Expired page with Return to Login, plus an empty login screen. User attributes these to signed-out access restriction. No Student /admin 403 or Advisor restricted-tab fallback is demonstrated. Data deviation: No demonstrated course-data deviation for this case; unobserved inputs are not claimed executed. Remaining: Confirm logged-in Student direct /admin 403, Advisor Student-tab fallback, unchanged queue, and fresh protected-page denial after each role signed out (which actor/URL for each).
Pass/Fail/Not Run	Not Run
Proposed review disposition	Partially verified; complete case remains unconfirmed
Tester	Cameron Ramesar (explicit user confirmation)
Execution date	27 October 2026 (user-confirmed)
Screenshot evidence reference	UAT-20-access-denied.png; UAT-20-logged-out.png
Defect ID if applicable	No confirmed functional mismatch/retest; see issue log

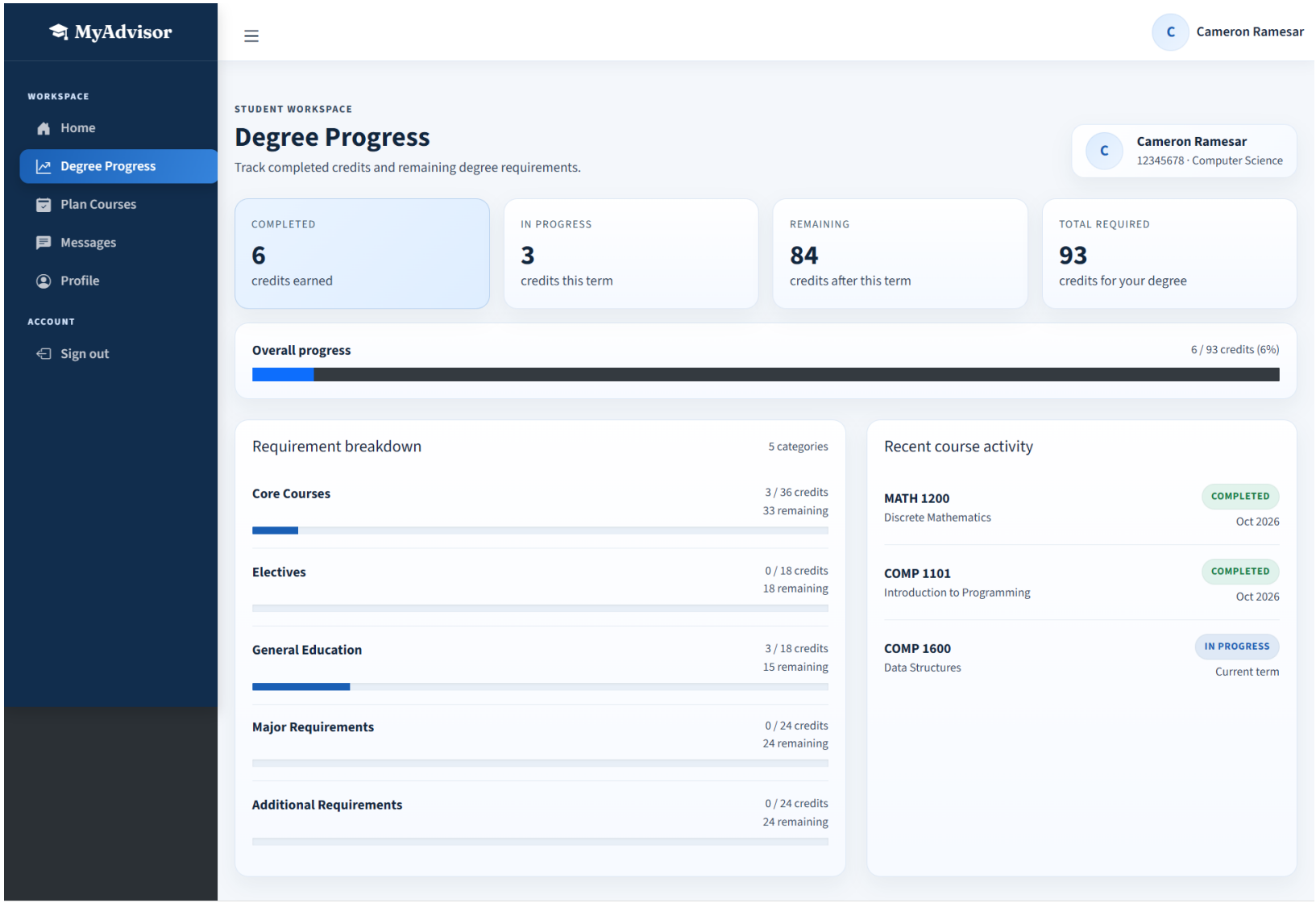
Signed-out/session-expired 401 presentation and login page do not establish logged-in Student /admin 403 or Advisor restricted-tab fallback. Refresh/persistence, queue uniqueness/removal, login attempts, actor-specific logout and feedback-reset substeps remain unverified where listed. Browser/version, actual URL, exact DB and tested build are not supplied. No manual retest or unconditional acceptance is claimed.

Figure 9.5.1. UAT-01



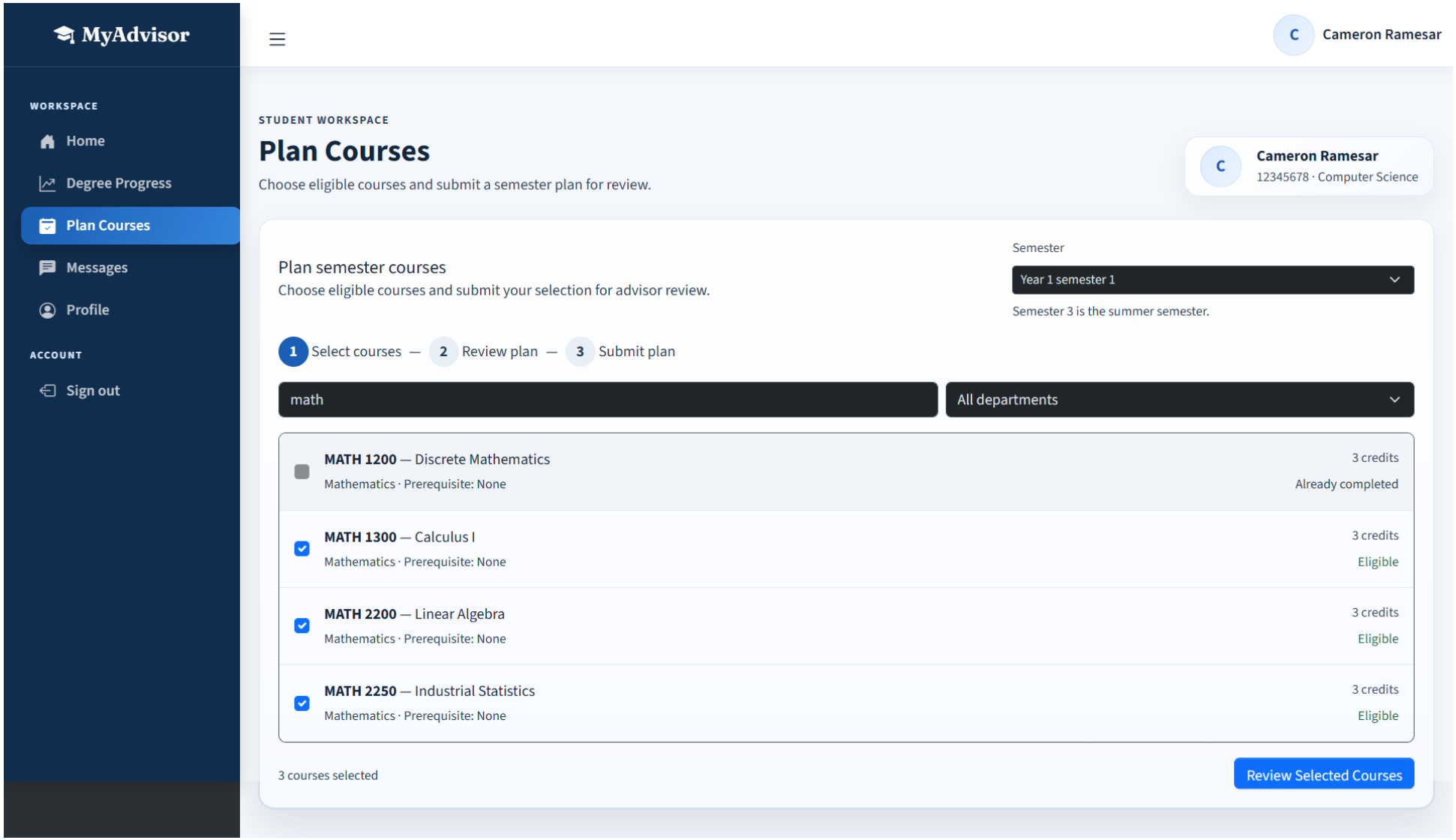
Student Home: authenticated state only (UAT-18 partial). Source: UAT-01.png. Captured 27 October 2026; demo identity shown, not submission ID.

Figure 9.5.2. UAT-02



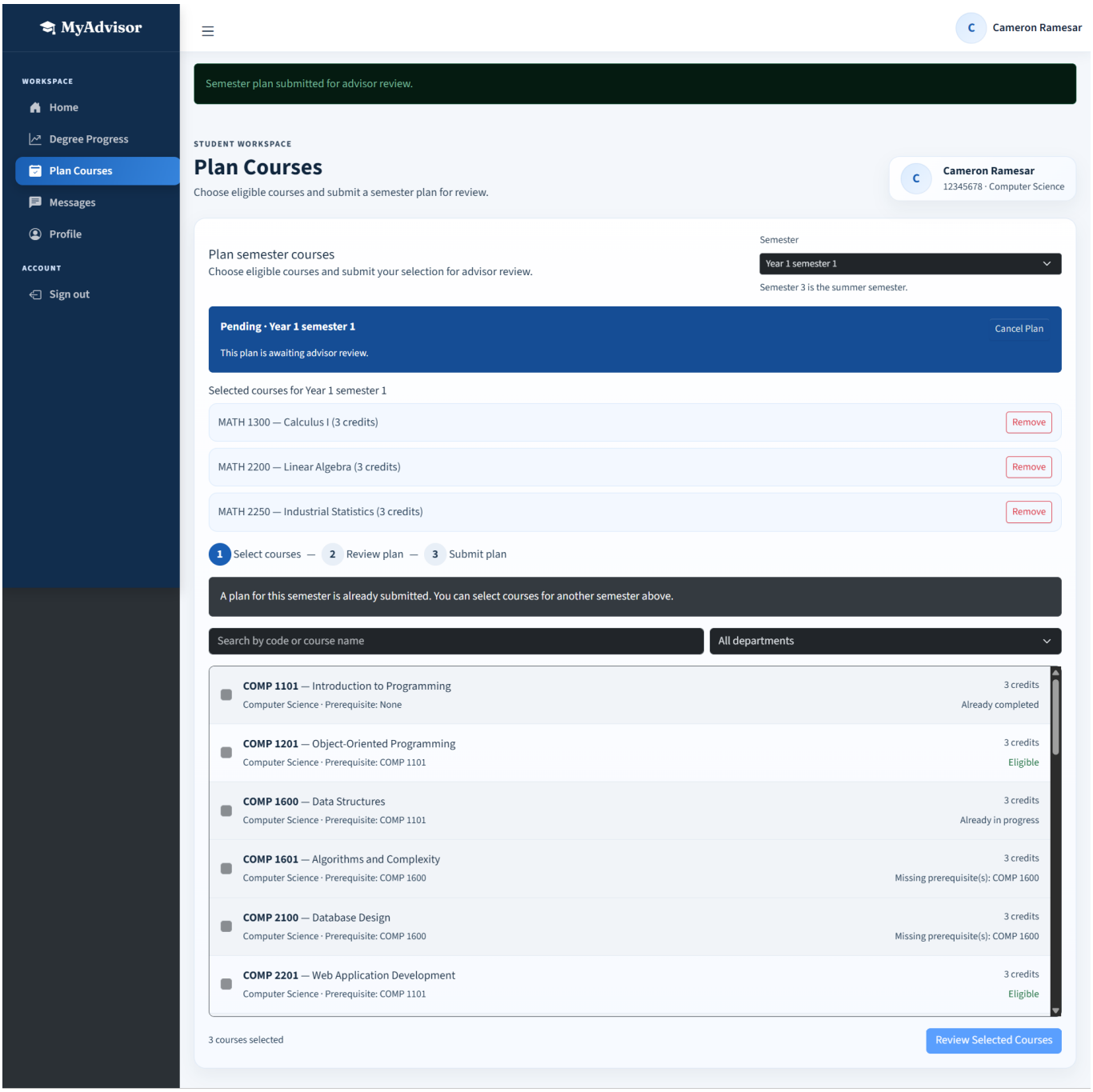
Degree Progress: 6 completed, 3 in progress, 84 remaining, 93 required; category total discrepancy visible (UAT-01 partial). Source: UAT-02.png. Captured 27 October 2026; demo identity shown, not submission ID.

Figure 9.5.3. UAT-03



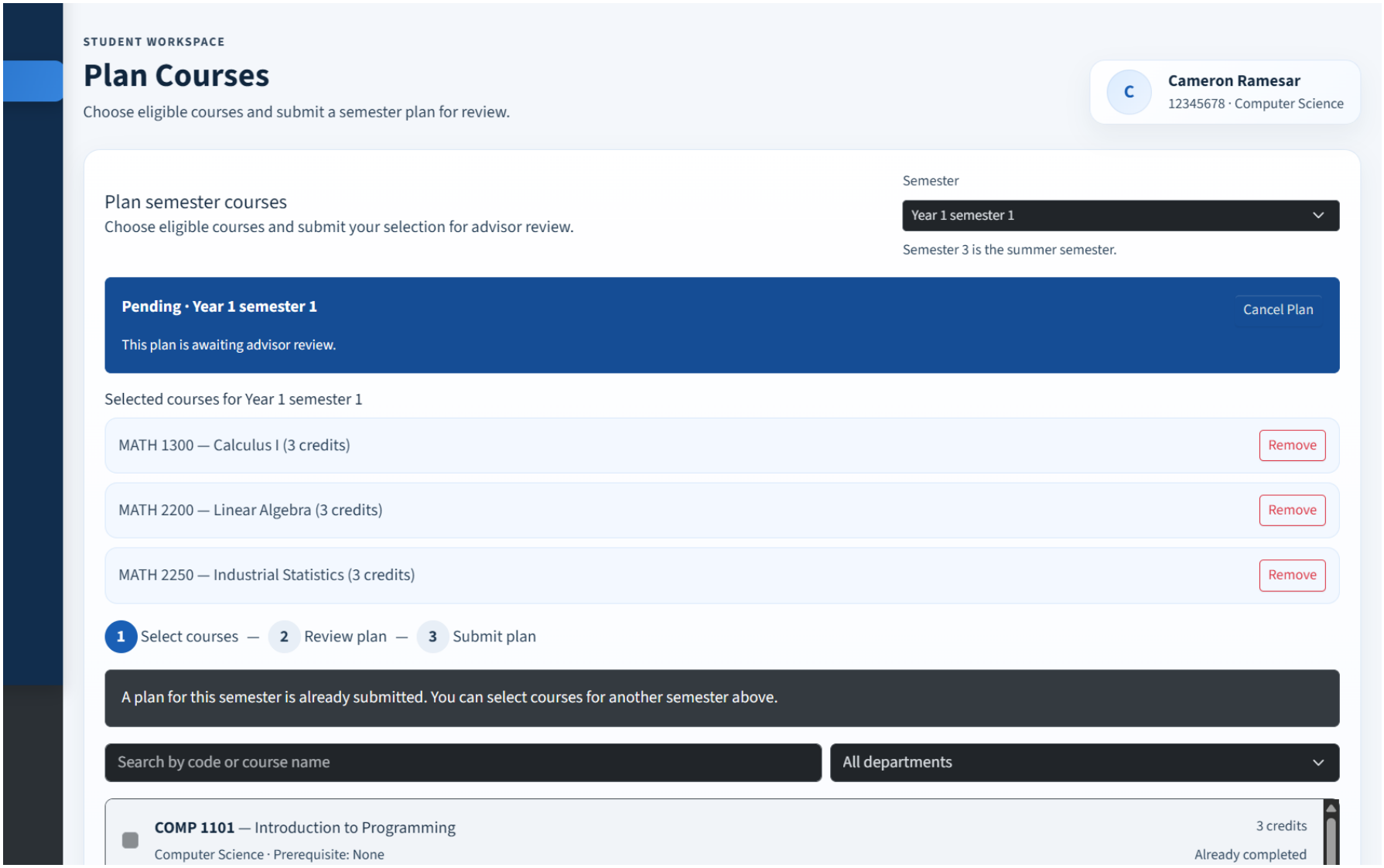
Mathematics search and three eligible selections, 9 credits (UAT-02/03 partial). Source: UAT-03.png. Captured 27 October 2026; demo identity shown, not submission ID.

Figure 9.5.4. UAT-04



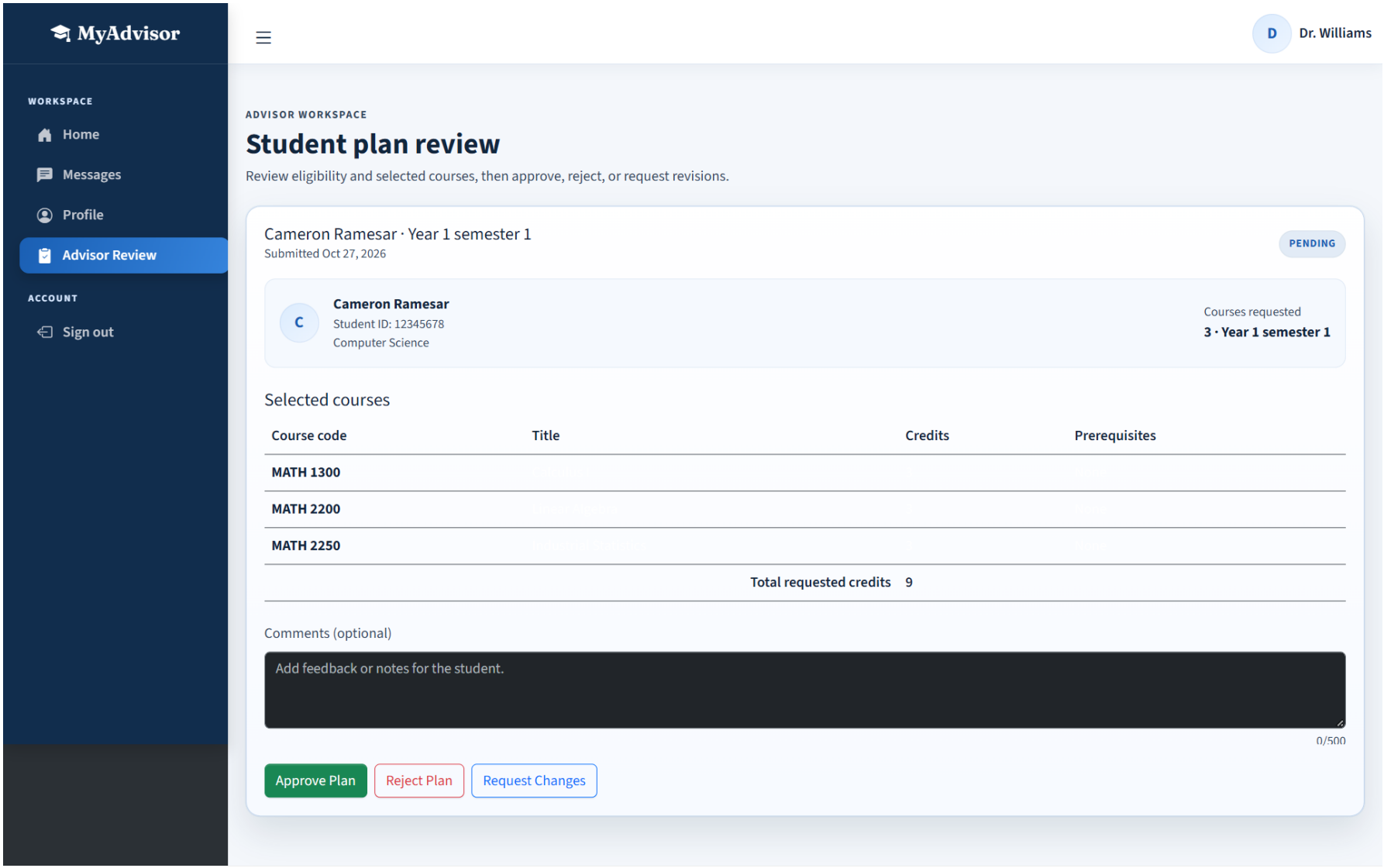
Successful three-course submission and Pending plan A (UAT-04 partial). Source: UAT-04.png. Captured 27 October 2026; demo identity shown, not submission ID.

Figure 9.5.4a. Enlarged original workflow detail



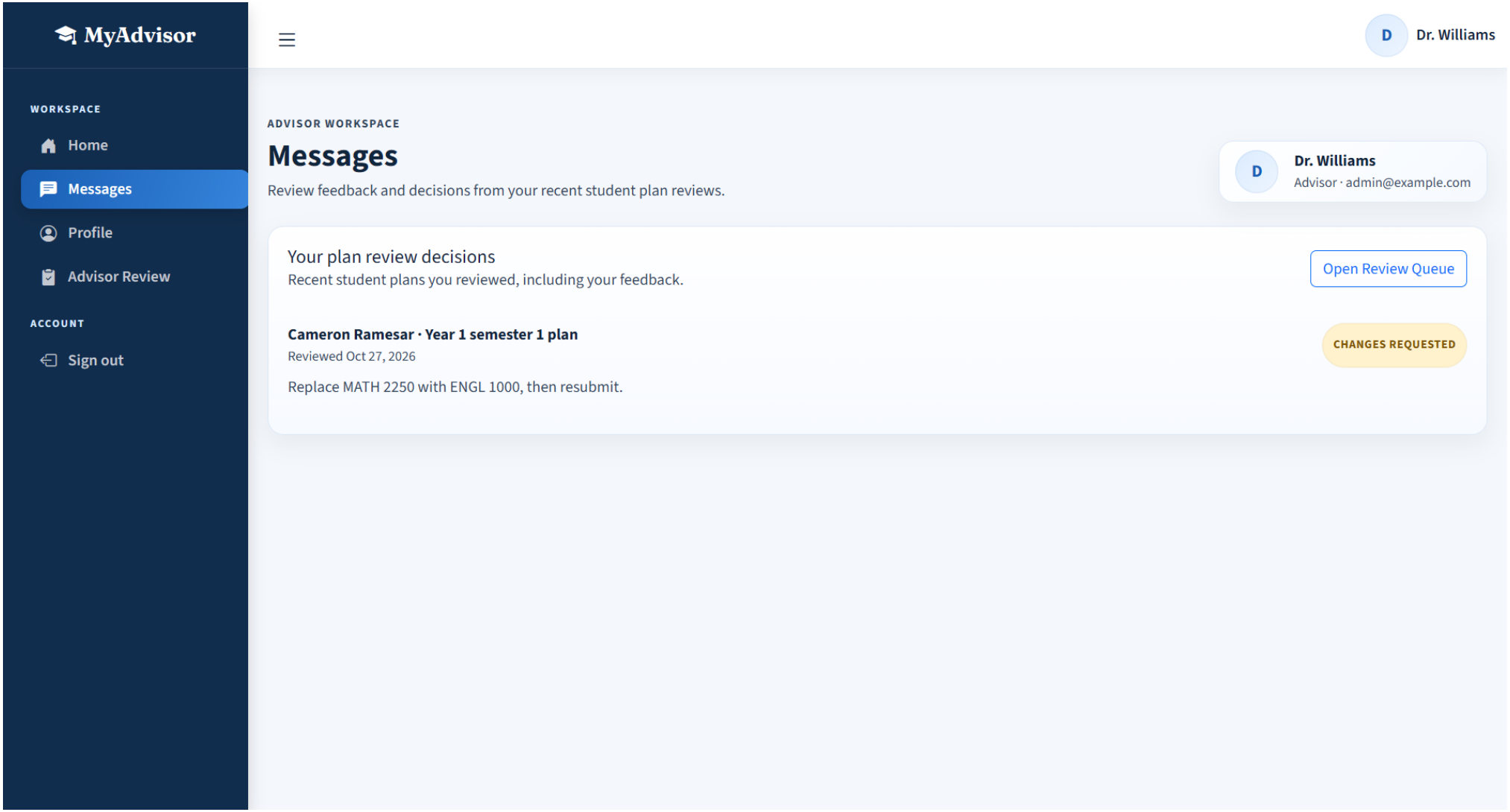
Pixel excerpt only; the preceding full original is preserved. Successful three-course submission and Pending plan A (UAT-04 partial).

Figure 9.5.5. UAT-05



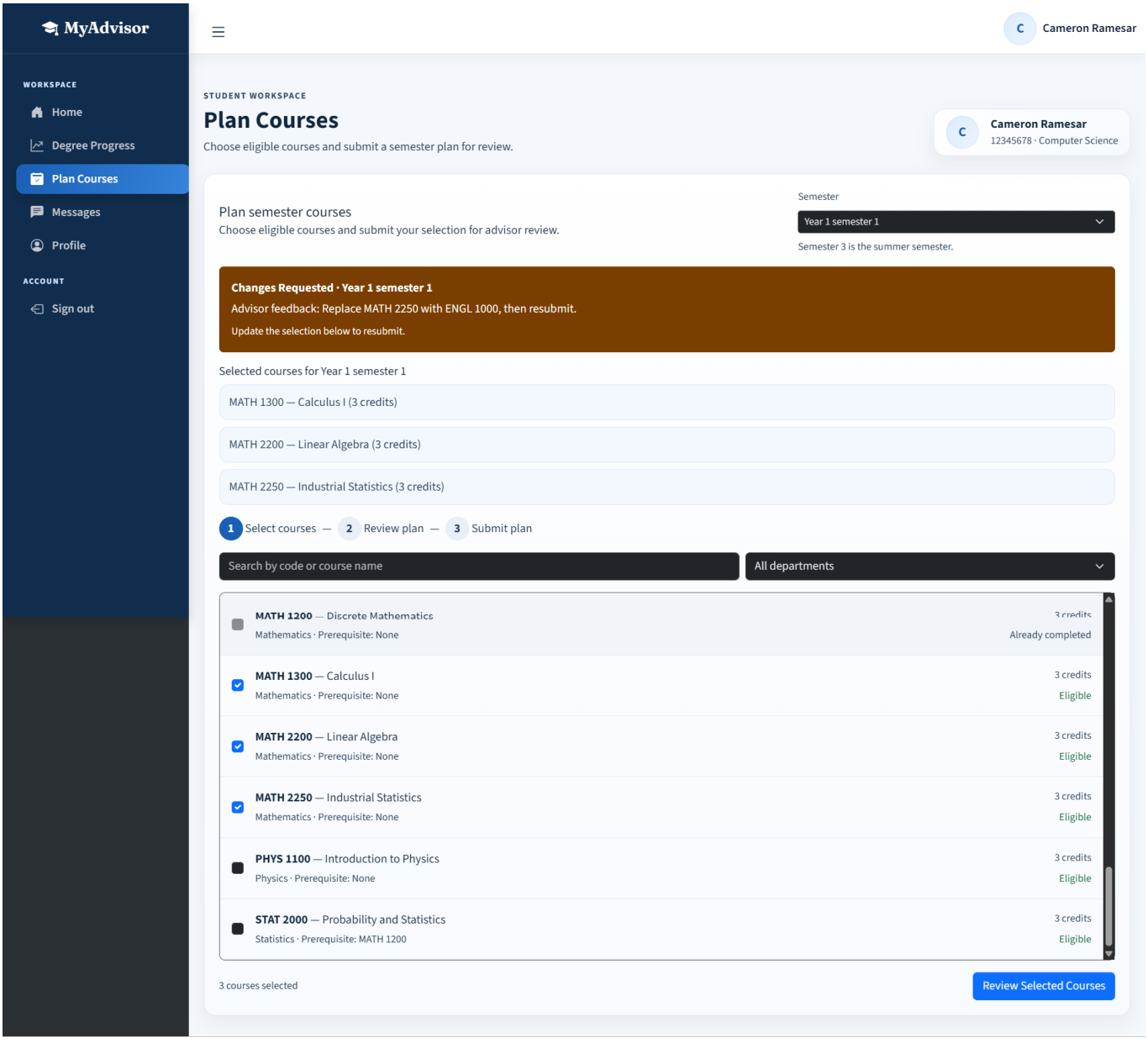
Advisor review objective: actual three-course/9-credit plan A (UAT-06 Pass; UAT-19 partial). Original table contrast defect remains visible. Source: UAT-05.png. Captured 27 October 2026; demo identity shown, not submission ID.

Figure 9.5.6. UAT-07-request-changes



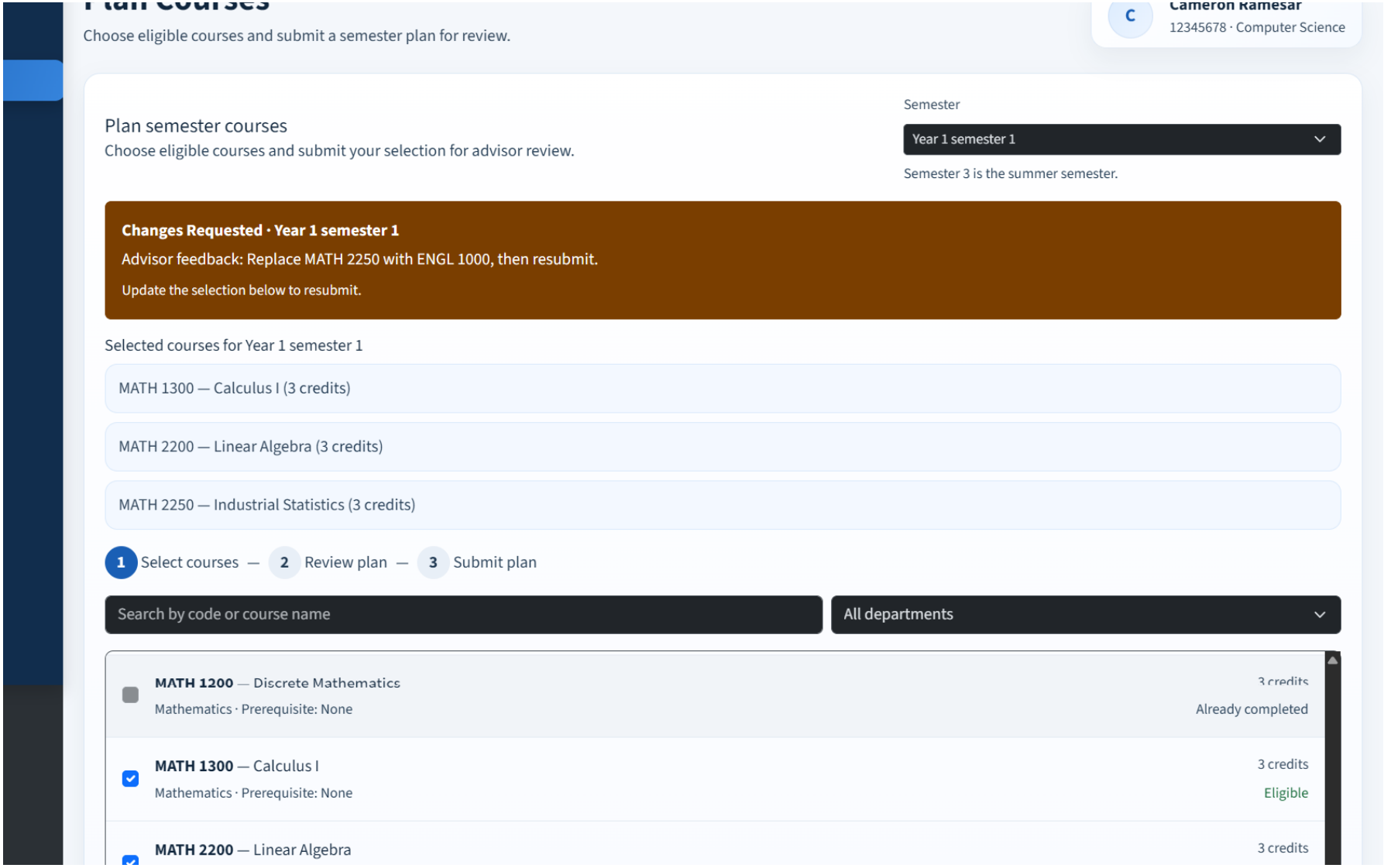
Advisor-side Changes Requested feedback (UAT-07 partial). Source: UAT-07-request-changes.png. Captured 27 October 2026; demo identity shown, not submission ID.

Figure 9.5.7. UAT-07-student-feedback



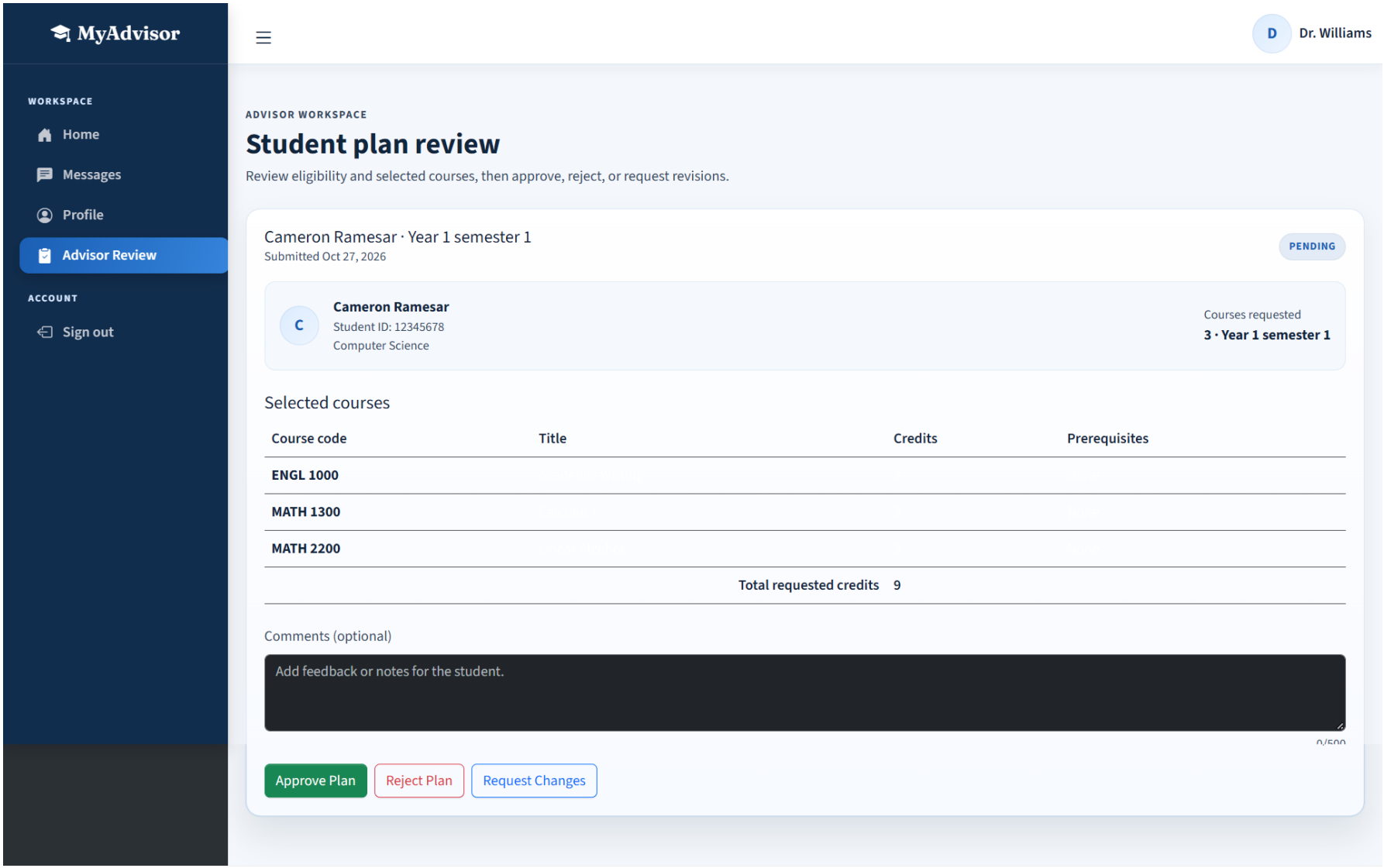
Student sees matching change-request feedback and editable selections (UAT-07 partial). Source: UAT-07-student-feedback.png. Captured 27 October 2026; demo identity shown, not submission ID.

Figure 9.5.7a. Enlarged original workflow detail



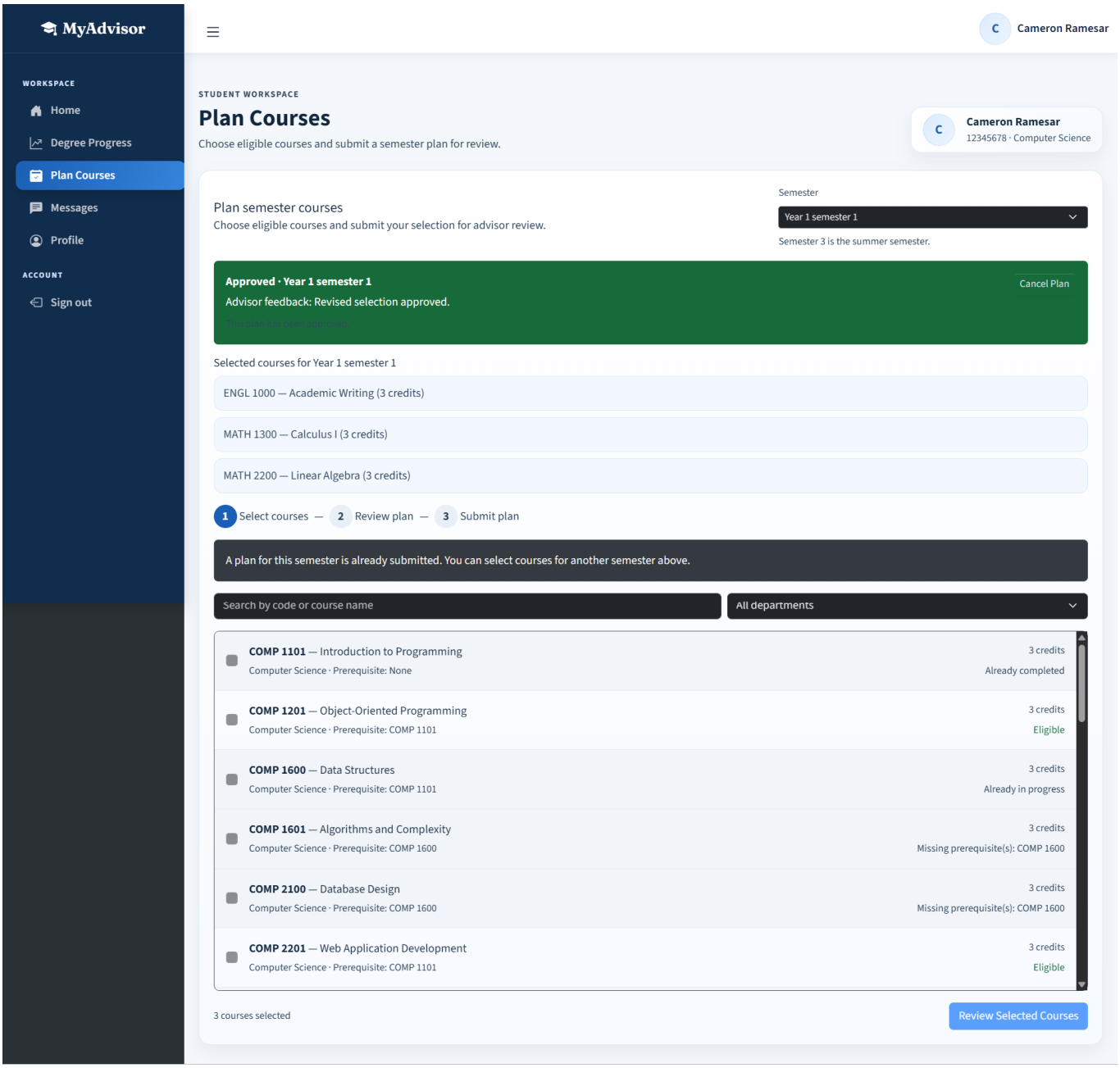
Pixel excerpt only; the preceding full original is preserved. Student sees matching change-request feedback and editable selections (UAT-07 partial).

Figure 9.5.8. UAT-08-resubmitted



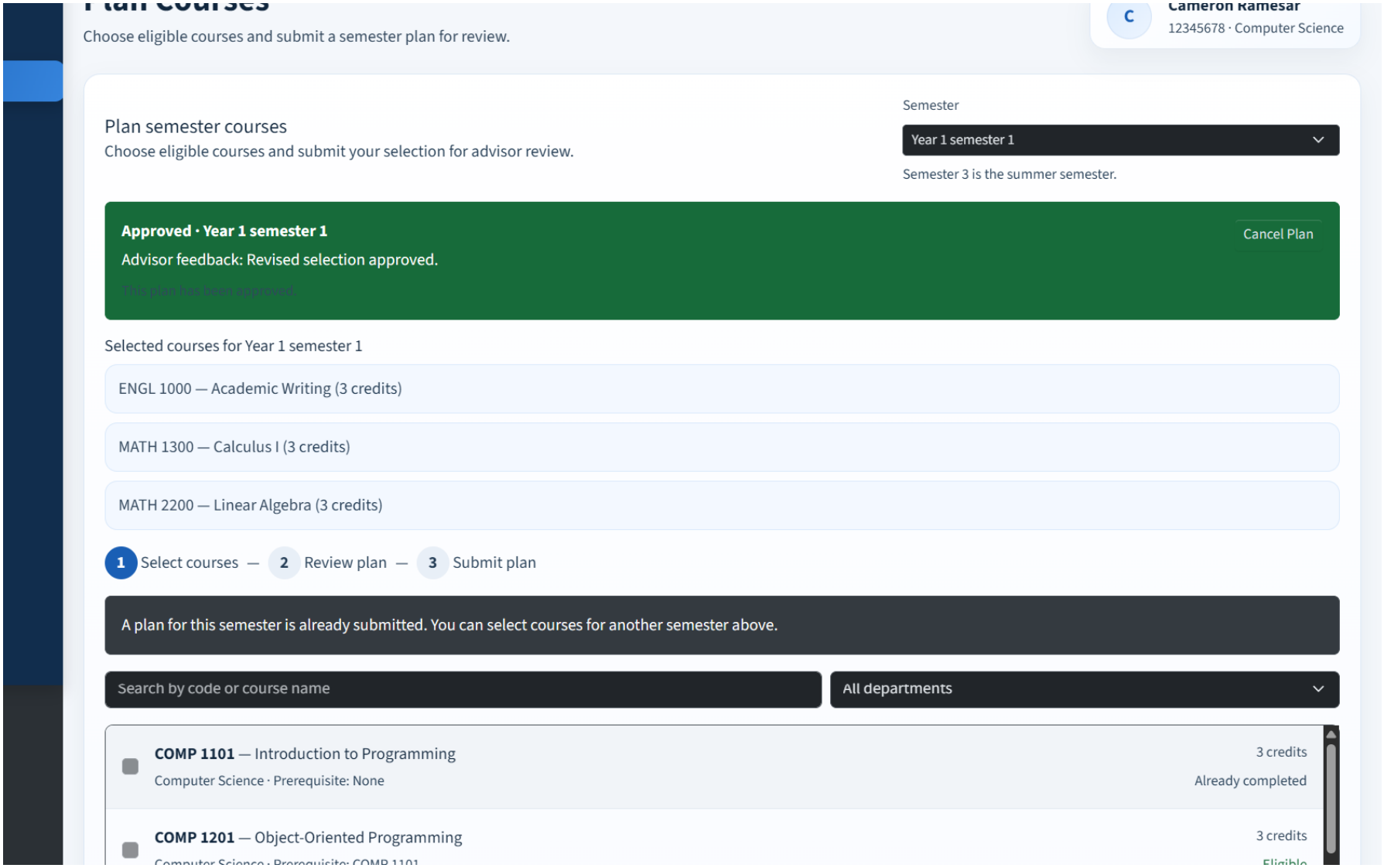
Revised plan A Pending with ENGL 1000/MATH 1300/MATH 2200, 9 credits (UAT-08 partial). Source: UAT-08-resubmitted.png. Captured 27 October 2026; demo identity shown, not submission ID.

Figure 9.5.9. UAT-09-approved



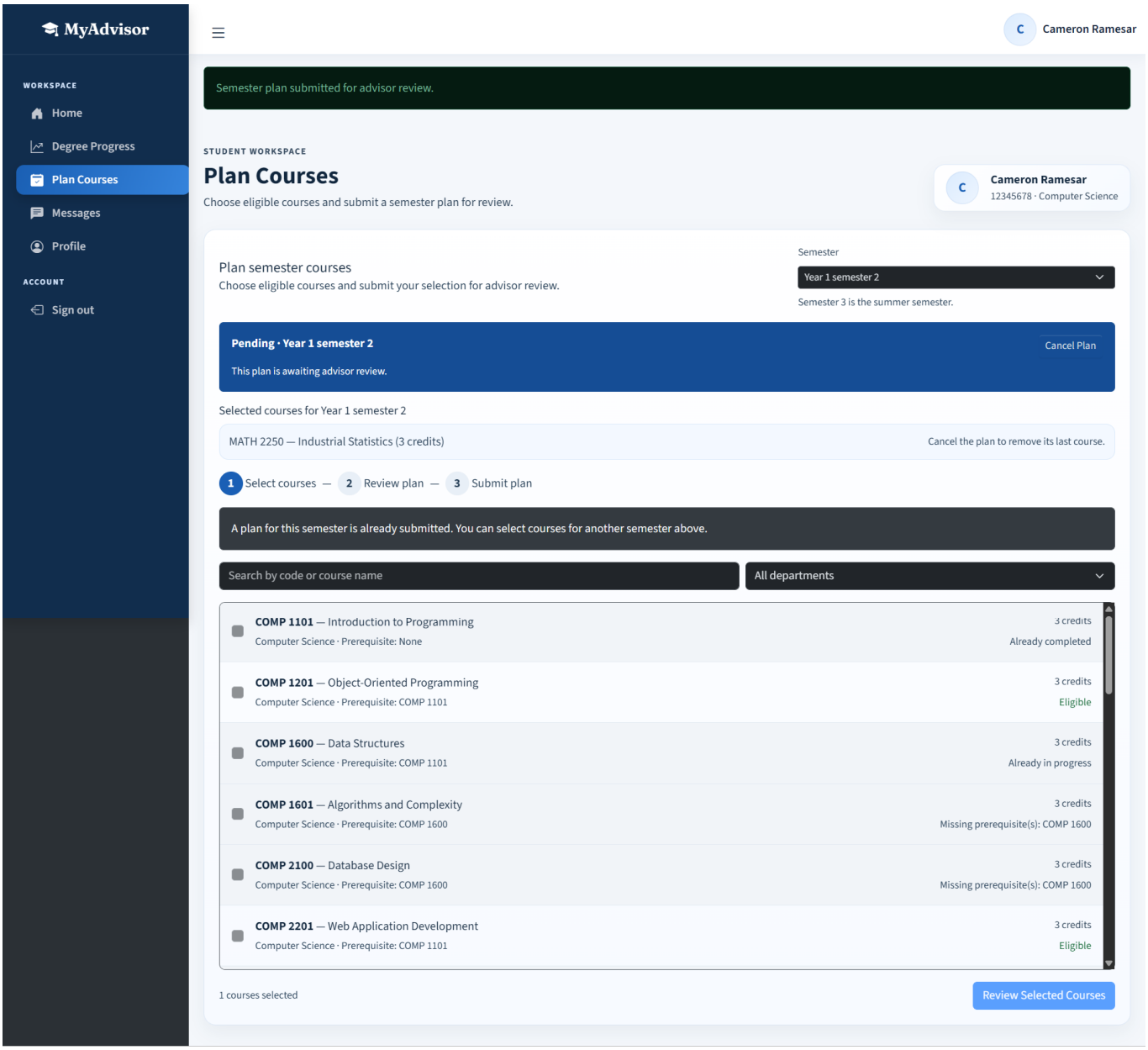
Approved plan A and revised-selection feedback (UAT-09 partial). Original low-contrast explanatory text retained. Source: UAT-09-approved.png. Captured 27 October 2026; demo identity shown, not submission ID.

Figure 9.5.9a. Enlarged original workflow detail



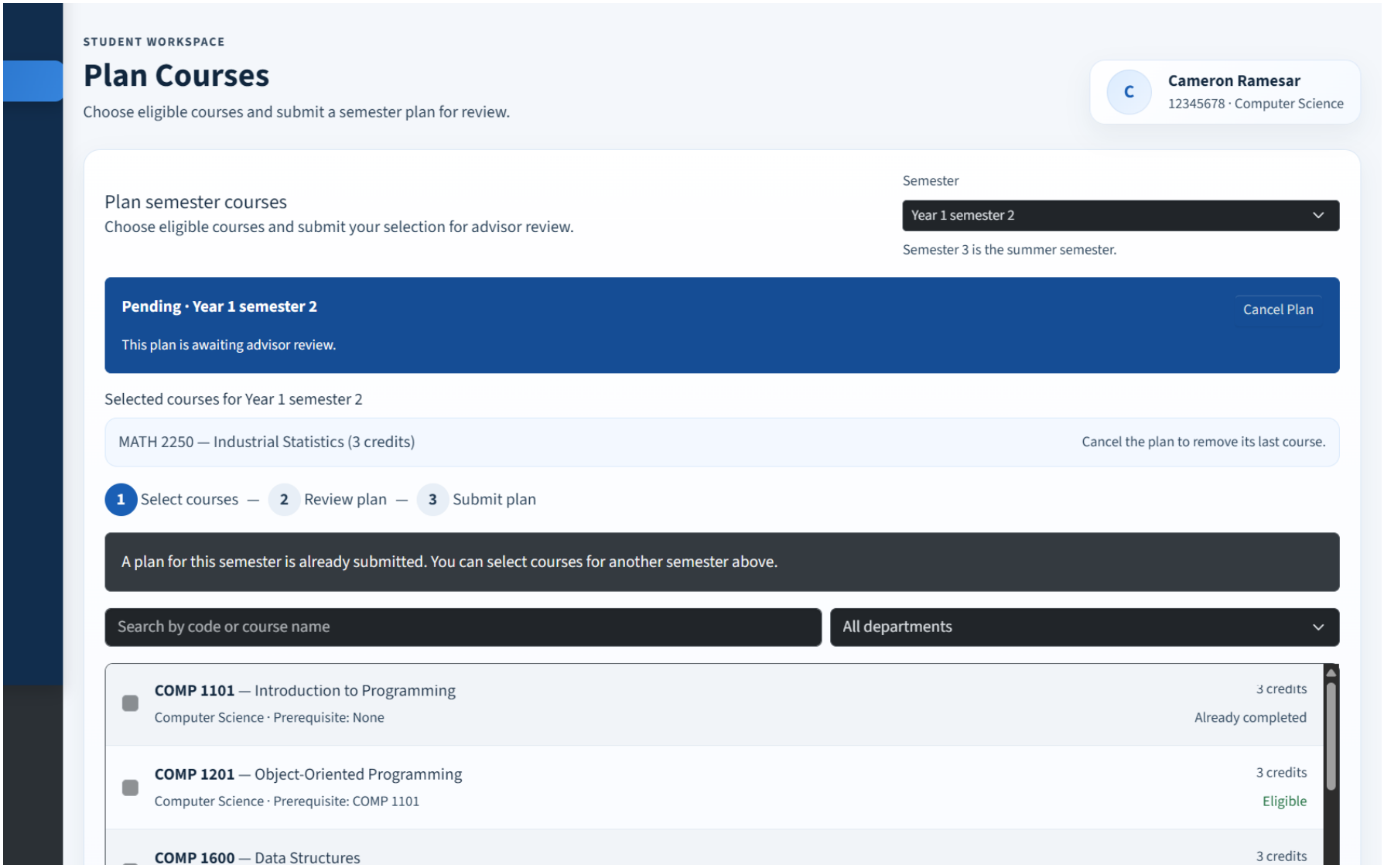
Pixel excerpt only; the preceding full original is preserved. Approved plan A and revised-selection feedback (UAT-09 partial). Original low-contrast explanatory text retained.

Figure 9.5.10. UAT-10-second-plan



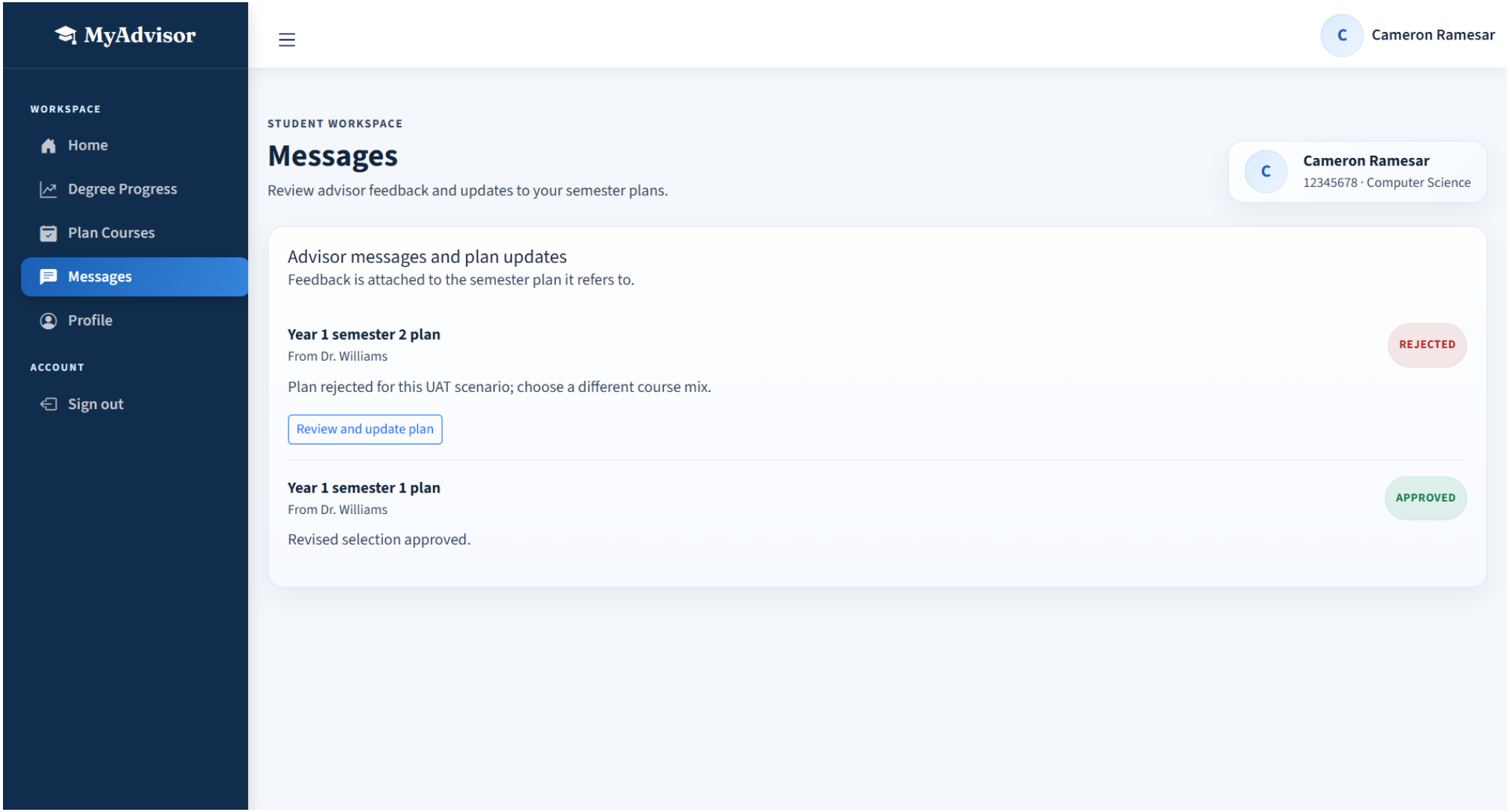
Semester B Pending with MATH 2250, 3 credits (UAT-10 partial). Source: UAT-10-second-plan.png. Captured 27 October 2026; demo identity shown, not submission ID.

Figure 9.5.10a. Enlarged original workflow detail



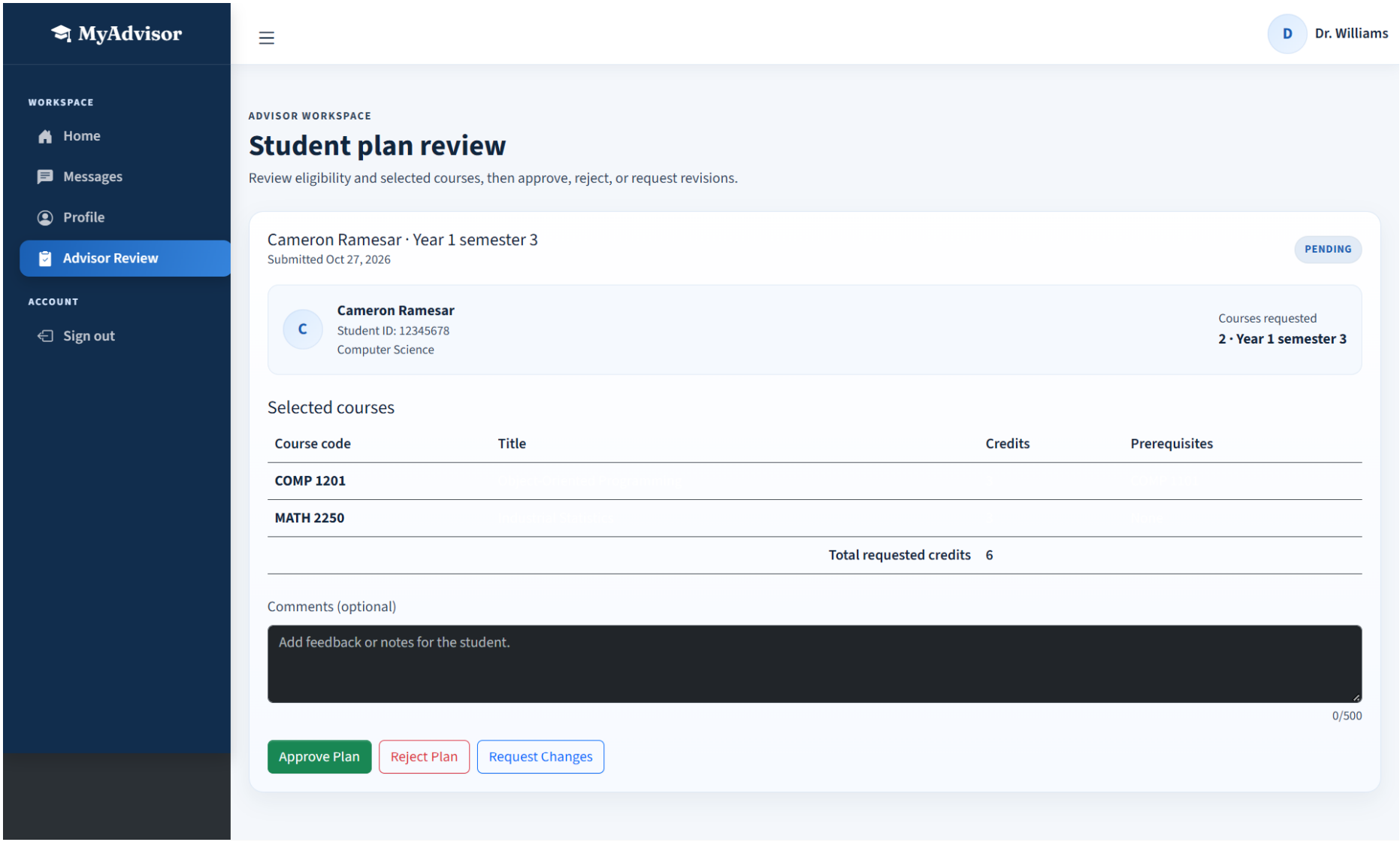
Pixel excerpt only; the preceding full original is preserved. Semester B Pending with MATH 2250, 3 credits (UAT-10 partial).

Figure 9.5.11. UAT-11-rejected



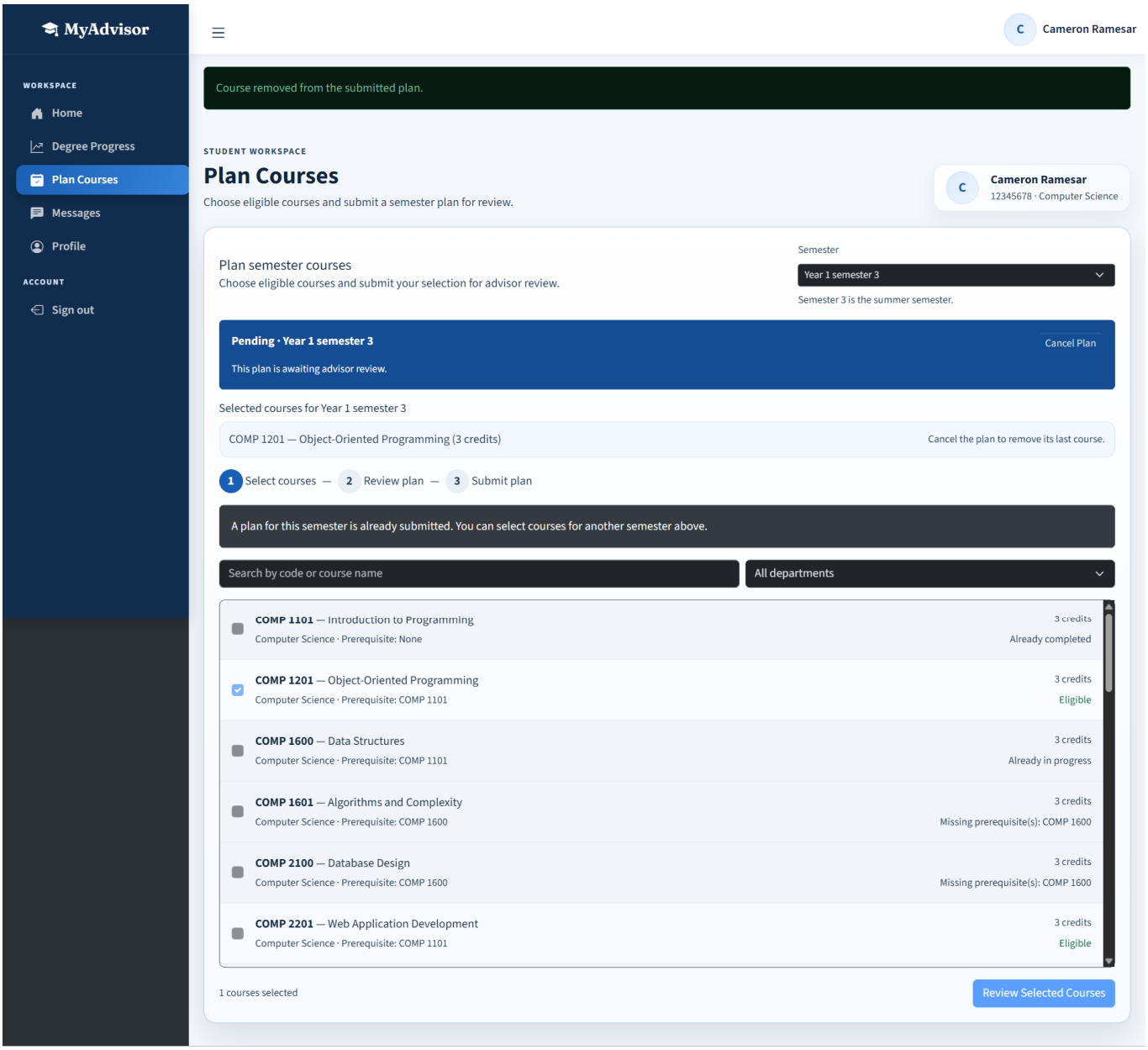
Messages show semester B rejection reason and retained A approval (UAT-11 partial). Source: UAT-11-rejected.png. Captured 27 October 2026; demo identity shown, not submission ID.

Figure 9.5.12. UAT-12-pending-plan



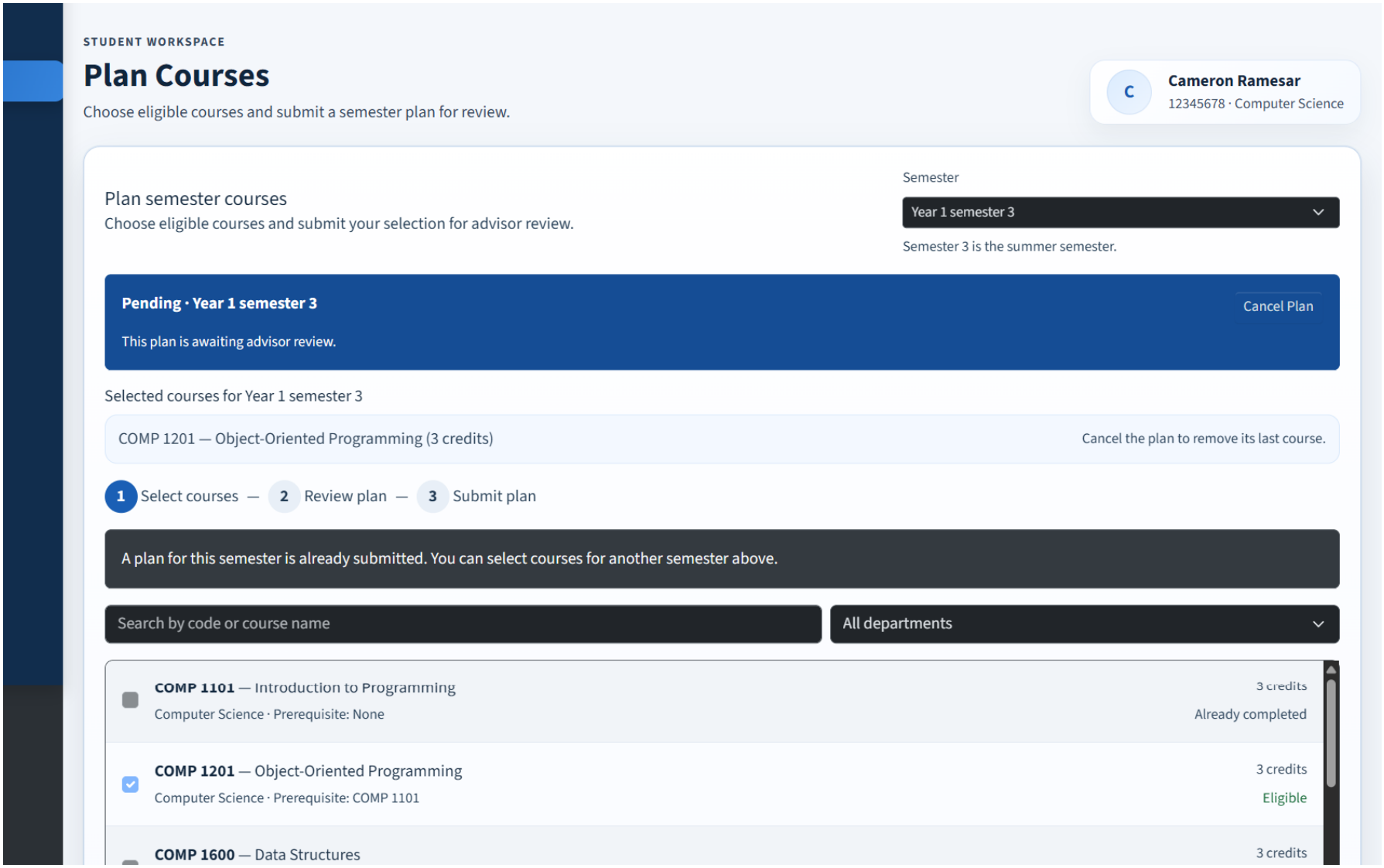
Advisor sees plan C with COMP 1201/MATH 2250, 6 credits (UAT-12 partial). Source: UAT-12-pending-plan.png. Captured 27 October 2026; demo identity shown, not submission ID.

Figure 9.5.13. UAT-13-course-removed



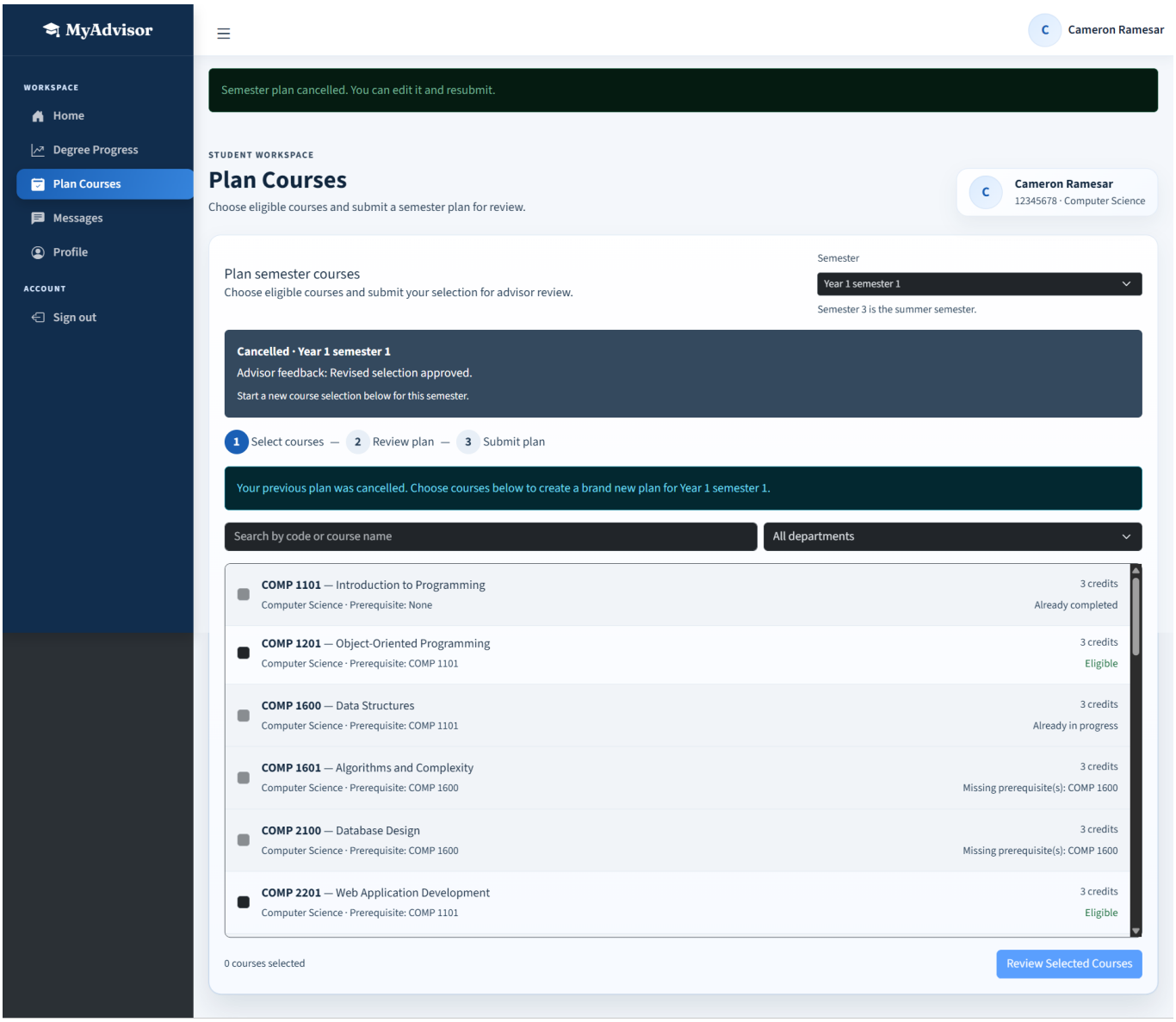
Pending C reduced to COMP 1201, 3 credits; final-course UI guard visible (UAT-13 partial). Source: UAT-13-course-removed.png. Captured 27 October 2026; demo identity shown, not submission ID.

Figure 9.5.13a. Enlarged original workflow detail



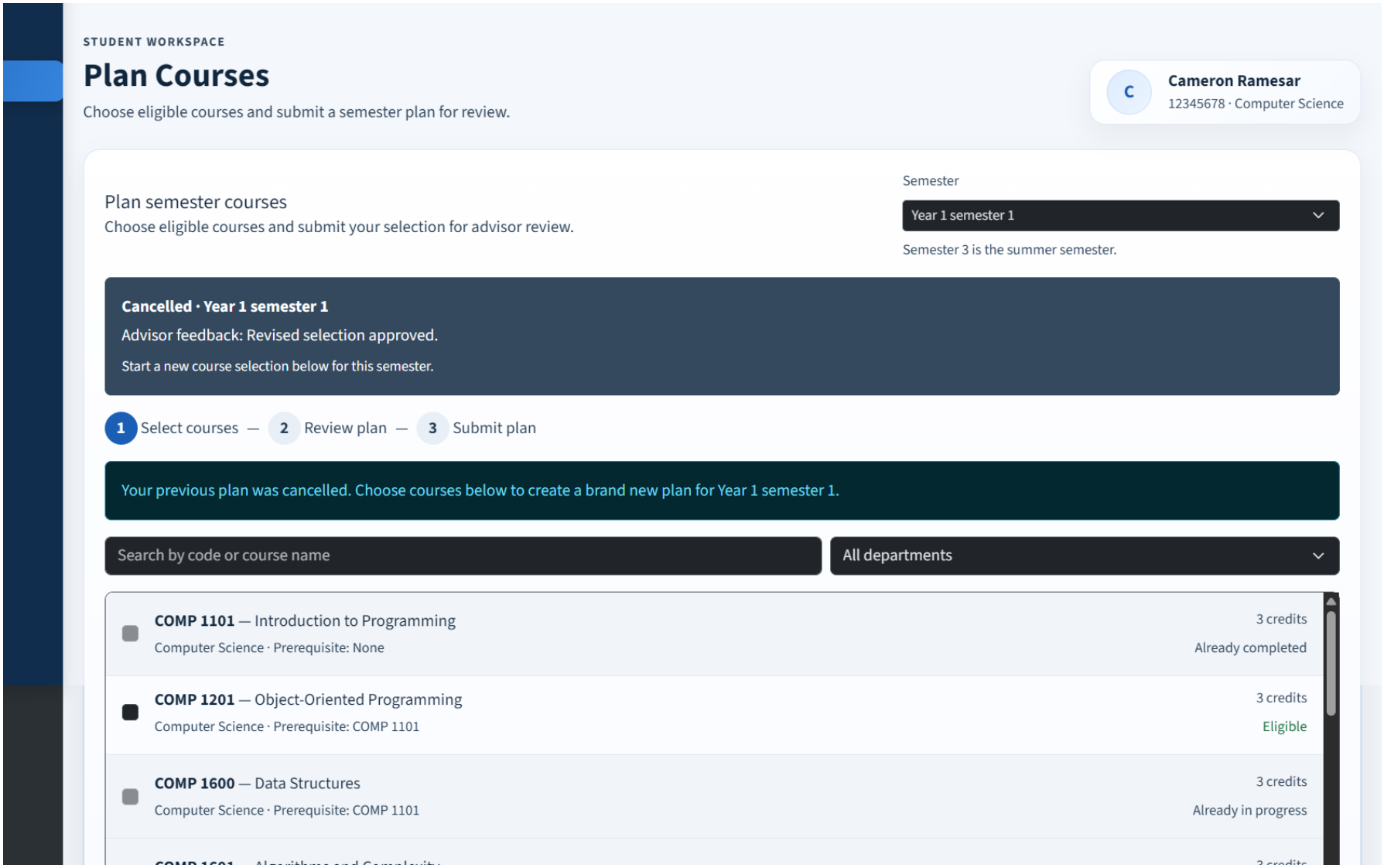
Pixel excerpt only; the preceding full original is preserved. Pending C reduced to COMP 1201, 3 credits; final-course UI guard visible (UAT-13 partial).

Figure 9.5.14. UAT-14-approved-cancelled



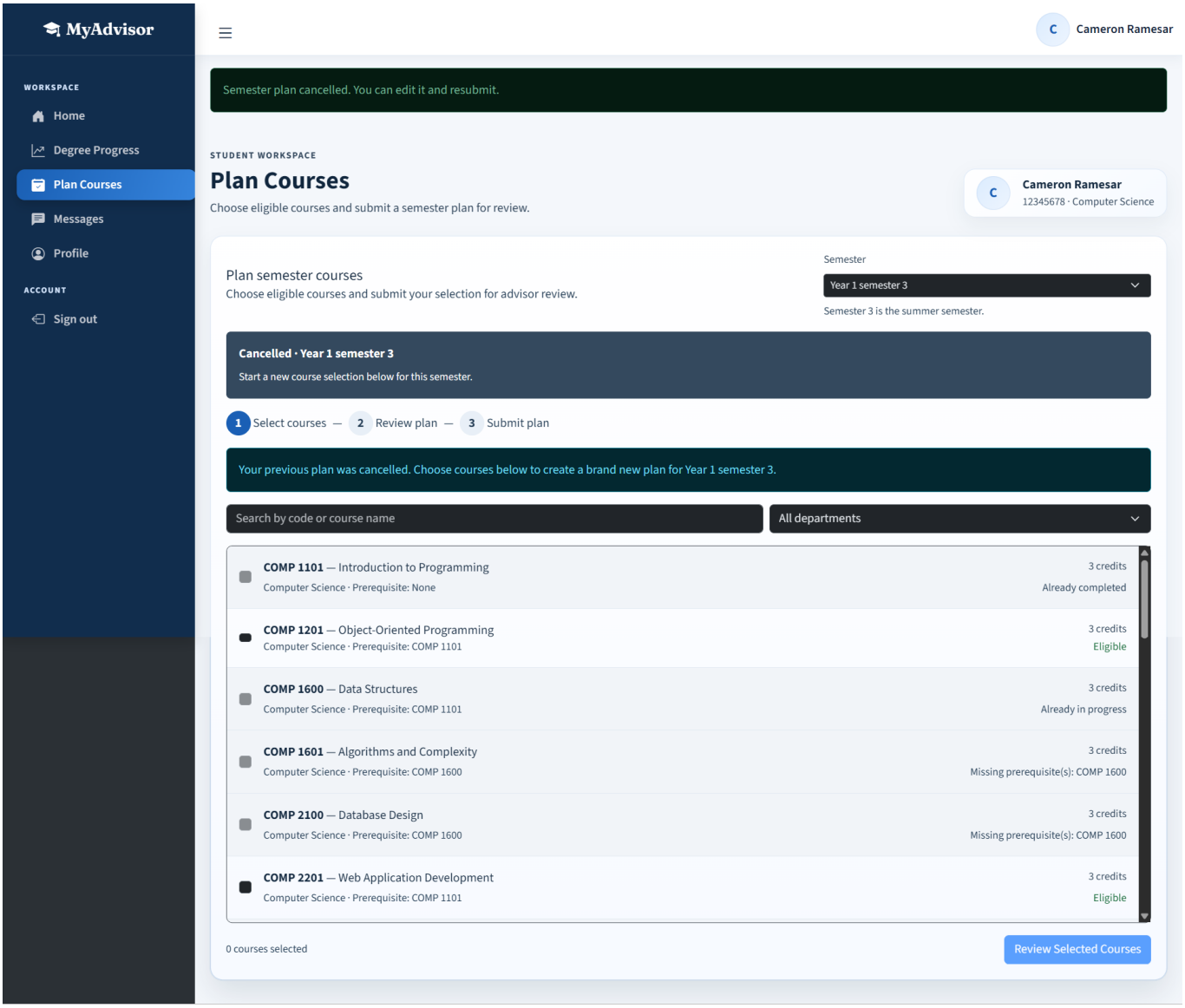
Approved A cancelled, feedback retained, selection cleared (UAT-14 partial). Source: UAT-14-approved-cancelled.png. Captured 27 October 2026; demo identity shown, not submission ID.

Figure 9.5.14a. Enlarged original workflow detail



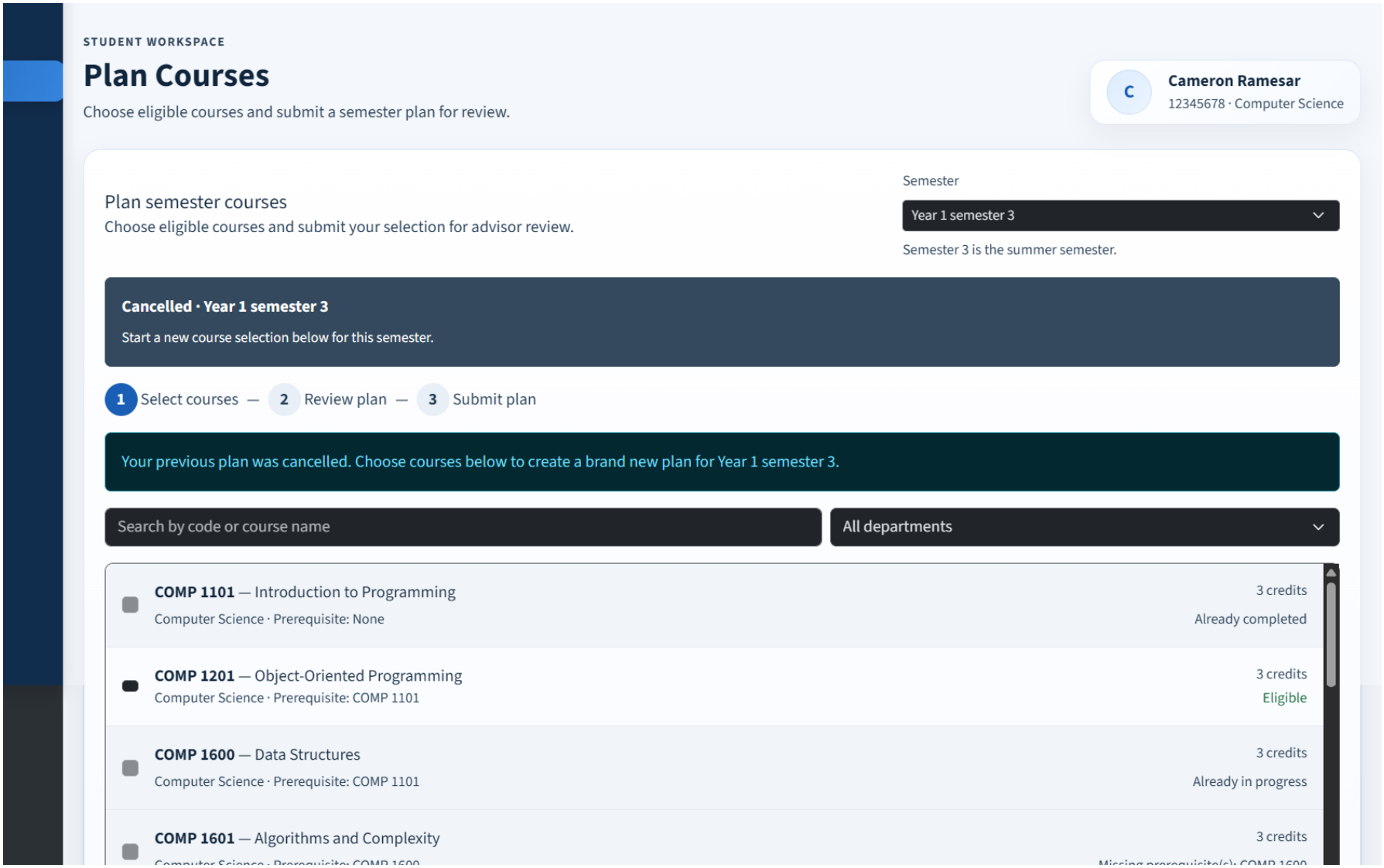
Pixel excerpt only; the preceding full original is preserved. Approved A cancelled, feedback retained, selection cleared (UAT-14 partial).

Figure 9.5.15. UAT-15-rejected-cancelled



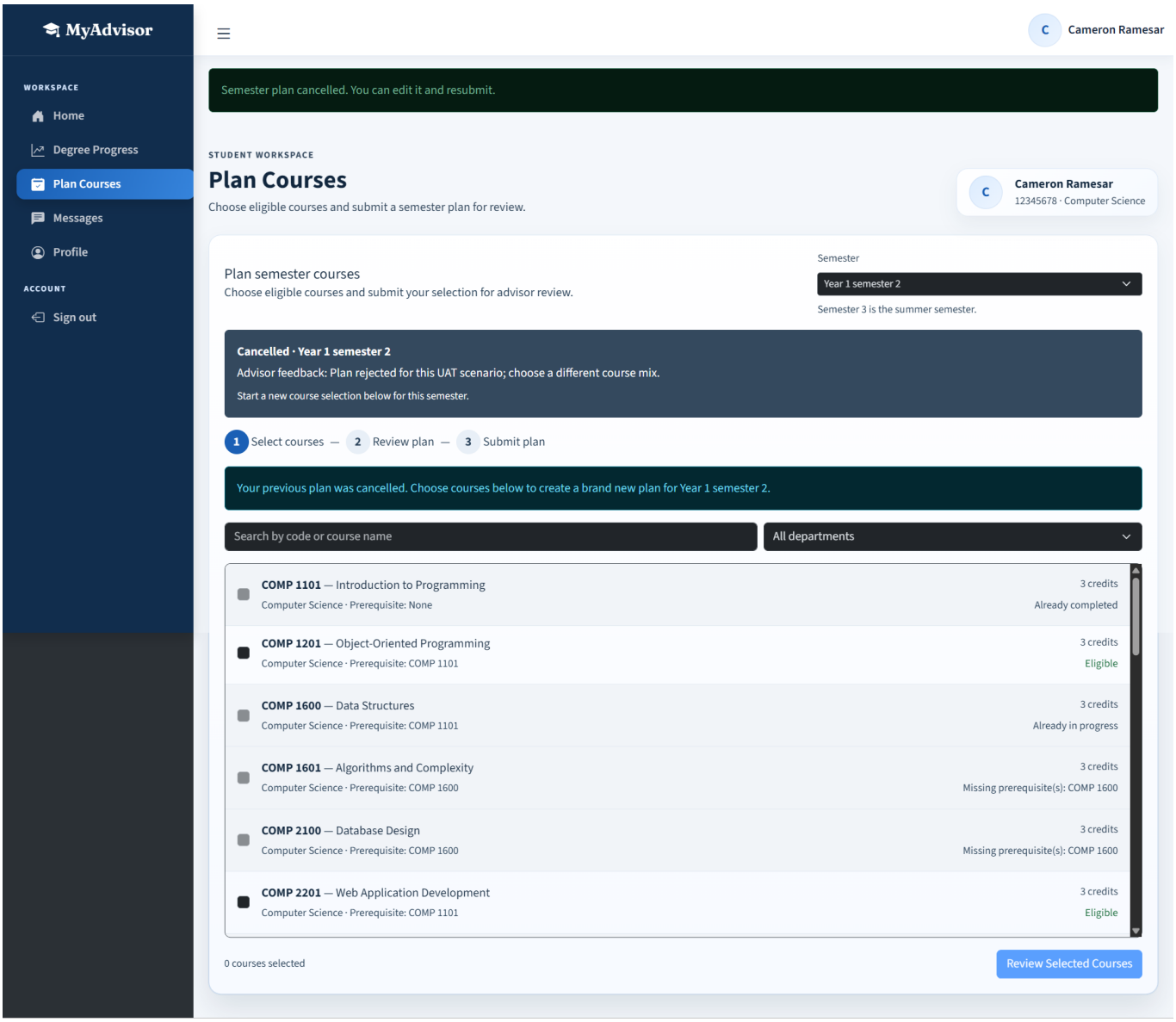
Semester C Cancelled and selection cleared (case UAT-16 partial); original filename is misleading. Source: UAT-15-rejected-cancelled.png. Captured 27 October 2026; demo identity shown, not submission ID.

Figure 9.5.15a. Enlarged original workflow detail



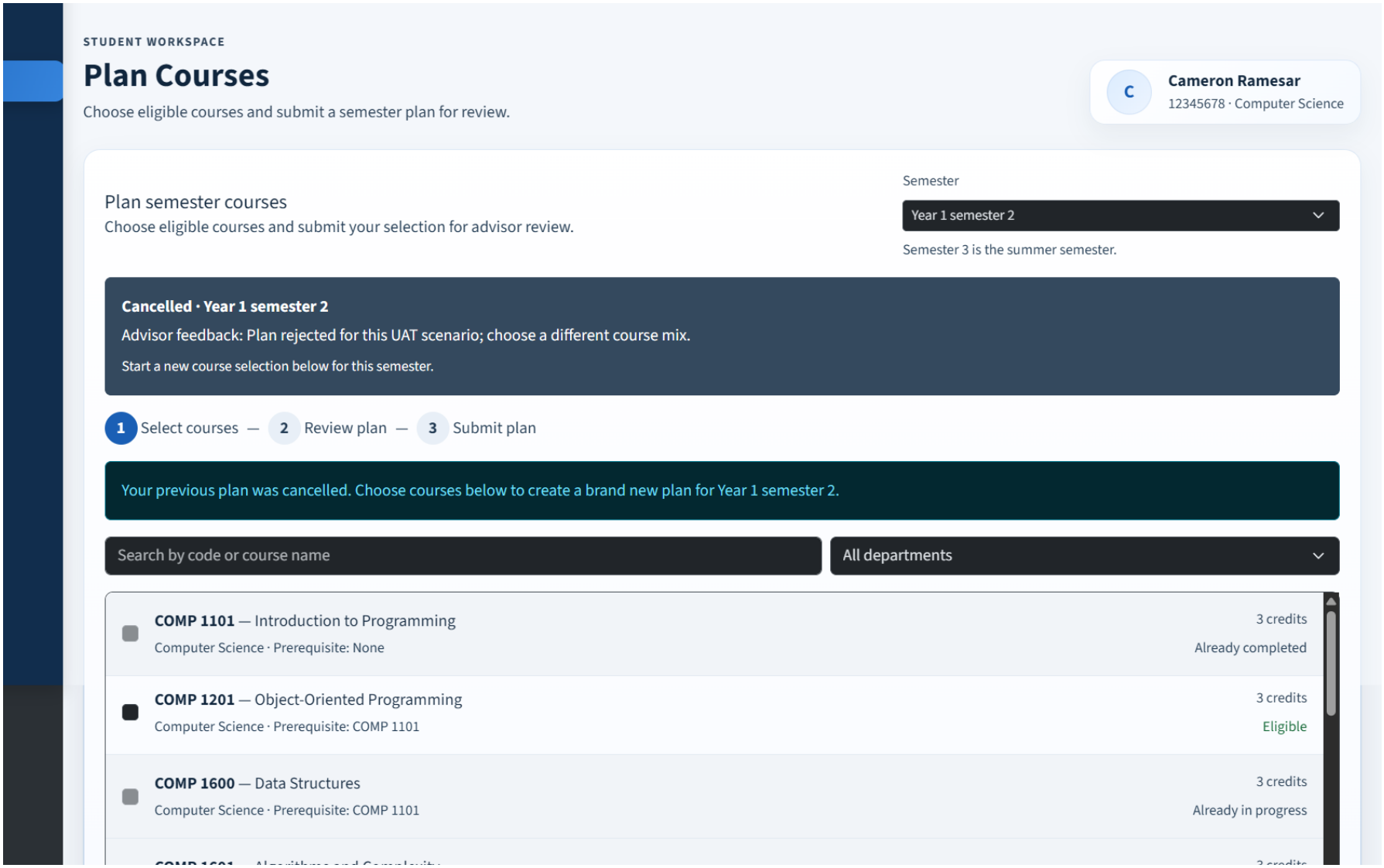
Pixel excerpt only; the preceding full original is preserved. Semester C Cancelled and selection cleared (case UAT-16 partial); original filename is misleading.

Figure 9.5.16. UAT-16-pending-cancelled



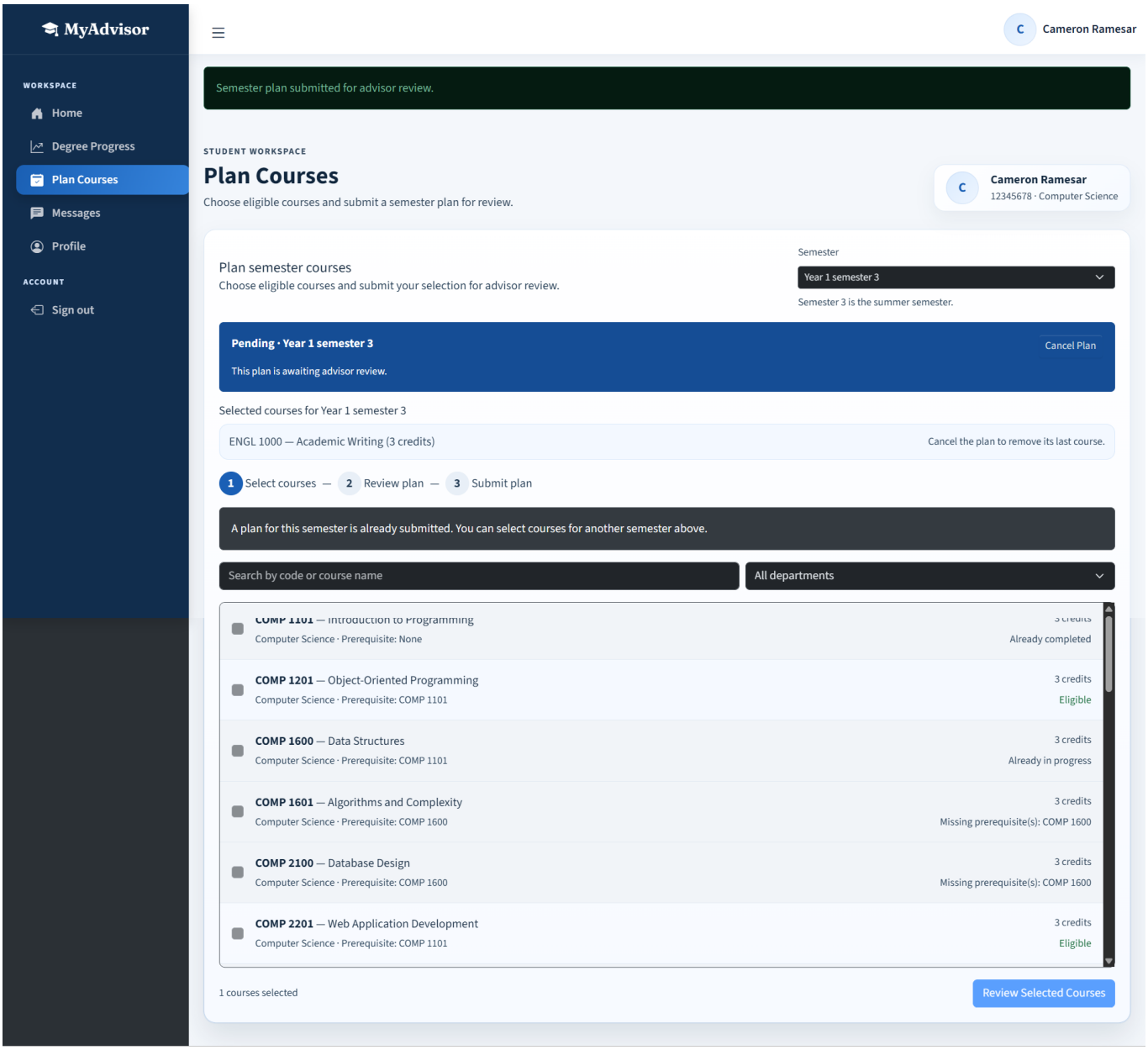
Semester B Cancelled, rejection feedback retained, selection cleared (case UAT-15 partial); original filename is misleading. Source: UAT-16-pending-cancelled.png. Captured 27 October 2026; demo identity shown, not submission ID.

Figure 9.5.16a. Enlarged original workflow detail



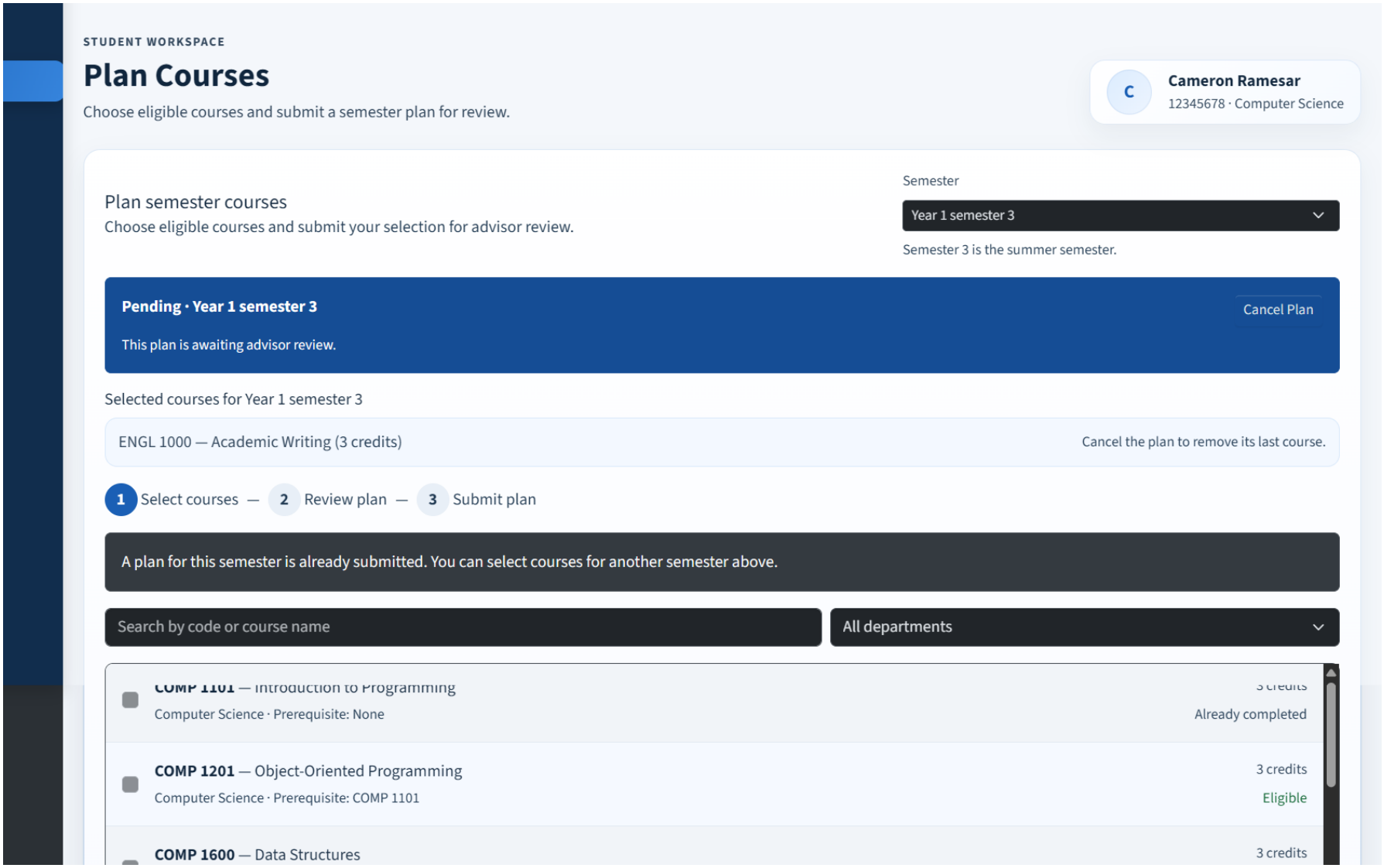
Pixel excerpt only; the preceding full original is preserved. Semester B Cancelled, rejection feedback retained, selection cleared (case UAT-15 partial); original filename is misleading.

Figure 9.5.17. UAT-17-new-plan



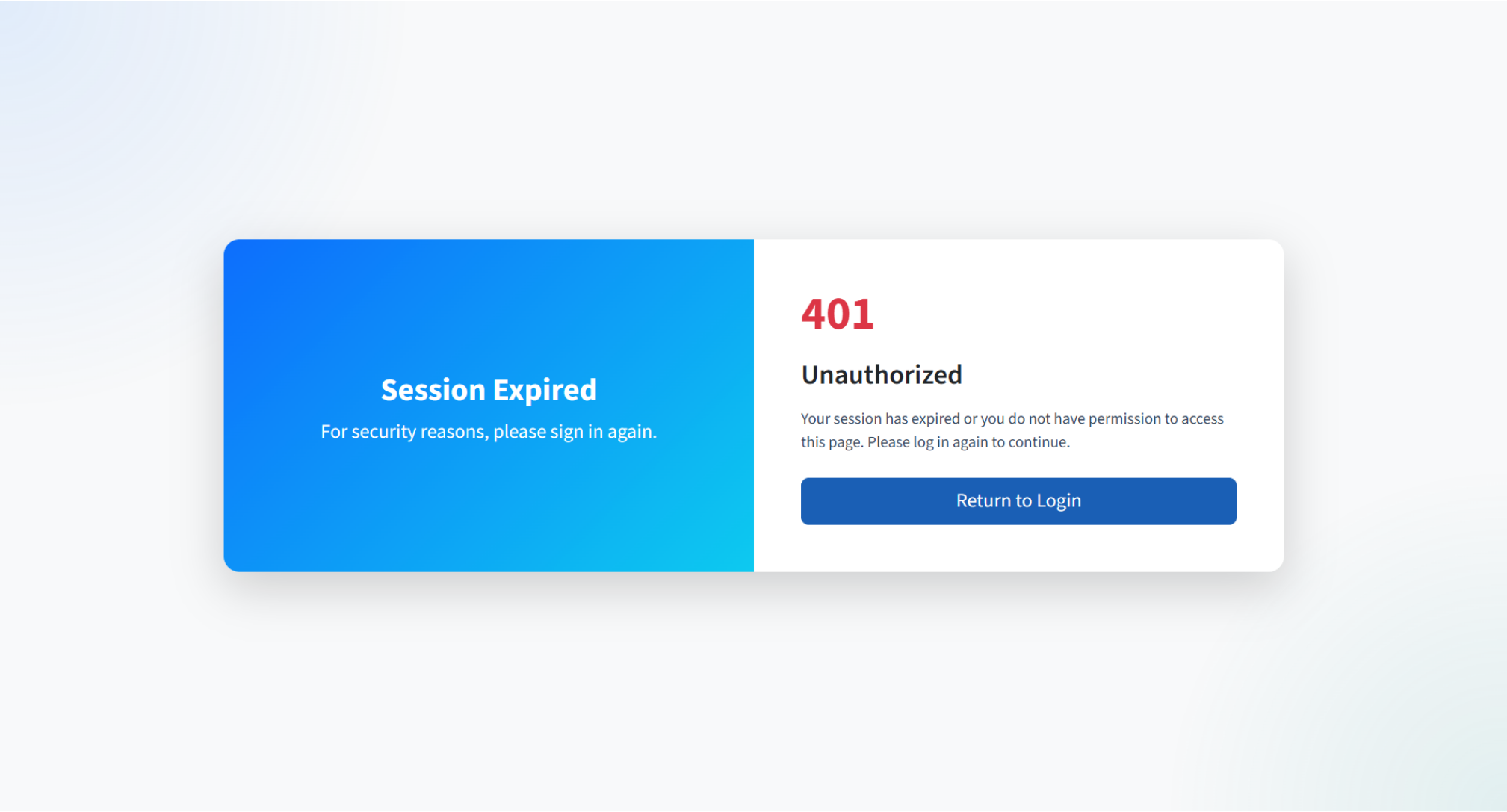
New C Pending after cancellation, ENGL 1000 only, 3 credits (UAT-17 partial). Source: UAT-17-new-plan.png. Captured 27 October 2026; demo identity shown, not submission ID.

Figure 9.5.17a. Enlarged original workflow detail



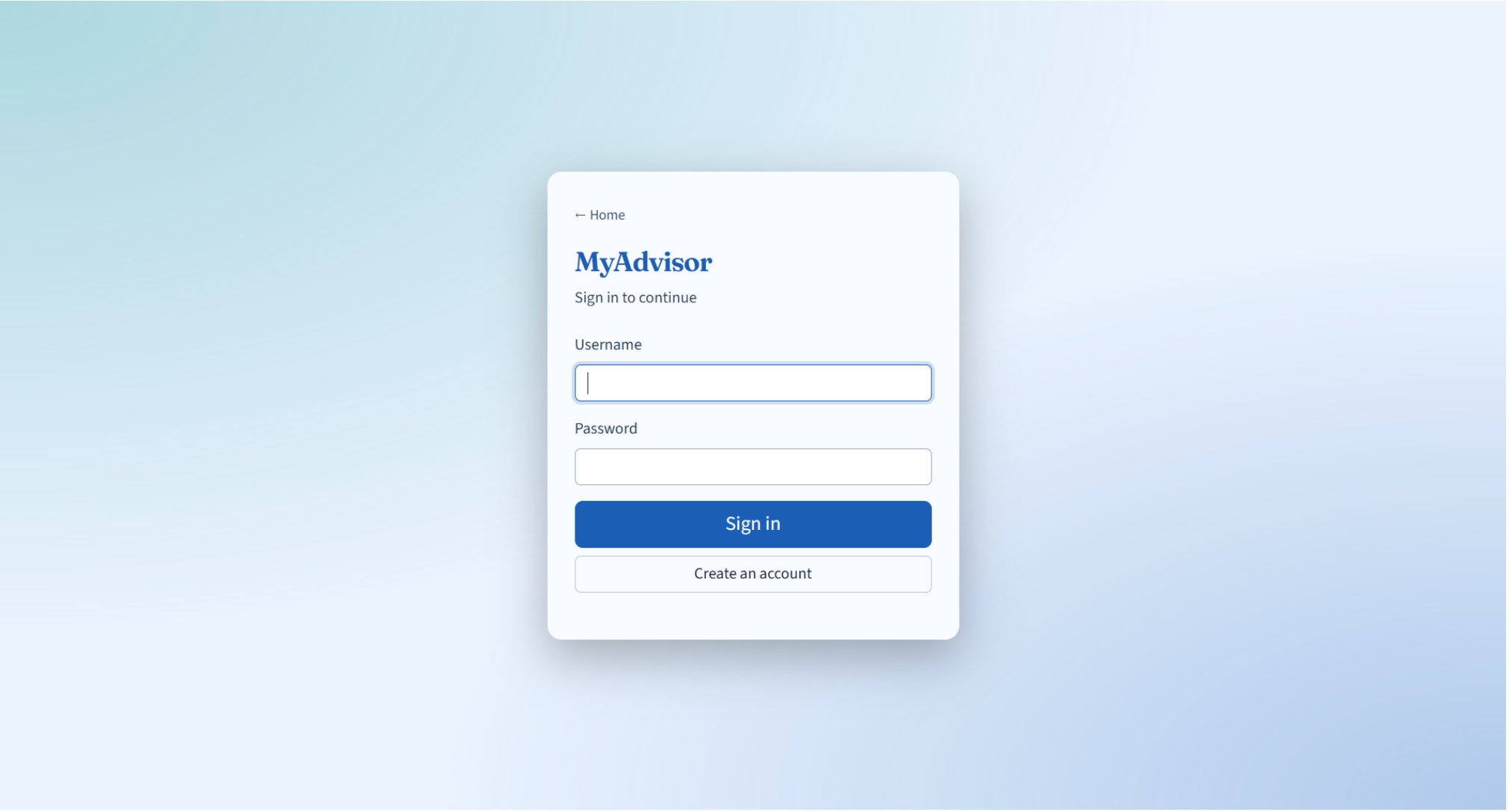
Pixel excerpt only; the preceding full original is preserved. New C Pending after cancellation, ENGL 1000 only, 3 credits (UAT-17 partial).

Figure 9.5.18. UAT-20-access-denied



Session Expired / 401 UI; does not establish logged-in Student 403 (UAT-20 partial). Source: UAT-20-access-denied.png. Captured 27 October 2026; demo identity shown, not submission ID.

Figure 9.5.19. UAT-20-logged-out



Empty login page after signed-out flow; actor-specific URL checks remain unverified (UAT-20 partial). Source: UAT-20-logged-out.png. Captured 27 October 2026; demo identity shown, not submission ID.

10. Requirements traceability

Updated **27 October 2026** from actual test source/markers, the 90-request collection, plans and inspected evidence. UT-01–23 are planned groups implemented by test_UT_01_... through test_UT_23_...; IT-01–20 use pytest.mark.it, including parameter marks. Request 01–90 numbers are actual collection identifiers. UAT-01–20 are the unique execution-sheet IDs. R01–R15 remain local requirement labels in the original plans, not official rubric identifiers.

WF1 — Track Degree Progress; R01 (totals/categories/boundaries), R02 (eligibility context), R11–12 (access), R15 (public response contracts)

Field	Recorded detail
Unit cases and actual source	UT-01–06 calculations/prerequisites in test_advising.py; UT-20–23 supporting auth in test_auth.py
Integration cases and actual source	IT-01/02/03/12 in Student workflow, notably test_degree_progress_through_authenticated_html_and_json, test_available_courses_use_completed_prerequisites_and_tracked_status
Postman request IDs	02, 07–10, 13–20, 25, 87–90 (shared auth boundaries included)
Manual UAT cases	UAT-01; supporting UAT-18/20
Existing execution evidence	Unit start, unit log; Student integration log; Postman 01–20; Degree Progress and Student Home, partial UAT only

WF2 — Plan Semester Course Selections; R02–05, R08–10, R11–14

Field	Recorded detail
Unit cases and actual source	UT-04–13 eligibility/selection/status, UT-18/19 cancellation/removal in test_advising.py; UT-20–23 shared auth
Integration cases and actual source	IT-03/04/06–12/14/15/18 in Student workflow, including test_valid_submission_persists_courses_and_can_be_retrieved, test_resubmission_reuses_plan_and_replaces_old_selection, test_allowed_cancellation_retains_history_and_replacement_gets_new_id; IT-19/20 DB constraints/rollback
Postman request IDs	03–04, 09, 11–12, 16–17, 20–50, 63–66, 68, 70, 72, 74–90 (shared/cross-role checkpoints included)
Manual UAT cases	UAT-02–05, UAT-08, UAT-10, UAT-12–17; supporting UAT-18/20
Existing execution evidence	Unit completed, Integration completed; Postman 21–40, 41–60, 61–80, 72–90; selection, Pending submission, partial UAT only

WF3 — Review and Approve Course Plans; R05–08, R11–14

Field	Recorded detail
Unit cases and actual source	UT-14–17 decision mapping/feedback/Pending guard in test_advising.py; UT-20–23 shared auth
Integration cases and actual source	IT-03–08/11–13/16/17 in Advisor workflow, including test_advisor_decision_persists_and_student_receives_only_own_feedback, test_student_resubmits_reviewed_plan_and_advisor_reviews_replacement, test_invalid_review_json_is_rejected_before_writes; IT-19/20 shared persistence boundaries
Postman request IDs	05–06, 10, 13–18, 42–47, 51–86, 88 (auth/Student delivery and state checkpoints included)
Manual UAT cases	UAT-06–09, UAT-11; Advisor queue checks in UAT-12/13/16/17; supporting UAT-19/20
Existing execution evidence	Advisor lifecycle, Advisor validation; Unit late tests; Postman review validation, decisions; Advisor Pending review, partial UAT only

Result interpretation and audit boundaries

The marked runs report **113 unit passes (602 deselected)** and **194 integration passes (521 deselected)**, each with one warning. The full suite is **715 passed, zero failures/errors, one warning**; selections overlap that full result. Unit and integration reports map every planned group to concrete function names and distinguish historical complete logs from latest aggregate/partial screenshots. No percentage of line/branch coverage was measured, so none is claimed. IT-19/20 source establishes constraint/rollback scope; their individual latest output is not visible in the screenshots.

Postman: **90 requests, 470 successful assertions, zero assertion failures/errors, one iteration**, 27 October 2026. Screenshots span 01–90; some edge rows are clipped. UAT: **1 Pass / 0 Fail / 19 Not Run**; 18 cases partially verified and UAT-05 required workflow unverified. All 19 screenshots map to actual manual outcomes in the matrix. WF2 now has screenshot-supported revision, removal, all three cancellations and replacement; WF3 has Changes Requested, Approved and Rejected outcomes/feedback. UAT-20 proves signed-out UI restriction only, not logged-in role checks. Cameron Ramesar confirms execution on 27 October 2026. DATA-01 and UI-01 remain open.

See the evidence inventory for every verified screenshot, readability limits, hashes and missing exports. The R-level matrices in unit plan and integration plan supplement this workflow matrix. Requirements without browser equivalents (malformed payload/schema/direct database rollback) remain automated/API scope; a mapping does not create a manual execution result.

Additional manual lifecycle traceability

Requirement scope	Actual manual evidence	Limitation
WF2 revision/resubmission	UAT-07/08: MATH 2250 replaced by ENGL 1000, 9 credits	Stored feedback clearance unconfirmed
WF3 feedback/decisions	UAT-07/09/11: Changes Requested, Approved, Rejected	Queue removal and Advisor confirmations unconfirmed
WF2 removal/cancellation/replacement	UAT-12?17: 6?3 credits, A/B/C Cancelled, C new ENGL 1000 Pending	Dismissal, persistence, Messages and queue checks as listed in matrix
Supporting access	UAT-20 401/login captures	Student 403 and Advisor fallback unverified

Tester clarification on 27 October 2026 leaves current totals unchanged: **1 Pass / 0 Fail / 19 Not Run**. Existing state/feedback evidence is retained; refresh persistence and remaining role checks are not inferred. See case-specific supplemental evidence requirements.

10.1 Requirement-level matrix R01–R14

These R identifiers are local planning identifiers derived from the requested scope, existing report and current source. They do not claim to be official assignment/audit requirement IDs. R13/R14 capture persistence and validation obligations needed to verify the workflows.

R01

Field	Recorded detail
Rule / objective	Degree totals, category breakdown and calculation boundaries
Unit cases	UT-01–UT-03
Integration cases	IT-02
Source basis / existing coverage	E01; dashboard service; report Track Degree Progress
UAT mapping / recorded outcome	UAT-01; Not Run
Postman request mapping / runtime outcome	19; executed; successful assertions in verified 27 October Runner run
Reporting evidence	Unit, Integration, Postman, UAT

R02

Field	Recorded detail
Rule / objective	Completed-only prerequisite eligibility and blocked tracked courses
Unit cases	UT-04–UT-06, UT-11
Integration cases	IT-03, IT-14
Source basis / existing coverage	E05,E08; prerequisite_ok/get_course_availability
UAT mapping / recorded outcome	UAT-02-03; Not Run
Postman request mapping / runtime outcome	20, 25; executed; successful assertions in verified 27 October Runner run
Reporting evidence	Unit, Integration, Postman, UAT

R03

Field	Recorded detail
Rule / objective	Nine valid semester names, trimming and invalid names
Unit cases	UT-07, UT-08
Integration cases	IT-14
Source basis / existing coverage	E06,E07; SEMESTER_OPTIONS/submit_plan
UAT mapping / recorded outcome	UAT-02, UAT-04, UAT-10, UAT-12; Not Run
Postman request mapping / runtime outcome	21, 26, 31; executed; successful assertions in verified 27 October Runner run
Reporting evidence	Unit, Integration, Postman, UAT

R04

Field	Recorded detail
Rule / objective	Nonempty, unique, existing course selections
Unit cases	UT-09–UT-11
Integration cases	IT-03, IT-14, IT-19
Source basis / existing coverage	submit_plan; PlanCourse uniqueness
UAT mapping / recorded outcome	UAT-03-04; Not Run
Postman request mapping / runtime outcome	22-25, 27, 29-31; executed; successful assertions in verified 27 October Runner run
Reporting evidence	Unit, Integration, Postman, UAT

R05

Field	Recorded detail
Rule / objective	Plan status and duplicate active-plan rules
Unit cases	UT-12, UT-13, UT-17
Integration cases	IT-05–IT-07, IT-15, IT-17
Source basis / existing coverage	RESUBMITTABLE_PLAN_STATUSES; submit_plan/review_plan
UAT mapping / recorded outcome	UAT-04-12; UAT-06 Pass, others Not Run
Postman request mapping / runtime outcome	39-40, 61-80; executed; successful assertions in verified 27 October Runner run
Reporting evidence	Unit, Integration, Postman, UAT

R06

Field	Recorded detail
Rule / objective	Advisor decision mapping and Pending-only review
Unit cases	UT-14, UT-15, UT-17
Integration cases	IT-05–IT-07, IT-13, IT-17
Source basis / existing coverage	review_plan; report advisor workflow
UAT mapping / recorded outcome	UAT-06-09, UAT-11, UAT-20; UAT-06 Pass, others Not Run
Postman request mapping / runtime outcome	51-78; executed; successful assertions in verified 27 October Runner run
Reporting evidence	Unit, Integration, Postman, UAT

R07

Field	Recorded detail
Rule / objective	Feedback type, raw length limit and trimming
Unit cases	UT-14, UT-16
Integration cases	IT-05, IT-16, IT-17
Source basis / existing coverage	review_plan and admin review route
UAT mapping / recorded outcome	UAT-07, UAT-09, UAT-11 (visible feedback only); Not Run
Postman request mapping / runtime outcome	54-61, 69, 74; executed; successful assertions in verified 27 October Runner run
Reporting evidence	Unit, Integration, Postman, UAT

R08

Field	Recorded detail
Rule / objective	Rejected/change-request resubmission resets existing plan
Unit cases	UT-12, UT-13
Integration cases	IT-06, IT-07, IT-20
Source basis / existing coverage	E02; resubmit_plan
UAT mapping / recorded outcome	UAT-07-08; Not Run
Postman request mapping / runtime outcome	61-73; executed; successful assertions in verified 27 October Runner run
Reporting evidence	Unit, Integration, Postman, UAT

R09

Field	Recorded detail
Rule / objective	Owned Pending/Approved/Rejected cancellation and replacement
Unit cases	UT-13, UT-18
Integration cases	IT-08, IT-09, IT-11
Source basis / existing coverage	E09,E10; cancel_plan; report additional use cases
UAT mapping / recorded outcome	UAT-14-17; Not Run
Postman request mapping / runtime outcome	78-86; executed; successful assertions in verified 27 October Runner run
Reporting evidence	Unit, Integration, Postman, UAT

R10

Field	Recorded detail
Rule / objective	Pending course removal, membership and last-course guard
Unit cases	UT-19
Integration cases	IT-10, IT-18
Source basis / existing coverage	E09; remove_submitted_course
UAT mapping / recorded outcome	UAT-13; Not Run
Postman request mapping / runtime outcome	48-50; executed; successful assertions in verified 27 October Runner run
Reporting evidence	Unit, Integration, Postman, UAT

R11

Field	Recorded detail
Rule / objective	Authentication, advisor role and student ownership isolation
Unit cases	UT-18, UT-19, UT-22
Integration cases	IT-02, IT-11–IT-13
Source basis / existing coverage	AuthDep/AdminDep; student identity from authenticated user
UAT mapping / recorded outcome	UAT-18-20 (browser scope only); Not Run
Postman request mapping / runtime outcome	02-18, 28, 33-47, 58-59; executed; successful assertions in verified 27 October Runner run
Reporting evidence	Unit, Integration, Postman, UAT

R12

Field	Recorded detail
Rule / objective	Credential checking, token/cookie lifecycle and logout
Unit cases	UT-20–UT-23
Integration cases	IT-01, IT-12
Source basis / existing coverage	AuthService; security utilities; auth/login/logout routes
UAT mapping / recorded outcome	UAT-18-20; Not Run
Postman request mapping / runtime outcome	07-12, 87-90; executed; successful assertions in verified 27 October Runner run
Reporting evidence	Unit, Integration, Postman, UAT

R13

Field	Recorded detail
Rule / objective	Committed plans/links, constraints and atomic persistence
Unit cases	UT-11, UT-12 (delegation only)
Integration cases	IT-03–IT-08, IT-10, IT-11, IT-15, IT-19, IT-20
Source basis / existing coverage	AdvisingRepository; SQLModel unique/FK constraints
UAT mapping / recorded outcome	UAT-04-17 (browser persistence subset); Not Run
Postman request mapping / runtime outcome	30-32, 47-88 (API-visible only; no direct DB constraint/rollback); executed; successful assertions in verified 27 October Runner run
Reporting evidence	Unit, Integration, Postman, UAT

R14

Field	Recorded detail
Rule / objective	HTTP parsing, business validation and unchanged state on errors
Unit cases	UT-08–UT-11, UT-15–UT-17 (business guards only)
Integration cases	IT-14, IT-16–IT-18
Source basis / existing coverage	Form parameters; admin JSON/form parsing; ValueError redirects
UAT mapping / recorded outcome	No direct malformed-payload browser case; Not Run
Postman request mapping / runtime outcome	22-30, 41, 54-60; executed; successful assertions in verified 27 October Runner run
Reporting evidence	Unit, Integration, Postman, UAT

Ranges are inclusive. The companion plan uses the same identifiers. A mapped case is planned coverage, not evidence of a pass.

10.2 R15 public JSON contracts

R15 is the later public-schema/allowlist/OpenAPI requirement. API-07/08 mappings are retained in evidence/workflow1-api.md; Postman response-contract checks include requests 18–20, 32 and 52. It has no direct manual browser schema case and does not create additional UT/IT/UAT results.

11. Defects, limitations and audit disposition

No Critical finding was verified. F01–F05 remain High and open. The following register reproduces the audit findings and proposed fixes; proposed fixes are not implemented or retested in this PDF task. Functional defects are independent of the 715 passing automated tests.

Severity describes impact/marking risk; it is not a grade. **No Critical finding is verified.** Open functional fixes below require a separate authorized correction step; no application fix or retest is claimed here.

F01

Field	Recorded detail
Severity	**High**
Rubric area	Implementation — Authorization; Test Plans/Implementation
File / evidence	app/routers/user_home.py; reproduction
Description and why it matters	Advisor JSON submit returns 403, but HTML POST /app/plan returns 303 and stores an Advisor-owned Pending plan; the same Advisor's approve API returns 200 and stores Approved. UI hiding does not enforce role separation.
Proposed smallest fix	Apply existing StudentDep to HTML submit/remove/cancel actions; add real-auth wrong-role/unknown-role no-write tests. A defensive prohibition on reviewing own plans may also be considered, but it is not a substitute for the Student entry-point guard.
Status	**Open — reproduced; application unchanged**

F02

Field	Recorded detail
Severity	**High**
Rubric area	Test Implementation; final API deliverable
File / evidence	Published postman directory; snapshot
Description and why it matters	Required MyAdvisor-Assignment2.postman_collection.json is absent remotely. The local static validator also needs that missing file. A marker cloning main cannot import the promised export.
Proposed smallest fix	Include the existing sanitized local v2.1 JSON in the reviewed submission commit; verify it is retrievable and imports. Do not substitute YAML.
Status	**Open — publication requires user instruction**

F03

Field	Recorded detail
Severity	**High**
Rubric area	Test Reporting; repository readiness
File / evidence	Remote snapshot; local comparison
Description and why it matters	Published main has six pytest images but no UAT PNGs, no Postman Runner PNGs, no current evidence matrix/inventory/traceability/supplemental sheet, and older versions of the five reports/progress/plans.
Proposed smallest fix	Publish the reviewed current docs/evidence together after corrections; verify every remote link.
Status	**Open — local work is not published**

F04

Field	Recorded detail
Severity	**High**
Rubric area	Test Reporting — Completed UAT report
File / evidence	reports/uat-execution.md, uat-final-report.md, supplemental checklist
Description and why it matters	Only one complete-case Pass; 18 partial cases and one unverified workflow. Refresh, navigation, login/logout and role substeps cannot be inferred from state screenshots.
Proposed smallest fix	Personally execute missing case-specific checks with actual dates/data/browser/build, record expected vs actual, preserve old evidence, and resolve or accept defects explicitly.
Status	**Open — incomplete acceptance**

F05 / DATA-01

Field	Recorded detail
Severity	**High**
Rubric area	Implementation WF1; Test Plans/Reporting
File / evidence	Service/model seed configuration; UAT-02; original wireframe
Description and why it matters	Verified 93 overall versus 120 category requirements; single-category course allocation supplies no overlap explanation. Correct curriculum values remain unspecified, affecting trustworthy progress guidance.
Proposed smallest fix	Obtain an authoritative degree/project requirement decision; then make only justified configuration/calculation and test/documentation changes.
Status	**Open — unresolved requirements issue, values unchanged**

F06 / UI-01

Field	Recorded detail
Severity	Medium
Rubric area	Implementation views; UAT
File / evidence	advisor_dashboard.html, app.css, UAT-05/08/12; UAT-09
Description and why it matters	Advisor table body details are almost white on near-white; UAT-09 approval explanatory text is dark on dark green. Existence/text rendering tests miss readability. The A1 blanket high-contrast description is not supported by these captures.
Proposed smallest fix	Inspect computed table/theme variables and style order; scope dark table text/hover text to .course-review-table, resolve the approved-notice paragraph cascade, then capture fresh browser evidence. No WCAG ratio is claimed without measurement.
Status	**Open — visually verified; current-browser retest needed**

F07

Field	Recorded detail
Severity	Medium
Rubric area	Persistence; integration fidelity
File / evidence	app/database.py; SQLite reproduction
Description and why it matters	Actual application SQLite engine reports FK PRAGMA 0 and commits an orphan PlanCourse in a fresh file. Integration fixture enables FK=ON, so its constraint passes do not describe default runtime enforcement. Service validation still protects normal tested selections; this is not proof of a user-facing arbitrary-row exploit.
Proposed smallest fix	Enable SQLite FK enforcement on every app connection and test the real engine factory/configuration; retain existing forced-constraint tests.
Status	**Open — reproduced; no DB/source change**

F08

Field	Recorded detail
Severity	Medium
Rubric area	Supporting A1 functionality; input/error handling
File / evidence	app/routers/register.py, auth_service.py; registration reproduction
Description and why it matters	Anonymous malformed email reaches internal RegularUserCreate validation, yields generic HTTP 500, and leaves DB unchanged. Browser email validation is not a server error contract.
Proposed smallest fix	Validate form data at the request boundary or catch validation errors and return a useful 422/form message; add a no-write invalid-email regression.
Status	**Open — reproduced**

F09

Field	Recorded detail
Severity	Medium
Rubric area	Security — browser writes
File / evidence	security.py, HTML mutation forms/routes
Description and why it matters	Production auth cookie is SameSite=None; no CSRF token/Origin enforcement was found. Cross-site form POSTs have no explicit application-level CSRF check. Actual exploitability depends on browser/deployment cookie policy; no exploit was executed.
Proposed smallest fix	If cross-site cookies are unnecessary, use an appropriate same-site policy; add token or validated Origin protection to authenticated writes and negative tests. Review GET logout separately.
Status	**Open — verified missing control; deployment impact conditional**

F10

Field	Recorded detail
Severity	Medium
Rubric area	Test Reporting — provenance
File / evidence	Old marked/Runner/UAT records; new evidence/audit/
Description and why it matters	Historical screenshots do not identify their executed commit/DB/browser completely. New full rerun now has XML/log/exit/source hashes and matches published app/test files, but cannot retroactively attest old Postman/UAT runs.
Proposed smallest fix	Record build and environment on new executions; preserve each run independently. Do not relabel old screenshots with a new build.
Status	**Partly addressed — fresh pytest evidence added; historical limits remain**

F11

Field	Recorded detail
Severity	Medium
Rubric area	Final deliverables
File / evidence	Original PDF cover; retained student transcript; current A2 PDF
Description and why it matters	Prompt 15 lacked a compiled A2 PDF and verified identity; Prompt 16 recovered the real identity and assembled this document.
Proposed smallest fix	Complete final document checks; retain open High findings and incomplete UAT.
Status	**Partly addressed — identity and local assembly verified; readiness blocked by F01–F05**

F12

Field	Recorded detail
Severity	Low
Rubric area	Documentation consistency/setup
File / evidence	README; Postman README; plans; evidence inventory
Description and why it matters	Clone instructions led to starter repo; Postman README said runtime pending despite recorded execution; historical H3 described current HTML 401 as 200; inventory called UAT-06 partial; two R-level mappings omitted proper cancellation/logout request ranges or UAT-06 Pass.
Proposed smallest fix	Correct verified statements and add current/historical pointers without changing original test inputs/oracles.
Status	**Corrected locally in audit; remote still stale**

F13

Field	Recorded detail
Severity	Low
Rubric area	API test assertion completeness
File / evidence	Postman request 19; collection scripts
Description and why it matters	Percentage range assertions would accept an incorrect in-range progress/category percentage; request 20 discovery alone trusts returned eligibility. Other pytest tests cover literal formulas and known prerequisites, so this is a collection coverage limit rather than evidence those formulas currently fail.
Proposed smallest fix	Add exact independently calculated percentage/category assertions and a known completed-prerequisite positive check in a later collection revision; rerun and retain new Runner results.
Status	**Open — tests retained unchanged**

F14

Field	Recorded detail
Severity	Low
Rubric area	Code/repository quality
File / evidence	Published comp3613a1/, runlog*.txt; .gitignore
Description and why it matters	Full older A1 tree and historical run logs duplicate the app and confuse execution targets. .gitignore protects .env, .venv, caches and database.db, but not every possible temp SQLite file or environment variant.
Proposed smallest fix	Keep top-level target explicit (README corrected); review archival/removal of redundant files; add narrow temp-DB/env patterns that preserve required evidence. No recursive deletion was performed.
Status	**Open cleanup suggestion; target clarified**

F15

Field	Recorded detail
Severity	Medium
Rubric area	Security configuration
File / evidence	.env.example, app/config.py, startup/login
Description and why it matters	Committed example uses a known placeholder signing key; app falls back to it without a production-strength check. Demo credentials are intentionally public. No private credential leak was found, and no production use of the placeholder is established.
Proposed smallest fix	Reject placeholder/weak keys in production, use a generated runtime secret, verify deployment configuration privately, and keep demo accounts confined to demo data.
Status	**Open configuration risk; production state Not verified**

F16

Field	Recorded detail
Severity	Low
Rubric area	Test Reporting/dependency maintenance
File / evidence	Existing screenshots and new pytest log
Description and why it matters	One TestClient/httpx deprecation remains. It is disclosed and does not fail assertions; hiding it or automatically upgrading dependencies would invalidate evidence expectations.
Proposed smallest fix	Retain warning in reports; evaluate a compatible dependency update separately and rerun affected suites if changed.
Status	**Open, disclosed**

11.1 Submission identity and PDF assembly update

F11 is partly addressed: this PDF compiles the A2 materials and the genuine Assignment 1 report-export transcript explicitly supplies the real student name and ID used on the cover; original PDF cover corroboration is recorded in validation. The audit’s earlier request to supply an unknown real ID is therefore superseded by recovered evidence. This update does not close the open authorization, UAT, requirements or publication issues.

11.2 Environment problems, corrections and retests

The Windows pytest temporary-directory/cache issue is an execution-environment problem. Marked screenshots demonstrate successful --basetemp runs; the audit’s fresh safe-basetemp, cache-disabled full run independently passed with exit 0. Historical HTML 401 handler behavior was corrected before the audit; retained integration evidence verifies current 401 status. Documentation errors F12 were corrected locally and are still stale on the published commit. None is misrepresented as a newly fixed application defect.

By contrast, F01 authorization bypass, F07 default-engine foreign keys, F08 malformed registration email and F06 contrast are reproduced application defects. No functional fix or successful retest for them is recorded. F05/DATA-01 is an unresolved requirement: degree total 93 versus categories 36+18+18+24+24=120, with each course stored under one category and no supported overlap/allocation rule. An authoritative curriculum/project decision is required before changing values.

11.3 Audit reproduction records (not new UAT failures)

authorization-reproduction.json

Recorded field	Actual audit observation
date	2026-10-27
scope	QA reproduction using existing F2 fixture, real advisor login, per-probe in-memory SQLite; not a UAT or pytest run
before_plan_count	0
advisor_json_submit_http	403
advisor_html_submit_http	303
created_plan_count	1
created_plan_owner_is_advisor	True
advisor_self_approval_http	200
stored_status	Approved

sqlite-integrity-reproduction.json

Recorded field	Actual audit observation
audit_date	2026-10-27
scope	Actual application SQLite engine in a new disposable file; no existing database opened
foreign_keys_pragma	0
orphan_plan_course_committed	True
test_fixture_difference	tests/integration/conftest.py explicitly enables foreign_keys=ON; app/database.py does not

registration-reproduction.json

Recorded field	Actual audit observation
audit_date	2026-10-27
scope	Anonymous register request against disposable in-memory F2 fixture; not browser UAT
invalid_email_http	500
response_is_generic_internal_error	True
database_unchanged	True

All 19 UAT screenshots preserve observed states. Incomplete checks remain Not Run, no missing evidence is silently replaced, and original low-contrast pixels remain visible. Wider accessibility, mobile behavior, PostgreSQL, concurrency and live deployment are unverified. Original screenshot build/environment attribution remains incomplete (F10). Exported passwords/tokens are blank; credentials and session data are excluded from submission prose.

11.4 Official rubric status at assembly

The following IDs are **audit labels**, not invented official rubric IDs. Wording and weights come from the supplied brief. Satisfied means supported within the stated local scope, not unconditional acceptance of every feature.

I1

Field	Recorded detail
Official component / requirement	Appropriate views implemented to facilitate minimum viable feature set
Weight	Implementation 25%
Classification	Partially satisfied
Evidence and marking risk	WF1–WF3 views and workflows exist; WF1 requirements discrepancy and Advisor table readability remain open (F05/F06).

I2

Field	Recorded detail
Official component / requirement	Authorization implemented
Weight	Same component
Classification	Partially satisfied
Evidence and marking risk	Real role/ownership guards work for JSON and Advisor routes; HTML Student actions allow an Advisor to submit and self-approve (F01).

I3

Field	Recorded detail
Official component / requirement	Views follow REST API conventions
Weight	Same component
Classification	Satisfied
Evidence and marking risk	Resource-based JSON paths, GET reads, POST creation with 201/Location, PUT replacement, DELETE cancellation/removal, structured 401/403/404/409/422 responses. Named Advisor POST actions are documented; legacy HTML forms use redirects. This does not close I2.

P1

Field	Recorded detail
Official component / requirement	Appropriate unit testing of modules in isolation
Weight	Test Plans 25%
Classification	Satisfied
Evidence and marking risk	UT-01–23 specify repository doubles, inputs, independent arithmetic, negative cases and boundaries.

P2

Field	Recorded detail
Official component / requirement	Integration tests covers main functionality of the application
Weight	Same component
Classification	Partially satisfied
Evidence and marking risk	IT-01–20 cover main Student/Advisor interactions and persistence. HTML Advisor-to-Student mutation denial and actual engine FK configuration are missing from the planned scope (F01/F07).

P3

Field	Recorded detail
Official component / requirement	UAT’s give reasonable tasks/success criteria that can be completed/validated by the user
Weight	Same component
Classification	Satisfied
Evidence and marking risk	Three original wireframe panels mapped to concrete browser tasks; 17 original cases plus 3 supporting authentication/access cases. Steps and expected outcomes are executable. Template-specific formatting remains Not verified.

T1

Field	Recorded detail
Official component / requirement	Tests implemented according to test plans
Weight	Test Implementation 25%
Classification	Partially satisfied
Evidence and marking risk	Every UT/IT group has source coverage, with explicit unit/integration boundaries. This is strong implementation traceability, but source-derived oracles leave F01/F05/F07 uncovered; manual execution is incomplete.

T2

Field	Recorded detail
Official component / requirement	Unit & Integration tests are runnable via the CLI
Weight	Same component
Classification	Satisfied
Evidence and marking risk	Existing marked-run screenshots and independent complete-suite execution, exit 0. All 113 unit and 194 integration nodes occur in the new full-run XML.

T3

Field	Recorded detail
Official component / requirement	All postman test scripts implemented for API testing
Weight	Same component
Classification	Satisfied locally
Evidence and marking risk	90 requests, 93 compiled script blocks, all 15 JSON API operations; meaningful state/read-back/error assertions. F13 qualifies calculation completeness.

T4

Field	Recorded detail
Official component / requirement	Postman collection runnable
Weight	Same component
Classification	Partially satisfied
Evidence and marking risk	Local v2.1 JSON plus existing successful Runner evidence. Required JSON is absent from published main (F02); a remote clone cannot follow the import instructions as published.

R1

Field	Recorded detail
Official component / requirement	Screenshots of passing unit and integration tests
Weight	Test Reporting 25%
Classification	Satisfied
Evidence and marking risk	Six readable original screenshots support 113/194 marked runs with one warning each. These six are also published.

R2

Field	Recorded detail
Official component / requirement	Screenshots of passing collection run on postman
Weight	Same component
Classification	Partially satisfied
Evidence and marking risk	Six local readable captures cover requests 01–90 and the same completed 470-assertion run. None is published in the inspected remote tree (F03).

R3

Field	Recorded detail
Official component / requirement	Completed User Acceptance Test report
Weight	Same component
Classification	Partially satisfied
Evidence and marking risk	Detailed honest execution/report records exist; 1 Pass, 18 partial cases, 1 unverified case. Completion/acceptance remains outstanding (F04).

D1

Field	Recorded detail
Official component / requirement	Submit 1 PDF document containing the listed deliverables
Weight	Final deliverable; no separate weight specified
Classification	Compiled locally; acceptance blocked
Evidence and marking risk	Prompt 16 produced the requested single A2 PDF with original artifacts, detailed plans, implementation, results and disclosed limitations. Completed UAT and material audit issues remain unresolved.

No component subtotal or fabricated numerical grade follows from these classifications. Biggest marking risks affect Authorization, runnable published API tests, completed UAT reporting, and the final package.

Assembly update to the historical table: D1 now has a single PDF at the requested path; content/readability checks are recorded in final-submission-checklist.md. Completed UAT, published collection/evidence and authorization remain partial. No numerical grade is inferred. F11 identity/assembly is partly addressed, not a blanket readiness sign-off.

12. Conclusion

MyAdvisor implements the original Student progress/planning and Advisor review workflows and adds substantial automated/API verification. Retained marked evidence shows 113 unit and 194 integration passes; the independently retained complete run shows 715 passes with one warning. The actual Postman iteration executed 90 requests and passed 470 assertions with no assertion failures or execution errors.

Manual evidence demonstrates submission, change requests, revision, approval, rejection, removal, cancellation and replacement states, but complete-case UAT remains 1 Pass, 0 Fail and 19 Not Run. The three-course Mathematics deviation and every documented incomplete check are preserved. Open High audit findings and missing published materials prevent submission-ready or full-acceptance claims. This PDF makes the work reviewable and does not resolve those issues.

13. References and GitHub repository

Submission repository: Camcoon/MyAdvisorWithTests. Public main was rechecked read-only at commit 864386fe10fa5e61869e07500d84e51ff7e46b9b. Local source root is the nested Assignment 2 directory; its origin still points to Camcoon/MyAdvisor, and its working tree includes pre-existing modifications/untracked work. No commit, push, remote change or deployment was performed.

Reference	Source and use
Assignment 2 brief and four-component marking rubric	docs/assignment2/official-assignment-brief.md — Student-supplied official text retained 27 October 2026; no claim of an independent university download.
Original Assignment 1 report and artifacts	docs/report.md; docs/report.pdf; docs/diagrams/use-case.png; docs/wireframes/Three-Panel Student Advisor Dashboard Mockup.png — Original diagram/wireframe preserved; original Mermaid ERD rendered without model changes.
Verified submission identity	docs/transcripts/current-report-export.md (11 October 2026), explicit Student statement; docs/report.pdf cover — Real ID used only from this evidence, never from seeded application data.
Detailed test plans	docs/assignment2/test-plans/uat-test-plan.md; unit-test-plan.md; integration-test-plan.md — Original inputs/oracles and UT/IT identifiers preserved.
Automated and API reports	docs/assignment2/reports/unit-test-report.md; integration-test-report.md; postman-test-report.md — Dated retained runs and evidence qualifications.
Final UAT records	docs/assignment2/reports/uat-execution.md; uat-final-report.md; uat-evidence-review.md; uat-supplemental-evidence.md — Authoritative 1/0/19 totals, actual observations and remaining conditions.
Evidence inventory and traceability	docs/assignment2/reports/evidence-inventory.md; requirements-test-traceability.md — Screenshot paths/case mapping and WF1–WF3 / R-level coverage.
Prompt 15 audit and retained outputs	docs/assignment2/final-quality-audit.md; evidence/audit/pytest-output.txt; pytest-results.xml; pytest-run.json; three reproduction JSONs — Open defects, current full-suite execution and source hashes.
Collection and API contracts	docs/assignment2/postman/MyAdvisor-Assignment2.postman_collection.json; postman/README.md; postman/validate_collection.cjs; api.md — 90 actual request definitions; v2.1 JSON; runtime and static results kept distinct.
Actual implementation	app/models, app/routers, app/services, app/repositories, app/dependencies; tests/unit and tests/integration — Concise examples reproduced from real files; no application or test source edits.

Course-linked templates (access attempted again 27 October 2026; contents inaccessible, not used as verified substantive sources):

Test Plans template and User Acceptance Tests Plan template. Check any additional course-specific template instructions personally before submission.

13.1 Evidence and submission availability

All original execution PNGs are embedded above: 3 Unit, 3 Integration, 6 Postman and 19 UAT. The full-suite raw text is embedded as labelled text; no full-suite screenshot exists. Local linked evidence is checked during assembly; references describe repository-relative source files, not attachments required to read the PDF. Required JSON/current evidence are absent or stale on the rechecked public commit. The final checklist records what remains to publish after correction and review.

Required final artifact: docs/assignment2/COMP3613_Assignment2_MyAdvisor.pdf. Checklist: docs/assignment2/final-submission-checklist.md. Validation artifacts and source/image manifest: docs/assignment2/pdf-review/.